NECESSARY CONDITIONS FOR $*$-EMBEDDABILITY IN PARTITION MONOIDS: NOTES ON THE PROJECTION CONDITION AND A NEW OBSTRUCTION

ABSTRACT. These notes concern regular $*$-semigroups that $*$-embed in a partition monoid $\mathcal{P}_X$. We (1) reformulate the known necessary condition "projections with a common lower bound commute" as "for every projection p, the set $\{x : px = xp = p\}$ is an inverse semigroup" and record some of its consequences; (2) show that the local (completely simple) structure is not constrained at all: every finite completely simple regular $*$-semigroup embeds in some $\mathcal{P}_n$, by an explicit construction, and $\mathcal{P}_4$ already contains non-orthodox completely simple $*$-subsemigroups; (3) prove a family of new necessary conditions indexed by digraphs, the simplest being the *triangle condition* and the *hexagon condition*: quasi-identities in six projection variables which hold in every $\mathcal{P}_X$, are equivalent for finite projection algebras, and fail in a 61-element regular $*$-semigroup that satisfies the projection condition; the witness is a quotient of $\mathcal{P}_3$, and the obstruction is visible in the projection algebra alone; (4) show that the triangle condition is a *master condition*: in every projection algebra it implies all the other digraph conditions, which therefore hold in every $\mathcal{P}_X$; (5) show that the projection condition and the triangle condition are nevertheless *not sufficient*: a 20-element regular $*$-semigroup (a Rees matrix semigroup over the trivial group on the path $a - c - b - d$, with its zero split into a 2×2 rectangular band) satisfies both, and all the digraph conditions, but embeds in no $\mathcal{P}_X$; the obstruction is a new quasi-identity of "wiring" type, valid in every $\mathcal{P}_X$, which mixes matched pairs and commuting pairs, and, from a census of the projection algebras of order 7, a fourth quasi-identity, closing a matched path $a \bowtie c \bowtie b$ to $a \bowtie b$ whenever some projection commutes with a and c and sees b as it sees c, with a 19-element counterexample to the sufficiency of the other three, and, from a census of order 8 (by one-point extensions, which makes orders 8 and 9 accessible), seven further candidate obstructions of order 8 (projection algebras with no embedding into $\mathcal{P}(\mathcal{P}_m)$ for $m \leq 8$) and quasi-identities of three new shapes, valid in $\mathcal{P}(\mathcal{P}_m)$ for $m \leq 7$, that they violate, four of which, the four-variable condition $bcb = b,\ ata \leq b,\ ctc \leq a \Rightarrow ata = btb$, the path-closing condition (R$'$) with two auxiliary projections, which subsumes the fourth quasi-identity, and the path-closing conditions (P_1) and (Q) of length 3, the latter subsuming the wiring condition, we prove in every $\mathcal{P}_X$, so that their witnesses are the fifth to ninth minimal non-embeddable projection algebras, and a tenth, of order 9, follows; for the family of graph semigroups $Q(G, \perp)$ to which the example belongs, and for a layered generalisation of it, the projection condition and the wiring condition are also sufficient, and they say that $\perp$ is a proper colouring whose bichromatic subgraphs are disjoint unions of complete bipartite graphs; (6) report computations: every regular $*$-semigroup of order ≤ 8 satisfying the projection condition embeds in a partition monoid (in $\mathcal{P}_5$ if it is not inverse), and every projection algebra of order ≤ 5 satisfying the projection and triangle conditions embeds in $\mathcal{P}(\mathcal{P}_4)$; and (7) make some general remarks (for finite S the quasi-identities of $\{\mathcal{P}_X\}$ decide embeddability; $\mathcal{P}_X$ embeds in the involuted semigroup of binary relations on 2^X).

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1. Setting and the basic facts about $\mathcal{P}_X$

Throughout, a *regular $*$-semigroup* is a semigroup with a unary operation satisfying $x^{**} = x$, $(xy)^* = y^*x^*$, $xx^*x = x$; its *projections* are the elements $p = p^2 = p^*$, and $\theta_p(q) = pqp$ is the projection algebra operation. For projections p, q we write $p \le q$ for the natural order ($p = pq = qp$, equivalently $p = qpq$). The set of projections of S is $\mathcal{P}(S)$.

1.1. **Partitions.** For a set X let $X' = \{x' : x \in X\}$ be a disjoint copy, and let $\mathcal{P}_X$ be the partition monoid on X: its elements are set partitions α of $X \sqcup X'$, the product $\alpha\beta$ is obtained by identifying the lower points of α with the upper points of β, taking connected components, and forgetting the middle layer; α^* interchanges X and X'. This is a regular $*$-semigroup.

A block of α is *transversal* if it meets both X and X', and *upper* (resp. *lower*) *nontransversal* if it lies in X (resp. X'). The *rank* of α is the number of transversal blocks. The *upper partition* $\ker\alpha$ of α is the partition of X into the upper parts of the blocks of α; dually the lower partition $\operatorname{coker}\alpha$.

A projection p of $\mathcal{P}_X$ is determined by the pair $(\varepsilon_p, \mathsf{T}(p))$, where ε_p is a partition of X and $\mathsf{T}(p)$ is a set of ε_p-classes: the blocks of p are $A \cup A'$ for $A \in \mathsf{T}(p)$ (the T-*blocks*) and B, B' for the remaining classes B (the N-*blocks*); we write $\mathsf{N}(p)$ for the set of these classes and

$$\operatorname{supp}(p) = \bigcup \mathsf{T}(p) \subseteq X.$$

Thus $\operatorname{rank}(p) = |\mathsf{T}(p)|$, and p has rank 0 iff $\mathsf{T}(p) = \emptyset$.

1.2. **Products.** Let $\alpha, \beta \in \mathcal{P}_X$. The classes of the join $\operatorname{coker}\alpha \vee \ker\beta$ (a partition of the middle copy of X) are the connected components of the middle layer; for such a class K let $U_K \subseteq X$ be the union of the upper parts of those transversal blocks of α whose lower parts lie in K, and

let $L_K \subseteq X'$ be the union of the lower parts of those transversal blocks of β whose upper parts lie in K. Then the blocks of $\alpha\beta$ are

- $U_K \cup L_K$ for the classes K with $U_K \neq \emptyset \neq L_K$ (the transversal blocks);
- the upper nontransversal blocks of α, and the sets U_K with $U_K \neq \emptyset = L_K$ (upper nontransversal);
- the lower nontransversal blocks of β, and the sets L_K with $L_K \neq \emptyset = U_K$ (lower nontransversal).

We shall use this only for projections, in the following forms. For projections p, q the middle join is $\varepsilon_p \vee \varepsilon_q$, and U_K (resp. L_K) is the union of the $\mathsf{T}(p)$-blocks (resp. $\mathsf{T}(q)$-blocks) contained in K.

Lemma 1.1. *Let p, q be projections of $\mathcal{P}_X$.*

(a) *The blocks of $\theta_p(q) = pqp$ are: all $\mathsf{N}(p)$-blocks, and the sets U_K for the classes K of $\varepsilon_p \vee \varepsilon_q$ meeting* $\operatorname{supp}(p)$*; the latter is a T-block of $\theta_p(q)$ iff K meets* $\operatorname{supp}(q)$*. In particular the blocks of $\theta_p(q)$ not contained in* $\operatorname{supp}(p)$ *are exactly the $\mathsf{N}(p)$-blocks. Consequently, if $e \leq p$ then* $\operatorname{supp}(e) \subseteq \operatorname{supp}(p)$ *and every block of e not contained in* $\operatorname{supp}(p)$ *is an $\mathsf{N}(p)$-block.*

(b) *Suppose $pq = qp$. Then every class K of $\varepsilon_p \vee \varepsilon_q$ is of one of the following types:* (i) *$K = U_K = L_K$;* (ii) *$K = U_K$ is a single $\mathsf{N}(q)$-block;* (iii) *$K = L_K$ is a single $\mathsf{N}(p)$-block;* (iv) *K is a single block of both p and q. The T-blocks of pq are the classes of type* (i)*; hence* $\operatorname{supp}(p) \cap \operatorname{supp}(q) = \operatorname{supp}(pq)$*, and every $\mathsf{N}(q)$-block contained in* $\operatorname{supp}(p)$ *is a union of $\mathsf{T}(p)$-blocks.*

(c) *Suppose $pqp = p$ and $qpq = q$. Then every class of $\varepsilon_p \vee \varepsilon_q$ meeting* $\operatorname{supp}(p) \cup \operatorname{supp}(q)$ *contains exactly one $\mathsf{T}(p)$-block and exactly one $\mathsf{T}(q)$-block. In particular* $\operatorname{rank} p = \operatorname{rank} q$*, and no $\mathsf{T}(p)$-block is an $\mathsf{N}(q)$-block.*

Proof. (a) $\theta_p(q) = (pq)(pq)^*$ is the projection whose T-blocks are the upper parts of the transversal blocks of pq and whose N-blocks are the upper nontransversal blocks of pq; now read off the description of pq. For the last sentence use $e = \theta_p(e)$.

(b) Since pq is symmetric, $U_K \cup L_K$ is a block only if $U_K = L_K$ (when both are nonempty), and the family of upper nontransversal blocks $\mathsf{N}(p) \cup \{U_K : L_K = \emptyset \neq U_K\}$ equals the family of lower ones $\mathsf{N}(q) \cup \{L_K : U_K = \emptyset \neq L_K\}$. Let K be a class. If $U_K \neq \emptyset \neq L_K$, an $\mathsf{N}(p)$-block inside K would be an N-block of pq, hence (as $U_K \neq \emptyset$) an $\mathsf{N}(q)$-block; but a set that is a block of both p and q is a class by itself, so K contains no $\mathsf{N}(p)$-blocks and dually no $\mathsf{N}(q)$-blocks: $K = U_K = L_K$, type (i). If $U_K \neq \emptyset = L_K$ then U_K is an $\mathsf{N}(q)$-block; K is a union of q-blocks containing the q-block U_K and p-blocks; as before the remaining p-blocks in K would be common blocks, so $K = U_K$, type (ii). Type (iii) is dual, and if $U_K = L_K = \emptyset$ then K consists of $\mathsf{N}(p)$- and $\mathsf{N}(q)$-blocks, which must coincide as above, type (iv). Finally, an $\mathsf{N}(q)$-block $B \subseteq \operatorname{supp}(p)$ lies in a class K; K is not of type (i) or (iii) (these consist of $\mathsf{T}(q)$-blocks only), nor of type (iv) (B meets $\operatorname{supp}(p)$), so $K = B$ is of type (ii).

(c) $pqp = p$ says that the upper projection of pq is p: its T-blocks $\{U_K : U_K \neq \emptyset \neq L_K\}$ are exactly the $\mathsf{T}(p)$-blocks, and its N-blocks $\mathsf{N}(p) \cup \{U_K : U_K \neq \emptyset = L_K\}$ are exactly the $\mathsf{N}(p)$-blocks. So every class meeting $\operatorname{supp}(p)$ contains exactly one $\mathsf{T}(p)$-block and at least one $\mathsf{T}(q)$-block; by symmetry, exactly one of each. A $\mathsf{T}(p)$-block which is an $\mathsf{N}(q)$-block would be a class by itself, containing no $\mathsf{T}(q)$-block. □

We say that projections p, q form a *matched pair*, $p \bowtie q$, if $pqp = p$ and $qpq = q$; by (c) this means that p and q have the same rank and their product has that rank (in the trace of the $\mathcal{D}$-class, $pq \neq 0$), and it gives a bijection $\mathsf{T}(p) \to \mathsf{T}(q)$ ("the block in the same class"). This is exactly the *friendly* relation $p \mathcal{F} q$ of East, Gray, Azeef Muhammed and Ruškuc (2025), defined

by $pqp = p$ and $qpq = q$; we keep the symbol $\bowtie$ and the word "matched" to emphasise the bijection of T-blocks, but the reader may read $\bowtie$ as $\mathcal{F}$ throughout. Note that $\mathcal{F}$ is reflexive and symmetric but not transitive, even in $\mathcal{P}(\mathcal{P}_X)$: in $\mathcal{P}_2$ the projection p with the single T-block $\{1,2\}$ is matched with $q = (\{1\}\ \mathsf{T},\ \{2\}\ \mathsf{N})$ and with $r = (\{2\}\ \mathsf{T},\ \{1\}\ \mathsf{N})$, but qrq has rank 0 (a T-block and its partner need not meet, they need only lie in the same class of the join).

2. The projection condition

The known necessary condition is

$$r = prp \ \&\ r = qrq \implies pq = qp \qquad (p, q, r \text{ projections}), \tag{Q1}$$

equivalently: an idempotent lying above a projection in the natural order is a projection. We record some reformulations and consequences.

Proposition 2.1. *For a regular $*$-semigroup S the following are equivalent.*

1. *S satisfies (Q1).*
2. *For every projection p, the set $S_p = \{x \in S : px = xp = p\}$ (the elements above p in the natural order) is an inverse semigroup.*
3. *Two projections commute if and only if they have a common lower bound in $\mathcal{P}(S)$.*
4. *The projection algebra satisfies the quasi-identity $\theta_p(r) = r = \theta_q(r) \Rightarrow \theta_p(q) = \theta_q(p)$.*

Proof. S_p is closed under multiplication and under $*$ (if $px = xp = p$ then $px^* = (xp)^* = p$), so it is a regular $*$-semigroup with zero p; it is inverse iff all its idempotents are projections, and its idempotents are exactly the idempotents $e \geq p$. This gives (1)$\Leftrightarrow$(2). For (3): if $pq = qp$ then pq is a projection below p and q; the converse is (Q1). For (4): for projections p, q, $\theta_p(q) = \theta_q(p)$ iff $pq = qp$ (if $pqp = qpq =: r$ then $pq = (pq)^2 = pr = r$ and $qp = rp = r$). □

Remark 2.2 (The condition in $\mathcal{P}_X$)**.** In $\mathcal{P}_X$, an element x lies above a projection p iff x maps every $\mathsf{T}(p)$-block onto itself by a "block bijection" (a partition of $A \sqcup A'$ into transversal blocks) and permutes the $\mathsf{N}(p)$-blocks among themselves in the same way, so that $(\mathcal{P}_X)_p \cong \prod_{A \in \mathsf{T}(p)} \mathcal{I}_A^* \times \mathcal{J}_p$, with $\mathcal{I}_A^*$ the dual symmetric inverse monoid on A and $\mathcal{J}_p$ an inverse monoid of partial block bijections between $\mathsf{N}(p)$-blocks. For p the projection with all blocks singletons and nontransversal, $(\mathcal{P}_X)_p$ is the symmetric inverse monoid $\mathcal{I}_X$. So (Q1) is exactly the statement that these "unitary" subsemigroups are inverse, which is why it resembles E-unitarity.

Some consequences of (Q1) which may be useful in the search for further conditions:

- For every projection r, the projections $\geq r$ pairwise commute; so every principal up-set of $\mathcal{P}(S)$ is a semilattice under multiplication.
- Two distinct projections in the same $\mathcal{D}$-class with nonzero product ($p \bowtie q$, $p \neq q$) have no common lower bound (they would commute and then $p = pqp = pq = qp = q$). In particular a completely 0-simple $*$-subsemigroup of S that contains its own zero is a 0-direct union of groups (a Brandt semigroup), and if S has a zero then (Q1) forces S to be inverse.
- If $x \geq p$ then xx^* and x^*x commute, being projections above p.

3. The local structure is unconstrained

A natural first guess for a second condition is a restriction on the "local" pieces of S, i.e. on its principal factors (the traces of its $\mathcal{D}$-classes), which are completely 0-simple regular $*$-semigroups. Some care is needed in saying what such a restriction could be. The zero of a principal factor is not in general an element of S, and the second consequence listed above shows that when it is – that is, when S contains a completely 0-simple $*$-subsemigroup together with its zero

– (Q1) already forces that subsemigroup to be a Brandt semigroup. So the zero pattern of a principal factor (which pairs p, q of its projections have $pq = 0$) is not local information: it is decided by how the $\mathcal{D}$-class sits inside S, and it does interact with the rest of S, as the hexagon condition of Section 4 shows. What remains genuinely local is the nonzero part of a principal factor – a set of projections of a $\mathcal{D}$-class with pairwise nonzero products, together with the corresponding sandwich entries and the maximal subgroup. These are exactly the completely simple $*$-subsemigroups of S (conversely, every completely simple $*$-subsemigroup of S lies in a single $\mathcal{D}$-class and arises this way), and any condition on them would have to be a condition on finite completely simple regular $*$-semigroups. The next theorem shows there is none: every one of them embeds.

Theorem 3.1. *Every finite completely simple regular $*$-semigroup embeds in a finite partition monoid. Explicitly, let $S = \mathcal{M}[G; I, I; P]$ be in normal form ($p_{ii} = 1$, $p_{ji} = p_{ij}^{-1}$, $(i, g, j)^* = (j, g^{-1}, i)$), let $G \leq \mathrm{Sym}(k)$ act on $\{1, \dots, k\}$ on the right, and let*

$$X = \{a_{i,s} : i \in I,\ s \leq k\} \sqcup \{c_{ij,s} : i \neq j \in I,\ s \leq k\}, \qquad |X| = k|I|^2.$$

Send (i, g, j) to the partition with transversal blocks $\{a_{i,s}, a'_{j,sg}\}$ ($s \leq k$), upper nontransversal blocks $\{c_{il,s}, a_{l,sp_{il}}, c_{li,sp_{il}}\}$ ($l \neq i$, $s \leq k$) together with singletons $\{c_{lm,s}\}$ for $i \notin \{l, m\}$, and lower nontransversal blocks defined in the same way from j. This is an injective $*$-homomorphism $S \to \mathcal{P}_X$.

Proof sketch. The transversal blocks of the image of $e_i = (i, 1, i)$ are the singletons $a_{i,s}$, so the images of the e_i are projections of rank k. In the middle layer of $e_i e_j$ the point $a_{i,s}$ lies in the e_j-block $\{c_{ji,t}, a_{i,s}, c_{ij,s}\}$ (with $s = tp_{ji}$), and $c_{ij,s}$ lies in the e_i-block $\{c_{ij,s}, a_{j,sp_{ij}}, c_{ji,sp_{ij}}\}$; no other blocks are attached (all remaining connectors are singletons for e_i or e_j), so the class of $a_{i,s}$ is $\{a_{i,s}, c_{ij,s}, c_{ji,sp_{ij}}, a_{j,sp_{ij}}\}$. Hence $e_i e_j$ has rank k and its matching $\mathsf{T}(e_i) \to \mathsf{T}(e_j)$ is $s \mapsto sp_{ij}$. Classes containing no $a_{i,\cdot}$ or $a_{j,\cdot}$ contribute nothing. The images of the group elements are the elements of the rank-k $\mathcal{H}$-classes of $\mathcal{P}_X$ determined by the permutations, and the multiplication $(i, g, j)(l, h, m) = (i, gp_{jl}h, m)$ is then read off block by block. Injectivity is clear. (All of this was also verified by computer for several nonabelian and non-flat sandwich matrices.) □

Since every finite group embeds in some $\mathrm{Sym}(k)$, every finite completely simple regular $*$-semigroup embeds. Two remarks on what this rules out:

Remark 3.2. Orthodoxy of the completely simple $*$-subsemigroups is not necessary either, even though $\mathcal{P}_n$ for small n looks "flat": the projections

$$p : \{0\}, \{1\} \text{ transversal},\ \{2, 3\} \text{ nontransversal};$$
$$q : \{1\}, \{3\} \text{ transversal},\ \{0, 2\} \text{ nontransversal};$$
$$r : \{0\}, \{3\} \text{ transversal},\ \{1, 2\} \text{ nontransversal}$$

of $\mathcal{P}_4$ satisfy $pqp = p$, $qrq = q$, $rpr = r$ (and the symmetric identities), but $pqr \neq pr$: the sandwich matrix of the rank-2 $\mathcal{D}$-class of $\mathcal{P}_4$ has a non-flat triangle, and the $*$-subsemigroup generated by p, q, r is a non-orthodox completely simple regular $*$-semigroup $\mathcal{M}[\mathbb{Z}_2; 3, 3; P]$ with $p_{12} \neq 1$.

Remark 3.3 (Normal $*$-bands)**.** For a strong semilattice Y of square $*$-bands $A_\alpha \times A_\alpha$ with structure maps $f_{\alpha\beta} \colon A_\alpha \to A_\beta$ ($\alpha \geq \beta$), (Q1) holds iff all $f_{\alpha\beta}$ are injective (a projection (c, c) of a lower component lies below (a, b) iff $f(a) = f(b) = c$). If Y has a least element 0, injectivity lets us regard every A_α as a subset of $A = A_0$, and then the band is the $*$-subsemigroup $\{(\alpha, a, b) : a, b \in A_\alpha\}$ of $Y \times (A \times A)$. Since semilattices and square bands embed in partition monoids and $\mathcal{P}_X \times \mathcal{P}_Y \leq \mathcal{P}_{X \sqcup Y}$, we get: *a finite normal $*$-band embeds in a partition monoid iff it*

satisfies (Q1). Several strong semilattices of completely simple $*$-semigroups with non-injective group maps were also found to embed by computer.

4. The triangle condition and its relatives

We now give a family of new necessary conditions. Each is a quasi-identity in $2k$ projection variables ($k \geq 3$), and they are independent of (Q1). The family is indexed by a pattern of matched pairs, which we encode as a digraph.

Definition 4.1. Let $k \geq 3$. A set E of ordered pairs (i, j) of distinct elements of $[k] = \{1, \ldots, k\}$ (a loopless digraph on $[k]$) is *admissible* if

(i) every vertex has an out-edge and an in-edge in E, and
(ii) for every $m \in [k]$ there is an edge $(i, j) \in E$ with $m \notin \{i, j\}$ (no vertex is incident to all edges).

Theorem 4.2 (Matching-pattern condition)**.** *Let E be an admissible digraph on $[k]$, and let $a_1, \ldots, a_k, b_1, \ldots, b_k$ be projections of a partition monoid $\mathcal{P}_X$ such that*

(H1) *$a_i a_j = a_j a_i =: w$ and $b_i b_j = b_j b_i =: z$ for all $i \neq j$, the products being independent of the pair $\{i, j\}$;*
(H2) *$a_i b_i = b_i a_i$ for all i;*
(H3) *$a_i b_j a_i = a_i$ and $b_j a_i b_j = b_j$ for all $(i, j) \in E$.*

Then $w = z$. Consequently $a_i = b_j$ for all $(i, j) \in E$, and if the bipartite graph $\{a_i — b_j : (i, j) \in E\}$ is connected then all $2k$ projections coincide.

The hypotheses say: the a_i are k pairwise commuting projections with common meet w, the b_i are k pairwise commuting projections with common meet z; a_i and b_i commute; and $a_i \bowtie b_j$ for $(i, j) \in E$. All of them are equations of the projection algebra (commutation of projections is expressible by $\theta_p(q) = \theta_q(p)$: if $pqp = qpq =: w$ then $pq \in H_w$ and $w \cdot pq = (pq)^2$, so pq is an idempotent of the group H_w, i.e. $pq = w = qp$), so each instance of the theorem is a quasi-identity of the projection algebras of partition monoids. The instances of most interest are:

- $k = 3$, $E =$ all six pairs: the *hexagon condition* (the six matched pairs form the hexagon $a_1\, b_2\, a_3\, b_1\, a_2\, b_3$). This was the original form of the theorem.
- $k = 3$, $E = \{(1, 2), (2, 3), (3, 1)\}$, a directed 3-cycle: the *triangle condition*. Only three matched pairs are assumed, $a_1 \bowtie b_2$, $a_2 \bowtie b_3$, $a_3 \bowtie b_1$, so this instance has strictly weaker hypotheses than the hexagon condition and implies it. It is the unique admissible digraph on three vertices with three edges (a derangement of [3] is a 3-cycle), hence the strongest instance with $k = 3$.
- $k = 4$, E a directed 4-cycle $\{(1, 2), (2, 3), (3, 4), (4, 1)\}$; and $k = 4$, E two disjoint 2-cycles $\{(1, 2), (2, 1), (3, 4), (4, 3)\}$ (the "double square").

Restricting to a subset of at least three indices only removes hypotheses, so an instance of the theorem follows from every instance obtained by deleting indices. Every admissible digraph contains, on some subset of its vertices, a directed cycle of length ≥ 3 or two vertex-disjoint 2-cycles (if all directed cycles are 2-cycles and any two of them share a vertex, they all share one vertex v, or three of them form a directed triangle; in the former case take an edge (p, q) avoiding v, follow out-edges from q until v is reached and in-edges from p back to v: the resulting closed walk through v contains no vertex twice, since a repetition would produce a cycle avoiding v, so it is a directed cycle of length ≥ 3). Both of these are admissible. Hence every instance of the theorem is implied by one of the *basic* instances: a directed ℓ-cycle C_ℓ ($\ell \geq 3$), or the double square D_2. In fact the triangle condition alone implies every non-star instance, in every projection algebra (Theorem 4.14), so only the case $E = C_3$ of the theorem needs a combinatorial

proof. For finite projection algebras the hexagon condition implies all these instances as well (Theorem 4.9 below), although the implications are not equational ones; in general projection algebras it does not even imply the triangle condition (Theorem 4.17).

Write $m_i := a_ib_i = b_ia_i$. The proof uses the following "transport" relations, which are formal consequences of (H1)–(H3); here $(i,j) \in E$:

$$a_im_ja_i = w, \qquad b_jm_ib_j = z, \qquad a_iza_i = m_i, \qquad b_jwb_j = m_j. \tag{1}$$

Indeed $a_im_ja_i = a_ia_jb_ja_i = a_j(a_ib_ja_i) = a_ja_i = w$; $b_jm_ib_j = b_jb_ia_ib_j = b_i(b_ja_ib_j) = b_ib_j = z$; $a_iza_i = a_ib_jb_ia_i = (a_ib_ja_i)b_i = m_i$; and $b_jwb_j = b_ja_ia_jb_j = (b_ja_ib_j)a_j = m_j$. By admissibility (i), every index i has an out-edge and an in-edge, so for every i there are j, j' with $a_im_ja_i = w$, $a_iza_i = m_i$, $b_im_{j'}b_i = z$ and $b_iwb_i = m_i$. (In an earlier version of these notes (1) was listed as a fourth hypothesis of the hexagon condition; that it is redundant was pointed out by Prover9, working from the axioms of projection algebras alone.)

For the hexagon condition Prover9 also shows that the hypotheses can be pared down to 21 equations: in (H1) it suffices to assume the six "cyclic" equations $a_2a_1a_2 = a_3a_2a_3 = a_1a_3a_1$ and $b_2b_1b_2 = b_3b_2b_3 = b_1b_3b_1$ (if w denotes the common value of the first three, then $a_1a_2a_1 = (a_1a_2a_1)^2 = a_1(a_2a_1a_2)a_1 = a_1wa_1 = a_1(a_3a_1a_3)a_1 = (a_1a_3a_1)^2 = w$, and similarly $a_3a_2a_3 = a_1a_3a_1 = w$, so the a_i pairwise commute with product w by the remark above); (H2) consists of the three equations $a_ib_ia_i = b_ia_ib_i$; and (H3) consists of twelve equations. The same reduction of (H1) applies to every instance; for the triangle condition the hypotheses are then $6+3+6 = 15$ equations. The Prover9 inputs are the files `hexagon.in`, `triangle.in` and `octagon.in`.

Proof. Write $A_i = \mathrm{supp}(a_i)$, $B_i = \mathrm{supp}(b_i)$, $W = \mathrm{supp}(w)$, $Z = \mathrm{supp}(z)$, and E_p for the partition of X underlying a projection p (its blocks are the T-blocks and N-blocks of p). We use Lemma 1.1 throughout.

Step 0: supports. By (H1) and Lemma 1.1(b), $A_i \cap A_j = W$ and $B_i \cap B_j = Z$ for $i \neq j$; by (H2), $A_i \cap B_i = \mathrm{supp}(m_i)$. In particular a point of X lies in at most one A_i unless it lies in W (and then in all of them), and similarly for the B_i and Z. Since $k \geq 3$,

$$\text{every } x \notin W \cup Z \text{ lies outside } A_i \cup B_i \text{ for some } i. \tag{2}$$

Step 1: E_w and E_z agree outside $W \cup Z$. By (1) and Lemma 1.1(a) (applied to $z = \theta_{b_i}(m_{j'})$, $w = \theta_{a_i}(m_j)$, $m_i = \theta_{a_i}(z)$, $m_i = \theta_{b_i}(w)$), for every i: the blocks of E_z not contained in B_i are exactly the $\mathsf{N}(b_i)$-blocks; the blocks of E_w not contained in A_i are exactly the $\mathsf{N}(a_i)$-blocks; and the blocks of E_{m_i} not contained in A_i (resp. B_i) are exactly the $\mathsf{N}(a_i)$-blocks (resp. $\mathsf{N}(b_i)$-blocks). Consequently, for $x \notin A_i \cup B_i$, the block of E_z containing x is the $\mathsf{N}(b_i)$-block of x, which is the block of E_{m_i} containing x, which is the $\mathsf{N}(a_i)$-block of x, which is the block of E_w containing x. By (2), E_w and E_z have the same block at every point of $X \setminus (W \cup Z)$; moreover such a common block is an $\mathsf{N}(a_i)$-block and an $\mathsf{N}(b_i)$-block, hence contained in $X \setminus (A_i \cup B_i) \subseteq X \setminus (W \cup Z)$. It follows that a block of E_w or of E_z is either a common block contained in $X \setminus (W \cup Z)$, or is contained in $W \cup Z$.

Step 2: matching of $\mathsf{T}(w)$ and $\mathsf{T}(z)$. Fix an edge $(i,j) \in E$. From (H1) and (1), $wzw = a_i(a_jza_j)a_i = a_im_ja_i = w$ and $zwz = b_j(b_iwb_i)b_j = b_jm_ib_j = z$, so $w \bowtie z$. By Lemma 1.1(c), every class of $E_w \vee E_z$ meeting $W \cup Z$ contains exactly one $\mathsf{T}(w)$-block and exactly one $\mathsf{T}(z)$-block. Common blocks are classes by themselves, so the class of a $\mathsf{T}(w)$-block Ω consists of blocks contained in $W \cup Z$: these are $\mathsf{T}(w)$-blocks, $\mathsf{N}(w)$-blocks contained in $Z \setminus W$, $\mathsf{T}(z)$-blocks and $\mathsf{N}(z)$-blocks contained in $W \setminus Z$. An $\mathsf{N}(z)$-block contained in W meets only $\mathsf{T}(w)$-blocks, so it meets only one of them (two would lie in a common class); hence every $\mathsf{N}(z)$-block contained in W lies inside a single $\mathsf{T}(w)$-block, and dually. Therefore the class of Ω is

$$\{\Omega\} \cup \{\mathsf{N}(z)\text{-blocks inside } \Omega\} \cup \{\mathsf{T}(z)\text{-blocks meeting } \Omega\} \cup \{\mathsf{N}(w)\text{-blocks inside those}\},$$

and since it contains exactly one $\mathsf{T}(z)$-block, *there is exactly one* $\mathsf{T}(z)$*-block* Z_Ω *meeting* Ω, and dually exactly one $\mathsf{T}(w)$-block W_Θ meeting a given $\mathsf{T}(z)$-block Θ.

Step 3: $\Omega \subseteq Z_\Omega$. Suppose $y \in \Omega \setminus Z_\Omega$. Then $y \notin Z$ (a $\mathsf{T}(z)$-block containing y would meet Ω, hence be Z_Ω), so the block N of E_z containing y is an $\mathsf{N}(z)$-block, and by Step 2, $N \subseteq \Omega \subseteq W \subseteq A_1 \cap \cdots \cap A_k$. For each i, N is a block of $E_z = \theta_{b_i}(m_{j'})$ (with $(j', i) \in E$), so by Lemma 1.1(a) either $N \subseteq B_i$ or N is an $\mathsf{N}(b_i)$-block. As $N \cap Z = \emptyset$ and $B_i \cap B_l = Z$, $N \subseteq B_i$ holds for at most one index $i = m$; by admissibility (ii) there is an edge $(i, j) \in E$ with $m \notin \{i, j\}$, so that N is both an $\mathsf{N}(b_i)$-block and an $\mathsf{N}(b_j)$-block. Since $N \subseteq A_i$ and $a_i b_i = b_i a_i$, Lemma 1.1(b) shows that N is a union of $\mathsf{T}(a_i)$-blocks; let τ be one of them. Now $a_i \bowtie b_j$ by (H3). The class of $\varepsilon_{a_i} \vee \varepsilon_{b_j}$ containing τ contains the b_j-block $N \supseteq \tau$, hence all $\mathsf{T}(a_i)$-blocks inside N; by Lemma 1.1(c) there is only one, so $N = \tau$. But then N is simultaneously a $\mathsf{T}(a_i)$-block and an $\mathsf{N}(b_j)$-block, which is impossible for a matched pair by Lemma 1.1(c). Hence $\Omega \subseteq Z_\Omega$.

Step 4: conclusion. Dually $Z_\Omega \subseteq W_{Z_\Omega}$ (the dual argument uses the same edge (i, j) with the roles of a and b exchanged: an $\mathsf{N}(w)$-block $N \subseteq Z \subseteq B_j$ is a union of $\mathsf{T}(b_j)$-blocks because b_j commutes with a_j, and $b_j \bowtie a_i$); since Ω meets Z_Ω and W_{Z_Ω} is the only $\mathsf{T}(w)$-block meeting Z_Ω, we get $W_{Z_\Omega} = \Omega$ and so $\Omega = Z_\Omega$. Thus $\mathsf{T}(w) = \mathsf{T}(z)$ and $W = Z$. Inside $W = Z$ both E_w and E_z consist of these T-blocks, and outside $W \cup Z$ they agree by Step 1. Hence $E_w = E_z$ and $\mathsf{T}(w) = \mathsf{T}(z)$, i.e. $w = z$.

Finally, $w = z$ is a common lower bound of a_i and b_j, so by (Q1) (which holds in $\mathcal{P}_X$) $a_i b_j = b_j a_i$ for all i, j; for $(i, j) \in E$ this gives $a_i = a_i b_j a_i = a_i b_j$, i.e. $a_i \leq b_j$, and dually $b_j \leq a_i$, so $a_i = b_j$. If the bipartite graph of E is connected, all $2k$ projections coincide. □

Remark 4.3. (1) The hypothesis $k \geq 3$ is needed: for $k = 2$ (two matched pairs $a_1 \bowtie b_2$, $a_2 \bowtie b_1$, the "square") the statement is false, and $\mathcal{P}_2$ already contains counterexamples (`digraph.py`). The admissibility conditions, on the other hand, are conditions of the proof rather than of the statement: for every X the exact condition is that E is not a star (Theorems 4.11 and 4.14 below; for instance the $k = 3$ "star" $\{(1, 2), (2, 1), (1, 3), (3, 1)\}$ – a star in the graph-theoretic but not in our sense, since it has two tails and two heads – the transitive tournament $\{(1, 2), (2, 3), (1, 3)\}$, the path $\{(1, 2), (2, 3)\}$ and the two disjoint pairs $\{(1, 2), (3, 4)\}$ are all valid in $\mathcal{P}_n$ for finite n; an earlier version of this remark wrongly reported counterexamples to the first two, due to a bug in the search program).

(2) Exhaustive search (allowing coincidences among the variables) confirms that the hexagon, triangle, directed 4-cycle, double-square and cyclic-octagon instances have no counterexamples among the projections of $\mathcal{P}_n$ for $n \leq 4$ (`digraph.py`, `octcheck.py`).

(3) For finite algebras the hexagon condition implies all the others (Theorem 4.9) and is equivalent to the conditions on three indices; in general it does not even imply the triangle condition (Theorem 4.17 below), while the conditions on four or more indices do not imply the hexagon condition even for finite algebras ($\mathcal{P}(Q_0)$). Whether the hexagon condition implies C_4, D_2 or the cyclic octagon condition in general is open. A separating model for "hexagon $\not\Rightarrow$ octagon" (or "hexagon $\not\Rightarrow$ triangle") cannot be small: in any regular $*$-semigroup with the hexagon hypotheses, a pair a_i, b_j $(i \neq j)$ that is *not* assumed matched can be neither matched (else the configuration contains a hexagon) nor commuting: in the cyclic octagon, if a_i and b_{i+2} commuted with product u, then $a_{i+2} u a_{i+2} = (a_{i+2} a_i)(b_{i+2} a_{i+2}) = w m_{i+2}$ would be a projection, so w and m_{i+2} would commute; but $w \bowtie m_l$ always ($w m_l w = wzw = w$ and $m_l w m_l = a_l z w z a_l = m_l$ by (1)), whence $w = m_{i+2}$, a degenerate case. Mace4 experiments by M. Kinyon find that the smallest projection algebra violating the cyclic octagon condition has 14 elements – the 8 vertices, w, z and the four m_i – and that it violates the hexagon condition as well; indeed, of the 3^8 ways to complete such a 14-element table, exactly one is a projection algebra, and in it every antipodal pair a_i, b_{i+2} is matched (`oct14.py`). So a separating example

needs further projections $a_ib_ja_i$ in "general position". On the partition side, the quotients $\mathcal{P}_4/\mu_1$ (57 projections) and $\mathcal{P}_5/\mu_1$ (303 projections) contain hexagon configurations with $w \neq z$ but no octagon configurations; $\mathcal{P}_5$ contains no octagon configuration with eight distinct vertices even when the products in (H1) are only required to commute, while $\mathcal{P}_6$ does.

4.1. **The hexagon condition implies the triangle condition (finite case).** Since the triangle condition assumes only three of the six matched pairs of the hexagon condition, it implies the hexagon condition trivially. The converse is true for finite projection algebras, but for a rather subtle reason: no fixed sextuple of terms in $a_1, \dots, b_3$ satisfies the hexagon hypotheses in every triangle configuration (see Remark 4.10), and the six projections to which the hexagon condition has to be applied are found by an iteration. Throughout this subsection $a_1, a_2, a_3, b_1, b_2, b_3$ is a triangle configuration ($a_1 \bowtie b_2$, $a_2 \bowtie b_3$, $a_3 \bowtie b_1$, and (H1), (H2)) in a projection algebra $P = \mathcal{P}(S)$, w and z are the products, and $m_i = a_ib_i$. Indices are mod 3. We call a projection x *bounded* if $x \leq a_i$ and $w, m_i \leq x$ for some i, or $x \leq b_j$ and $z, m_j \leq x$ for some j; e.g. the six vertices are bounded.

We use freely the transport relations (1), which by the argument of Lemma 4.4 hold for all $i \neq j$.

Lemma 4.4. *$b_jm_ib_j = z$ and $a_jm_ia_j = w$ for all $i \neq j$.*

Proof. Put $e = b_jm_ib_j$. Writing $m_i = b_ia_i$ and $b_jb_i = z$ gives $e = za_ib_j$; writing $m_i = a_ib_i$ gives $e = b_ja_iz$. Hence $e = ze = ez$, so $e \leq z$. Conversely $zez = zm_iz = za_iz = z$ because $z \leq b_{i+1} \bowtie a_i$ (here index i has an out-edge); so $z \leq e$ and $e = z$. Dually for w. □

Lemma 4.5 (bounded elements)**.** *Let $e \leq a_i$, $e' \leq a_j$ with $i \neq j$, and $w \leq e, e'$. Then $ee' = e'e = w$. Dually two bounded elements under distinct b's commute with product z. If $e \leq a_i$, $f \leq b_i$ and $m_i \leq e, f$, then $ef = fe = m_i$.*

Proof. $ee'e = e(a_ia_j)e'(a_ja_i)e = ewe'we = ewe = w$, and symmetrically $e'ee' = w$; two projections with $pqp = qpq = w$ commute with product w (Section 4). Likewise $efe = e(a_ib_i)f(b_ia_i)e = em_ifm_ie = m_i = fef$. □

Thus any six bounded projections, one under each vertex, satisfy the commutation hypotheses (H1), (H2) of the hexagon condition with the same w and z; only the six cross matchings are at issue. They will come from the following *walk*: for a projection $x \leq a_1$ put

$$x \xrightarrow{\theta_{b_3}} x_1 \xrightarrow{\theta_{a_2}} x_2 \xrightarrow{\theta_{b_1}} x_3 \xrightarrow{\theta_{a_3}} x_4 \xrightarrow{\theta_{b_2}} x_5 \xrightarrow{\theta_{a_1}} x_6 =: \Phi(x) \leq a_1,$$

alternately projecting onto the free partner ($a_i \mapsto b_{i-1}$) and transporting through a matched pair ($b_j \mapsto a_{j-1}$, where $a_{j-1} \bowtie b_j$). The six elements $x_1, \dots, x_6$ lie under $b_3, a_2, b_1, a_3, b_2, a_1$ respectively.

Lemma 4.6 (invariance)**.** *The walk preserves boundedness: if $x \leq a_i$ is bounded then so is $\theta_{b_{i-1}}(x)$, and if $y \leq b_j$ is bounded then so is $\theta_{a_{j-1}}(y)$.*

Proof. Let $y = \theta_b(x)$ with $b = b_{i-1}$. Since $z \leq b$, $yzy = b\,x(bzb)x\,b = b(xzx)b$, and $xzx = x(a_iza_i)x = xm_ix = m_i$, so $yzy = bm_ib = z$, i.e. $z \leq y$. Similarly $ym_{i-1}y = b(xm_{i-1}x)b = b(x(a_im_{i-1}a_i)x)b = b(xwx)b = bwb = m_{i-1}$. The transport step is the same computation with the roles of a and b exchanged: for $x' = \theta_a(y)$, $a = a_{j-1}$, one has $x'wx' = a(ywy)a = a(y(b_jwb_j)y)a = am_ja = w$ and $x'm_{j-1}x' = a(y(b_jm_{j-1}b_j)y)a = aza = m_{j-1}$. □

Lemma 4.7 (monotonicity)**.** *In every projection algebra, $x \leq y$ implies $\theta_p(x) \leq \theta_p(y)$.*

Proof. In the binary notation $u * v = u\theta_v$, the axioms imply $(x * y) * (z * y) = x * (z * y)$ and $(x * y) * (y * z) = (x * y) * z$ (both are found by Prover9 in a few steps from (P2)–(P5)). If $x \leq y$,

i.e. $x * y = x$, then $(x * p) * (y * p) = x * (y * p) = (x * y) * (y * p) = (x * y) * p = x * p$, i.e. $\theta_p(x) \leq \theta_p(y)$. □

Lemma 4.8 (stabilisation)**.** *Suppose $\mathcal{P}(S)$ is finite. If $x_0 \to x_1 = \theta_{p_1}(x_0) \to \cdots \to x_n = \theta_{p_n}(x_{n-1}) = x_0$ is a closed walk of projections, then $x_{j-1} \bowtie x_j$ for all j.*

Proof. Put $s_j = x_{j-1}p_j$, so that $s_j^* s_j = x_j$ and $s_j s_j^* = \theta_{x_{j-1}}(p_j) \leq x_{j-1}$, and $s = s_1 s_2 \cdots s_n$. Since $s_j^* s_j s_{j+1} = x_j s_{j+1} = s_{j+1}$, we get $s^* s = x_n = x_0$, and $ss^* \leq s_1 s_1^* = \theta_{x_0}(p_1) \leq x_0$ (using $s_j(\cdot)s_j^*$ maps projections $\leq s_j^* s_j$ to projections $\leq s_j s_j^*$). Now $t \mapsto s^* t s$ is an order isomorphism from the projections below ss^* onto the projections below $s^* s$; as $ss^* \leq s^* s$ and $\mathcal{P}(S)$ is finite, $ss^* = s^* s$. Hence $\theta_{x_0}(p_1) = x_0$, so $x_0 x_1 x_0 = (x_0 p_1 x_0)(p_1 x_0) = x_0$, while $x_1 x_0 x_1 = x_1$ always ($\theta_{\theta_p(q)}(q) = \theta_p(q)$ is an axiom consequence). Rotating the cycle gives the other pairs. □

Theorem 4.9. *Let P be a finite projection algebra (more generally, $P = \mathcal{P}(S)$ with S a regular $*$-semigroup having finitely many projections) satisfying the hexagon condition. Let $k \geq 3$, let E be a digraph on $[k]$ in which every vertex has an in-edge and an out-edge, and let $a_1, \ldots, a_k, b_1, \ldots, b_k$ satisfy (H1)–(H3) of Theorem 4.2 for E. Then $w = z$. In particular, for finite projection algebras the hexagon condition implies the triangle condition and the matching-pattern conditions of all digraphs with in- and out-edges at every vertex (on any number $k \geq 3$ of vertices, e.g. C_ℓ, D_2 and the cyclic octagon); on three vertices the converse holds too, since every E on $[3]$ has fewer hypotheses than the hexagon, so for finite projection algebras the hexagon condition, the triangle condition and all the other non-star conditions on three indices (Theorem 4.11) are equivalent. The conditions on $k \geq 4$ indices are strictly weaker: $\mathcal{P}(Q_0)$ satisfies the C_4, D_2 and cyclic-octagon conditions (`mace4parse.py`), as no fourth pair can be added to its hexagon.*

Proof. By a theorem of Imaoka (see East and Azeef Muhammed, arXiv:2301.04845, Section 4) every projection algebra is $\mathcal{P}(S)$ for some regular $*$-semigroup S. We first treat a triangle configuration $a_1, \ldots, b_3$ in P; at the end we explain why only the transport relations were used. The operator Φ is a composite of maps θ_p, hence monotone by Lemma 4.7, and $\Phi(a_1) \leq a_1$. Therefore $a_1 \geq \Phi(a_1) \geq \Phi^2(a_1) \geq \cdots$ is a descending chain, which is eventually constant since P is finite: there is $x^* = \Phi^n(a_1)$ with $\Phi(x^*) = x^*$. By Lemma 4.6 (and induction from the bounded element a_1) all elements of the walk from x^* are bounded, and this walk $x^* \to x_1 \to \cdots \to x_6 = x^*$ is closed, so by Lemma 4.8 consecutive elements are matched. Put

$$A_3 = x_1 \leq b_3, \quad A_1 = x_3 \leq b_1, \quad A_2 = x_5 \leq b_2, \qquad B_2 = x_2 \leq a_2, \quad B_3 = x_4 \leq a_3, \quad B_1 = x_6 \leq a_1.$$

By Lemma 4.5 the A_j pairwise commute with product z, the B_i with product w, and $A_i B_i = B_i A_i$; the six consecutive matchings $x_1 \bowtie x_2 \bowtie \ldots \bowtie x_6 \bowtie x_1$ are exactly the matchings $A_i \bowtie B_j$, $i \neq j$. So $(B_1, B_2, B_3 \,|\, A_1, A_2, A_3)$ satisfies the hexagon hypotheses with products w and z, and the hexagon condition gives $w = z$.

For a general E with in- and out-edges at every vertex, fix three indices i, j, l and run the same walk $a_i \to b_j \to a_l \to b_i \to a_j \to b_l \to a_i$. Lemmas 4.4, 4.5, 4.7 and 4.8 are general, and the proof of Lemma 4.6 used, at a step from a vertex of index i to a vertex of index j, only the transport relations $a_i z a_i = m_i$, $b_j m_i b_j = z$ (an out-edge at i) and $a_i m_j a_i = w$, $b_j w b_j = m_j$ (an in-edge at j), and dually. So the walk from a_i stays bounded, and the rest of the argument is unchanged. □

Remark 4.10. (1) The proof was found with substantial help from Prover9 and Mace4, run by M. Kinyon. Mace4 models with $w \neq z$ showed that the derived projections forming the hexagon vary from model to model (the walk from a_1 stabilises after 0, 1 or 2 turns in the models found), and Prover9 proved the ground equations behind Lemmas 4.5, 4.6 and 4.7, from whose output the hand proofs were extracted.

(2) The implication cannot be obtained by a single application of the hexagon condition to a fixed sextuple of terms in $a_1, \dots, b_3$: for any such sextuple, one of its hexagon hypotheses $t = t'$ fails in some model of the triangle hypotheses in which $w = z$ (Mace4 finds them), while $w = z$ fails in others; since the theory "axioms + triangle hypotheses" is Horn, its initial model satisfies neither $t = t'$ nor $w = z$. For infinite projection algebras the hexagon condition does not imply the triangle condition (Theorem 4.17); in the example the walk closes after one turn but Lemma 4.8 fails, and Mace4 also finds finite models of the triangle hypotheses in which the chain $a_1 > \Phi(a_1) > \Phi^2(a_1)$ descends strictly for two turns before closing (sizes 13, 21, and at least 30 for three turns; `chain.py`).

(3) Applied to $\mathcal{P}(\mathcal{P}_X)$ for finite X, which satisfies the hexagon condition by Theorem 4.2, the theorem shows that condition (ii) of admissibility is not needed in Theorem 4.2 for finite X: in- and out-edges at every vertex suffice. (Indeed Steps 1 and 2 of the proof of Theorem 4.2 only use (i) – in Step 1 the $\mathsf{N}(b_i)$-block and the $\mathsf{N}(a_i)$-block of a point outside $A_i \cup B_i$ coincide by Lemma 1.1(b), since $a_i b_i = b_i a_i$ – and only Step 3 used (ii).) Theorem 4.11 below goes further.

(4) The argument shows more generally that in a finite projection algebra, for any bounded $x \le a_1$, the walk from x ends in a closed 6-cycle which is a hexagon configuration with products w and z. Two further closed cycles always exist, through z and through w ($z \to m_2 \to z \to m_3 \to z \to m_1 \to z$ and dually); they are not hexagons, which is why the bounds in Lemma 4.6 involve m_i as well as w or z.

Theorem 4.11 (non-star patterns)**.** *Let E be a digraph on $[k]$, $k \ge 3$. Call E a* star *if it is empty, or all its edges have the same tail (an out-star), or all have the same head (an in-star); equivalently, E is not a star iff it has two distinct tails and two distinct heads.*

(a) *If E is not a star, then in every projection algebra the hypotheses (H1)–(H3) for E imply $w \bowtie z$ and the four transport relations*

$$a_i z a_i = m_i, \qquad b_i w b_i = m_i, \qquad z a_i z = z, \qquad w b_i w = w \tag{3}$$

at every non-isolated index i (hence also $b_j m_i b_j = z$ and $a_j m_i a_j = w$ for all $j \ne i$); moreover each isolated index i can be replaced by the pair $a_i' := \theta_{a_i}(b_{j_0})$, $b_i' := \theta_{b_i}(a_{j_0})$, where j_0 is any non-isolated index, after which (H1)–(H3) still hold with the same w, z and (3) holds at every index. Consequently, in a finite projection algebra satisfying the hexagon condition, and in particular in $\mathcal{P}(\mathcal{P}_X)$ for finite X, (H1)–(H3) for E force $w = z$.

(b) *If E is a star, then $\mathcal{P}_2$ contains projections satisfying (H1)–(H3) for E with $w \ne z$.*

So for finite X the matching-pattern condition of Theorem 4.2 holds in $\mathcal{P}_X$ for exactly the non-star digraphs E. The smallest instances beyond the hexagon are the path *$a_1 \bowtie b_2$, $a_2 \bowtie b_3$, the* 2-cycle-plus-spare *$a_1 \bowtie b_2$, $a_2 \bowtie b_1$ (both on three indices, with (H1), (H2)), and the* two disjoint pairs *$a_1 \bowtie b_2$, $a_3 \bowtie b_4$ on four indices.*

Proof. (a) For an index i with an out-edge (i, l) we have $a_i z a_i = a_i b_l b_i a_i = (a_i b_l a_i) b_i = m_i$ and $z a_i z = z b_l a_i b_l z = z$; dually an in-edge (l, i) gives $b_i w b_i = m_i$ and $w b_i w = w$. By the argument of Lemma 4.4, $z a_i z = z$ gives $b_j m_i b_j = z$ for all $j \ne i$, and $w b_i w = w$ gives $a_j m_i a_j = w$ for all $j \ne i$.

Step 1: $b_i w b_i = m_i$ at every tail i. Let $i \ne i'$ be tails. Since $w \le a_i$, $w b_i w = w a_i b_i a_i w = w m_i w = w(a_i z a_i) w = wzw$, and likewise $w b_{i'} w = wzw$. Hence, $b_i w b_i$ being a projection,

$$b_i w b_i = b_i (w b_i w) b_i = b_i (w b_{i'} w) b_i = b_i a_i a_{i'} b_{i'} a_{i'} a_i b_i = m_i\, m_{i'}\, m_i = a_i b_i\, (b_{i'} a_{i'} b_{i'})\, b_i a_i = a_i z a_{i'} z a_i = a_i z a_i = m_i,$$

using $b_i b_{i'} = z$, $z a_{i'} z = z$ and $a_i z a_i = m_i$. Dually $a_j z a_j = m_j$ at every head j, since there are two distinct heads: $z a_j z = z m_j z = z(b_j w b_j) z = zwz = z b_{j'} w b_{j'} z = z m_{j'} z$, so $a_j z a_j = a_j (z m_{j'} z) a_j = m_j m_{j'} m_j = b_j a_j (a_{j'} b_{j'} a_{j'}) a_j b_j = b_j w b_{j'} w b_j = b_j w b_j = m_j$.

Step 2: $w \bowtie z$. Choose a tail i and a head j with $i \neq j$ (possible since there are two of each). Then $wzw = a_i a_j z a_j a_i = a_i m_j a_i = a_i a_j b_j a_j a_i = w b_j w = w$ by Step 1 and the in-edge at j, and $zwz = b_j b_i w b_i b_j = b_j m_i b_j = b_j b_i a_i b_i b_j = z a_i z = z$ by Step 1 and the out-edge at i.

Step 3: the remaining relations. At a tail i we now have $a_i z a_i = m_i$, $z a_i z = z$ (out-edge) and $b_i w b_i = m_i$ (Step 1), and $w b_i w = w$ follows from $w \bowtie z$ by monotonicity (Lemma 4.7): $z \leq b_i$ gives $w = wzw \leq w b_i w \leq w$. Dually at a head. So (3) holds at every non-isolated index, and Lemma 4.4 gives $b_j m_i b_j = z$, $a_j m_i a_j = w$ for all $j \neq i$.

Step 4: isolated indices. Let i be isolated and j_0 non-isolated. Since $w \leq a_i$, monotonicity gives $z = zwz \leq z a_i z \leq z$, so $z a_i z = z$, and dually $w b_i w = w$; by Lemma 4.4 again, $b_l m_i b_l = z$ and $a_l m_i a_l = w$ for $l \neq i$. The other two relations of (3) may fail at i (Mace4 finds models with $a_i z a_i < m_i$, and so does $\mathcal{P}_2$), but the two sandwiches are equal:

$$u_i := a_i z a_i = b_i w b_i = m_i m_{j_0} m_i.$$

Indeed $z = b_i m_{j_0} b_i$ (Lemma 4.4 at j_0), and writing $m_{j_0} = a_{j_0} b_{j_0}$, $b_{j_0} b_i = z$ in one copy and $m_{j_0} = b_{j_0} a_{j_0}$ in the other, $z = b_i a_{j_0} z = z a_{j_0} b_i$; hence $a_i z a_i = a_i(b_i a_{j_0} z)(z a_{j_0} b_i) a_i = m_i(a_{j_0} z a_{j_0}) m_i = m_i m_{j_0} m_i$, and dually $b_i w b_i = m_i(b_{j_0} w b_{j_0}) m_i = m_i m_{j_0} m_i$. Thus $u_i = \theta_{m_i}(m_{j_0}) \leq m_i$, and for $l \neq i$ we get $b_l u_i b_l = z$: it is $\leq b_l m_i b_l = z$ by monotonicity, and $z(b_l u_i b_l) z = z u_i z = z a_i z a_i z = z$; dually $a_l u_i a_l = w$.

Put $a_i' = a_i b_{j_0} a_i$ and $b_i' = b_i a_{j_0} b_i$. Then $a_i' \leq a_i$ and $w \leq a_i'$, since $a_i' w a_i' = a_i(b_{j_0} w b_{j_0}) a_i = a_i m_{j_0} a_i = w$; dually $z \leq b_i' \leq b_i$. By Lemma 4.5, a_i' commutes with every a_l ($l \neq i$), and with a_l' if l is also isolated, with product w, and dually for b_i': (H1) holds for the new family, with the same w and z. For (H2),

$$a_i' b_i' a_i' = a_i b_{j_0}(m_i a_{j_0} m_i) b_{j_0} a_i = a_i b_{j_0}\big(b_i(a_i a_{j_0} a_i) b_i\big) b_{j_0} a_i = a_i(b_{j_0} u_i b_{j_0}) a_i = a_i z a_i = u_i,$$

and dually $b_i' a_i' b_i' = b_i w b_i = u_i$; so $a_i' b_i' = b_i' a_i' = u_i =: m_i'$. (H3) is unaffected, as i lies on no edge. Finally (3) holds at i for the new pair: $a_i' z a_i' = a_i b_{j_0}(a_i z a_i) b_{j_0} a_i = a_i(b_{j_0} u_i b_{j_0}) a_i = u_i = m_i'$, dually $b_i' w b_i' = m_i'$, and $z a_i' z = z$, $w b_i' w = w$ as before since $w \leq a_i'$, $z \leq b_i'$. The relations at the other indices only involve a_l, b_l, m_l, w, z for $l \neq i$ and are unchanged.

Step 5. After the replacement all four relations hold at every index, and with them the proof of Lemma 4.6, hence of Theorem 4.9, goes through for any three indices of E: the walk from a_i ends in a closed 6-cycle which is a hexagon configuration with products w and z, and the hexagon condition gives $w = z$.

(b) In $\mathcal{P}_2$ let F be the projection with the single block $\{1, 2, 1', 2'\}$ (rank 1) and e the projection with blocks $\{1, 2\}$, $\{1', 2'\}$ (rank 0); then $e \leq F$, $eF = Fe = e$, $F \bowtie F$. For an out-star with tail v (or for $E = \emptyset$) take $a_v = F$, $a_i = e$ for $i \neq v$, and $b_j = F$ for all j: then $w = e$, $z = F$, all pairs commute, $a_v \bowtie b_j$ for all j, and $w \neq z$. For an in-star with head v take $a_i = F$ for all i, $b_v = F$ and $b_j = e$ for $j \neq v$. □

The transport relations are all that the matchings of E contribute; once they are available, matchings can be manufactured at will, because of the following elementary fact.

Lemma 4.12. *For any two projections p, q of a regular $*$-semigroup, $\theta_p(q) \bowtie \theta_q(p)$.*

Proof. pq is idempotent, being $(pq)(pq)^*(pq)$. Hence $(pqp)(qpq)(pqp) = p(qp)^4 = p(qp) = pqp$ and symmetrically $(qpq)(pqp)(qpq) = qpq$. □

Call a family $a_1, \ldots, a_k, b_1, \ldots, b_k$ satisfying (H1), (H2) *transported* if the four relations (3) hold at every index. By Theorem 4.11(a) and its proof, a configuration for a non-star digraph becomes transported after the isolated indices are replaced, and in a transported family $w \bowtie z$, $b_j m_i b_j = z$ and $a_j m_i a_j = w$ for all $i \neq j$ (Lemma 4.4), and the bounded-elements Lemma 4.5 applies.

Lemma 4.13 (edge creation)**.** *Let $a_1, \dots, a_k, b_1, \dots, b_k$ be transported, $k \geq 2$, and $i \neq j$. Put $a'_i = \theta_{a_i}(b_j)$, $b'_j = \theta_{b_j}(a_i)$, and replace a_i by a'_i and b_j by b'_j. The new family is again transported, with the same w, z and $m_1, \dots, m_k$; $a'_i \bowtie b'_j$; and every pair a_l, b_m with $l \neq i$, $m \neq j$ is unchanged.*

Proof. $a'_i \leq a_i$, and $w, m_i \leq a'_i$: indeed $a'_i w a'_i = a_i(b_j w b_j)a_i = a_i m_j a_i = w$ and $a'_i m_i a'_i = a_i(b_j m_i b_j)a_i = a_i z a_i = m_i$. Dually $b'_j \leq b_j$ and $z, m_j \leq b'_j$. By Lemma 4.5, a'_i commutes with every a_l ($l \neq i$) with product w, and with b_i with product m_i; dually for b'_j; so (H1), (H2) hold with the same w, z, m_l. The transport relations at i: $a'_i z a'_i = a_i(b_j m_i b_j)a_i = m_i$, $b_i w b_i = m_i$ is unchanged, $z a'_i z = z$ since $z = zwz \leq z a'_i z \leq z$ by monotonicity ($w \leq a'_i$), and $w b_i w = w$ is unchanged; dually at j; the other indices are untouched. Finally $a'_i \bowtie b'_j$ by Lemma 4.12. □

Theorem 4.14 (the triangle condition is a master condition)**.**
(a) *Every transported family on $k \geq 3$ indices yields, on any three of its indices, a triangle configuration with the same w and z. Consequently, in* every *projection algebra, the triangle condition implies the matching-pattern condition of every non-star digraph.*
(b) *For every set X, the matching-pattern condition of E holds in $\mathcal{P}_X$ iff E is not a star ($|X| \geq 2$ for the "only if").*

Proof. (a) Apply Lemma 4.13 three times on the indices $1, 2, 3$: create $(1, 2)$, replacing a_1, b_2; then $(2, 3)$, replacing a_2, b_3; then $(3, 1)$, replacing a_3, b_1. Each step leaves the pairs created before untouched, since they involve different elements: after the three steps $a'_1 \bowtie b'_2$, $a'_2 \bowtie b'_3$, $a'_3 \bowtie b'_1$, and (H1), (H2) hold with the original w, z by the lemma. For the consequence, take a configuration for a non-star E, replace its isolated indices (Theorem 4.11), obtaining a transported family with the same w, z, and build the triangle configuration; the triangle condition gives $w = z$.

(b) The triangle condition holds in $\mathcal{P}_X$ for every X by Theorem 4.2, so the non-star conditions hold by (a); the stars fail already in $\mathcal{P}_2 \subseteq \mathcal{P}_X$ (Theorem 4.11(b)). □

Thus the matchings of a pattern serve only to establish the transport relations; and among all matching-pattern conditions the triangle condition is the strongest one valid in partition monoids, in the sense that it implies all the others in every projection algebra, while the path and the 2-cycle conditions are equivalent to it (they imply it trivially, having fewer hypotheses). In the proof of Theorem 4.2 only the triangle is now needed; the other admissible digraphs follow from it algebraically. The hexagon condition is strictly weaker than the triangle condition in general (Theorem 4.17 below), although the two are equivalent on finite projection algebras (Theorem 4.9).

Remark 4.15. (1) Theorem 4.11 was found in three stages, each driven by Mace4 and Prover9 experiments of M. Kinyon. First, on three indices, exhaustive search in $\mathcal{P}_n$, $n \leq 4$, showed that two matched pairs force $w = z$ iff they form a directed path $\{(1, 2), (2, 3)\}$ (up to relabelling) or the 2-cycle $\{(1, 2), (2, 1)\}$, while two pairs with a common tail $\{(1, 2), (1, 3)\}$ or a common head $\{(2, 1), (3, 1)\}$, the pair $\{(1, 2), (3, 2)\}$, and a single pair all admit counterexamples in $\mathcal{P}_2$ (`digraph3.py`, `digraph4.py`); the path is the case of a *full* vertex (in- and out-edge), where $w \bowtie z$ is immediate ($wzw = a_i(a_{j_0} z a_{j_0})a_i = a_i m_{j_0} a_i = w$), and for such E with every vertex on an edge the argument of Steps 3 and 5 was the first version of the theorem.

(2) The 2-cycle-plus-spare required the treatment of isolated indices (Step 4). There is no projection algebra of size ≤ 12 with $a_1 \bowtie b_2$, $a_2 \bowtie b_1$, (H1), (H2), $w \neq z$ and none of the pairs $(a_1, b_3), (a_2, b_3), (a_3, b_1), (a_3, b_2)$ matched; there is exactly one of size 13, 45 of size 14 and 2054 of size 15 (`square.in`, isofiltered). All of them violate the hexagon condition in the same way: the walk from a_1 closes after a single turn, $a_1 \to b_3 a_1 b_3 \to a_2 \to b_1 \to a_3 b_1 a_3 \to b_2 \to a_1$, and $\{a_1, a_2, a_3 b_1 a_3 \mid b_1, b_2, b_3 a_1 b_3\}$ is a hexagon with $w \neq z$ generating a copy of $\mathcal{P}(Q_0)$

(`sqfilter2.py`; the size-13 algebra is $\mathcal{P}(Q_0)$ with a_3, b_3 adjoined above it, and 1934 of the size-15 algebras contain it). The elements $a_3b_1a_3 = \theta_{a_3}(b_1)$ and $b_3a_1b_3 = \theta_{b_3}(a_1)$ appearing here are exactly the replacements a_3', b_3' of Step 4, and in 49 of the models $a_3za_3 < m_3$, which showed that the transport relations themselves cannot be recovered at an isolated index; the identity $a_3za_3 = b_3wb_3 = m_3m_1m_3$ and the computation of $a_3'b_3'a_3'$ were first checked on these models (`sqstar.py`) and then proved by hand.

(3) With four indices the full-vertex condition is no longer the boundary: the pattern $\{(1,2),(3,4)\}$ of two *disjoint* matched pairs has no full vertex, yet admits no counterexample in $\mathcal{P}_n$ for $n \leq 4$ nor in the 57-element projection algebra of $\mathcal{P}_4/\mu_1$, which violates the hexagon condition (`nofull.py`, `nofull3.py`, `p4mu_disjoint.py`), and the walk argument is unavailable for it as it stands, since a closed walk alternating between out-vertices $\{1,3\}$ and in-vertices $\{2,4\}$ only produces the $k = 2$ square. Mace4 finds a projection algebra with $w \neq z$ for this pattern (the 14-element algebra of Remark 4.3(3), in which all twelve cross pairs are matched, and, excluding the extra pairs that would create a full vertex, a unique one of size 18 with $a_1 \bowtie b_4$, $a_3 \bowtie b_2$ as well; `disjoint.in`, `disjoint2.in`), so the pattern is not a quasi-identity of all projection algebras, but both models violate the hexagon condition and in both $w \bowtie z$ and all sixteen transport relations hold. Prover9 then proved $wzw = w$ from the two disjoint pairs in 0.12 seconds (`disjoint3.in`, a 165-step proof); Steps 1 and 2 of the proof above are the hand version, and they apply verbatim whenever E has two tails and two heads, which is what unifies the full-vertex and the disjoint-pair cases. Patterns with fewer than two tails or two heads are the stars, which fail in $\mathcal{P}_2$ by (b).

(4) The non-star conditions do not follow from the hexagon condition in infinite projection algebras (Theorem 4.17), but they do follow from the triangle condition, in every projection algebra (Theorem 4.14), and hence hold in $\mathcal{P}_X$ for every X. A direct combinatorial proof in $\mathcal{P}_X$ would have to handle the case of Step 3 of the proof of Theorem 4.2 in which the bad index is the middle vertex of a path; the algebraic reduction to the triangle avoids it.

4.2. **The infinite case: the unrolled hexagon.** The finiteness hypothesis of Theorems 4.9 and 4.11 cannot be dropped. The counterexample is obtained by "unrolling" the hexagon of $\mathcal{P}(Q_0)$ along its cycle of matchings $a_1 \bowtie b_2 \bowtie a_3 \bowtie b_1 \bowtie a_2 \bowtie b_3 \bowtie a_1$: each vertex is replaced by a descending chain, all matchings are taken level by level except the last one, which shifts the level by one.

Recall the structure of $\mathcal{P}(Q_0)$ (Section 4.3): it consists of six projections $a_1, a_2, a_3, b_1, b_2, b_3$ and five *low* elements w, z, m_1, m_2, m_3 (the images of the five partitions of $\{1,2,3\}$), with $a_ia_j = w$, $b_ib_j = z$ $(i \neq j)$, $a_ib_i = b_ia_i = m_i$, $a_i \bowtie b_j$ $(i \neq j)$; $\theta_\ell(x) = \ell$ for every low ℓ and every x (so the low elements form a right-zero subalgebra, $\ell * \ell' = \ell'$, and any two of them are matched); and $\theta_{a_i}(w) = w$, $\theta_{a_i}(z) = m_i$, $\theta_{a_i}(m_i) = m_i$, $\theta_{a_i}(m_j) = w$ $(j \neq i)$, dually $\theta_{b_i}(z) = z$, $\theta_{b_i}(w) = m_i$, $\theta_{b_i}(m_i) = m_i$, $\theta_{b_i}(m_j) = z$.

Definition 4.16. The *unrolled hexagon* is the set

$$H_\infty = \{w, z, m_1, m_2, m_3\} \;\cup\; \{e_{i,n}, f_{i,n} : i \in \{1,2,3\},\ n \in \mathsf{N}\}$$

with the operation $x * y = \theta_y(x)$ defined as follows, where ℓ denotes a low element, $i, j \in \{1,2,3\}$, and $M = \{(1,2),(3,2),(3,1),(2,1),(2,3)\}$ is the set of *level-preserving* pairs (all (i,j) with $i \neq j$ except the *crossing* pair $(1,3)$):

(i) $\theta_\ell(x) = \ell$ for all x; $\theta_{e_{i,n}}(\ell)$ and $\theta_{f_{i,n}}(\ell)$ are $\theta_{a_i}(\ell)$ and $\theta_{b_i}(\ell)$ as in $\mathcal{P}(Q_0)$ (independent of n);

(ii) $\theta_{e_{i,m}}(e_{i,n}) = e_{i,\max(n,m)}$, $\theta_{f_{i,m}}(f_{i,n}) = f_{i,\max(n,m)}$; $\theta_{e_{j,m}}(e_{i,n}) = w$ and $\theta_{f_{j,m}}(f_{i,n}) = z$ for $i \neq j$;

(iii) $\theta_{f_{i,m}}(e_{i,n}) = \theta_{e_{i,m}}(f_{i,n}) = m_i$;

(iv) for $(i,j) \in M$: $\theta_{f_{j,m}}(e_{i,n}) = f_{j,\max(n,m)}$ and $\theta_{e_{i,m}}(f_{j,n}) = e_{i,\max(n,m)}$;
(v) for the crossing pair: $\theta_{f_{3,m}}(e_{1,n}) = f_{3,\max(m,n-1)}$ and $\theta_{e_{1,m}}(f_{3,n}) = e_{1,\max(m,n+1)}$.

Thus $e_{i,0} > e_{i,1} > \cdots$ and $f_{i,0} > f_{i,1} > \cdots$ are chains ("level" n), $w, m_i \le e_{i,n}$ and $z, m_i \le f_{i,n}$, no low element is below another one and no chain element is below a low one. Two chain elements are matched iff they are equal, or of the form $e_{i,n}, f_{j,n}$ with $(i,j) \in M$, or $e_{1,n+1}, f_{3,n}$; a low element is matched with every low element and with no chain element. Two elements commute iff they are on the same chain, or are $e_{i,n}, e_{j,m}$ (product w), $f_{i,n}, f_{j,m}$ (z), $e_{i,n}, f_{i,m}$ (m_i), or a low element and an element above it. Level 0 is a copy of $\mathcal{P}(Q_0)$ with the matching $a_1 \bowtie b_3$ broken in one direction: $\theta_{f_{3,0}}(e_{1,0}) = f_{3,0}$ but $\theta_{e_{1,0}}(f_{3,0}) = e_{1,1} < e_{1,0}$.

Theorem 4.17. (a) *H_∞ is a projection algebra satisfying* (Q1).
(b) *H_∞ satisfies the hexagon condition and the matching-pattern condition of every non-star digraph on $k \ge 4$ vertices.*
(c) *$(e_{1,0}, e_{2,0}, e_{3,0} \mid f_{1,0}, f_{2,0}, f_{3,0})$ is a triangle configuration with $w \ne z$. More generally, H_∞ violates the matching-pattern condition of every digraph on three vertices other than the hexagon, and of every star.*

Consequently the hexagon condition does not imply the triangle condition, nor any other non-star condition on three indices, in the variety of projection algebras; and a regular $*$-semigroup S with $\mathcal{P}(S) \cong H_\infty$ (which exists by Imaoka's theorem) satisfies (Q1) *and the hexagon condition but embeds in no partition monoid.*

Proof. (a) By construction, the *type* of a product (low, or which chain) depends only on the types of the factors, and the level of a product of two chain elements of levels n, m is $\max(n + c, m + c')$ with $c, c' \in \{-1, 0, 1\}$, while a product involving a low element is low. Hence for a term $t(x_1, \ldots, x_r)$ of depth d and an assignment of types to the variables, either t is low for all levels, or its value lies on a fixed chain at level $\max_{v \in S}(n_v + c_v)$ for some set S of the variables and constants $|c_v| \le d$. Two such expressions of the same type agree on N^r as soon as they agree on $\{0, \ldots, 2d+1\}^r$, since evaluating at $n_v = 2d+1$, $n_u = 0$ ($u \ne v$) tells whether $v \in S$ and recovers c_v. The axioms (P1)′–(P5)′ have depth ≤ 4, so it suffices to check them on all triples of level ≤ 9; this was done by computer (`hinf2.py`, levels ≤ 12). For (Q1): by the description of the order, two elements with a common lower bound are comparable, or of the form $e_{i,n}, f_{i,m}$, or $e_{i,n}, e_{j,m}$, or $f_{i,n}, f_{j,m}$, or a low element and an element above it, and all these pairs commute.

(b) We first classify the tuples $(x_1, \ldots, x_k)$, $k \ge 2$, in which all products $x_i * x_j$ ($i \ne j$, both orders) are equal to one element w', using the description of commuting pairs: (α) all x_i on one chain, w' on that chain; (β) each x_i is w or an e-element, the e-elements on pairwise distinct chains, $w' = w$; (β') dually with z and f-elements; (γ) each x_i is m_p or on e_p or on f_p, at most one on e_p and at most one on f_p, $w' = m_p$. (Two low elements have equal products in both orders only if they are equal; a low ℓ and a chain element c only if $\ell \le c$; and $e_{i,n}, f_{j,m}$ with $i \ne j$ never.)

The hexagon. Let $(x_1, x_2, x_3 \mid y_1, y_2, y_3)$ satisfy the hexagon hypotheses with products w', z', and suppose $w' \ne z'$. If some x_i is low, then $w' = x_i$, the y_j ($j \ne i$) are low, being matched with x_i, so $z' = y_j$, and x_l ($l \ne i$) is low, being matched with some y_j, $j \notin \{i, l\}$; so $x_1 = x_2 = x_3 = w'$ and $y_j = z'$ for $j \ne i$; finally y_i is matched with the low element x_l, so y_i is low, $y_i = z'$, and (H2) at i gives $w' = z'$. So all six are chain elements. Case (γ) is then impossible (three chain elements in $e_p \cup f_p$ with at most one on each). In case (α), say all $x_i \in e_p$: the elements matched with an element of e_p are that element itself and the $f_{q,m}$ with $(p,q) \in M \cup \{(1,3)\}$, $q \ne p$; if $y_j = f_{q,m}$, then by (H2) y_j commutes with $x_j \in e_p$, which requires $q = p$; so every y_j lies on e_p and equals both x_i, $i \ne j$, whence all six elements coincide. In case (β), $x_i = e_{\sigma(i),n_i}$ with σ injective and $w' = w$. A y_j equal to some x_i would be matched with the other $x_{i'}$, on

a different e-chain, impossible; so $y_j = f_{\tau(j),m_j}$ with $(\sigma(i),\tau(j)) \in M \cup \{(1,3)\}$ for all $i \neq j$. If τ is not injective, z' is on an f-chain, all $\tau(j)$ are equal, $= q$, and the three values $\sigma(i)$ lie in the two-element set $\{i : (i,q) \in M \cup \{(1,3)\}\}$, impossible; so τ is a permutation and $z' = z$. By (H2), x_i and y_i commute, which fails if $\sigma(i) \neq \tau(i)$; so $\sigma = \tau$. The six matchings $x_i \bowtie y_j$ then involve each ordered pair $(\sigma(i),\sigma(j))$, $i \neq j$, exactly once; the five level-preserving ones give $n_i = m_j$ and the crossing one gives $n_{i_0} = m_{j_0} + 1$. The five equalities connect all six levels ($K_{3,3}$ minus a perfect matching is a 6-cycle, and deleting one more edge leaves a path), so all levels are equal, a contradiction. Case (β') is dual.

Non-star patterns on $k \geq 4$ vertices. Every non-star digraph contains two disjoint edges, a directed path of length two, or a 2-cycle, and its condition on $[k]$ follows from the condition of $\{(1,2),(3,4)\}$, $\{(1,2),(2,3)\}$ or $\{(1,2),(2,1)\}$ on $[4]$ (restrict to four indices containing the two edges). In each of these patterns the two edges have distinct tails and distinct heads. Suppose $(x_1,\dots,x_4 \mid y_1,\dots,y_4)$ satisfies (H1)–(H3) for one of them with $w' \neq z'$. Since $k = 4 > 3$, in case (β) some $x_i = w$, and in case (γ) at least two x_i equal m_p. Recall that the elements matched with a chain element c are c itself and chain elements on a chain of the other letter, and that the elements commuting with $e_{p,n}$ are those of e_p, f_p, of the other e-chains, and w, m_p.

Case (α), all $x_i \in e_p$. The head y_b of each edge is matched with its tail $x_a \in e_p$ and commutes with $x_b \in e_p$, hence $y_b = x_a$: the y's contain two elements of e_p (the two tails, of distinct indices). This rules out cases (β), (γ) (at most one element on e_p) and (β') for the y's, and case (α) on a chain other than e_p. So all $y_j \in e_p$, $w' = e_{p,N}$, $z' = e_{p,N'}$, at most one x has level $< N$ and at most one y has level $< N'$. One of the two heads has level N' and equals a tail, whose level is $\leq N$; so $N' \leq N$, and symmetrically $N \leq N'$; so $w' = z'$.

Case (β), some $x_i = w$. Then y_i is w or an e-element. So the y's are not in case (β'); in case (β), $z' = w = w'$. In case (α) they lie on one chain e_q; each tail is matched with its head, an element of e_q, hence equals it, and two tails on one e-chain contradict (β). In case (γ) they lie in $e_q \cup f_q \cup \{m_q\}$ with at most one on e_q; every $x_i = w$ needs $y_i \in e_q$, so exactly one x_i is w and the others lie on the three distinct e-chains, and the two of them not on e_q need y's in $e_q \cup f_q \cup \{m_q\}$ commuting with them, i.e. again on e_q; impossible.

Case (γ), at least two $x_i = m_p$. An element commutes with m_p iff it is above m_p, i.e. lies in $e_p \cup f_p \cup \{m_p\}$. So the y's are not in case (γ) for $q \neq p$ (and for $q = p$, $z' = m_p = w'$), nor in cases (β), (β') (two of them would lie on e_p, resp. f_p). In case (α) they lie on e_p or on f_p; the tails equal their heads, so two x's lie on that chain, contradicting (γ).

The cases (α) on f_p and (β') are dual. Hence $w' = z'$.

(c) The level-0 vertices satisfy (H1), (H2), $e_{1,0} \bowtie f_{2,0}$, $e_{2,0} \bowtie f_{3,0}$, $e_{3,0} \bowtie f_{1,0}$ (all in M), and $w \neq z$. For E on $[3]$ missing the ordered pair (p,q), choose a permutation σ with $(\sigma(p),\sigma(q)) = (1,3)$ and put $x_i = e_{\sigma(i),0}$, $y_j = f_{\sigma(j),0}$: all matchings required by E are in M, hence hold at level 0. For an out-star with tail 1 on $[k]$ take $x_1 = e_{1,0}$, $x_i = w$ $(i \geq 2)$, and all $y_j = e_{1,0}$: then $w' = w$, $z' = e_{1,0}$, all required pairs commute, and $x_1 \bowtie y_j$ trivially. In-stars are dual, and the empty pattern is contained in a star.

The final assertion follows because $\mathcal{P}(\mathcal{P}_X)$ satisfies the triangle condition for every X (Theorem 4.2), and the projection algebra of a $*$-subsemigroup of $\mathcal{P}_X$ is a subalgebra of $\mathcal{P}(\mathcal{P}_X)$. □

Remark 4.18. (1) Together with Theorems 4.9, 4.11 and $\mathcal{P}(Q_0)$ this gives the complete picture of the implications between the matching-pattern conditions: on finite projection algebras the hexagon condition is equivalent to every non-star condition on three indices and implies every non-star condition on more indices, but not conversely ($\mathcal{P}(Q_0)$ satisfies all the conditions on $k \geq 4$ indices); on arbitrary projection algebras the non-star conditions on three indices other than the hexagon are strictly stronger than the hexagon condition (H_∞). On arbitrary projection algebras the triangle, path and 2-cycle conditions are equivalent and imply all the others (Theorem 4.14).

Whether the hexagon condition implies the conditions on $k \geq 4$ indices in general is open; an "unrolled octagon" would separate them, but there the products of non-adjacent vertices have to be new elements (Remark 4.3(3)) and we have not constructed it. All these conditions were verified on H_∞ by computer as well (`hinf2_run.py`, `hinf2_pat.py`).

(2) In H_∞ the walk of Section 4.1 from $e_{1,0}$ closes after one turn, $e_{1,0} \to f_{3,0} \to e_{2,0} \to f_{1,0} \to e_{3,0} \to f_{2,0} \to e_{1,0}$, but its consecutive elements $e_{1,0}, f_{3,0}$ are not matched: in the semigroup S the element $s = e_{1,0}f_{3,0}e_{2,0}f_{1,0}e_{3,0}f_{2,0}$ has $s^*s = e_{1,0}$ and $ss^* = e_{1,1} < e_{1,0}$. So it is the Dedekind-finiteness used in Lemma 4.8, not the descending chain condition, that fails here. Mace4 finds finite models of the triangle hypotheses with $w \neq z$ in which the chain $a_1 > \Phi(a_1) > \Phi^2(a_1)$ descends strictly for two turns before closing (sizes 13 and 21; three turns need at least 30 elements; `chain.py`), which suggests that an example of the other kind, with an infinite strictly descending walk, exists as well.

(3) The non-star conditions all hold in $\mathcal{P}_X$ for every X (Theorem 4.14); H_∞ is not a partition monoid. Before this was known, the natural place to look for counterexamples was among *periodic* projections of $\mathcal{P}_{\mathbb{Z}\times[m]}$, invariant under the level shift $(n, i) \mapsto (n + 1, i)$, since these can carry the level-shifting matchings of H_∞ inside an honest partition monoid. Such a projection is described by a partition of $[m]$ into types c, each with a modulus $d_c \geq 0$ and a potential $\phi_c : c \to \mathbb{Z} \pmod{d_c}$ up to a constant – its blocks are $\{(n + \phi(i), i) : i \in c\}$ for $d_c = 0$ and $\{(n, i) : i \in c,\ n \equiv \phi(i) + r \pmod{d_c}\}$ for $d_c > 0$ – together with a T/N label per type, and the products $\theta_p(q)$ are computed exactly by a union-find with potentials on $[m]$ (`periodic.py`; the description of $\theta_p(q)$ in Lemma 1.1(a) makes this a finite computation). Exhaustive search over all periodic projections with $m \leq 3$ points per level, potentials in $\{-1, 0, 1\}$ and moduli ≤ 2 (460 projections for $m = 3$), and with $m = 4$, potentials in $\{-1, 0, 1\}$ and moduli ≤ 2 (5064 projections), finds no counterexample to the path condition, the 2-cycle condition, or (for $m \leq 3$) the two-disjoint-pairs, $K_{2,2}$ and C_4 conditions, while reproducing the counterexamples to the star patterns and, as a check, none to the triangle and hexagon conditions (`periodic3.py`). These searches were run before Theorem 4.14 was found; that theorem explains their outcome, and the direct combinatorial attack on Step 3 of the proof of Theorem 4.2 at the middle vertex of a path (where one only obtains that the offending $\mathsf{N}(z)$-block $N \subseteq B_2$ is a union of blocks that are simultaneously $\mathsf{T}(a_1)$- and $\mathsf{T}(b_2)$-blocks, a union of $\mathsf{T}(a_3)$-blocks, and contained in a single $\mathsf{T}(a_2)$-block) is no longer needed.

4.3. **The witness.**

Theorem 4.19. *Let ρ be the relation on $\mathcal{P}_3$ defined by $\alpha\,\rho\,\beta$ iff $\alpha = \beta$, or $\operatorname{rank}\alpha, \operatorname{rank}\beta \leq 1$ and $\ker\alpha = \ker\beta$, $\operatorname{coker}\alpha = \operatorname{coker}\beta$, or $\operatorname{rank}\alpha = \operatorname{rank}\beta = 2$ and $\alpha\,\mathcal{H}\,\beta$. Then ρ is a $*$-congruence, $Q := \mathcal{P}_3/\rho$ has 67 elements and satisfies (Q1), and Q embeds in no partition monoid. The same holds for the 61-element $*$-subsemigroup Q_0 of Q generated by the six projections of rank 2, and for the 103-element quotient $\mathcal{P}_3/\mu$ where μ identifies only the elements of rank ≤ 1 with equal kernel and cokernel.*

Proof. That ρ and μ are $*$-congruences, that the quotients satisfy (Q1), and that Q_0 has 61 elements were checked by direct computation (see Section 6; ρ and μ are among the ten $*$-congruences of $\mathcal{P}_3$). Write a_i for the image of the rank-2 projection of $\mathcal{P}_3$ whose transversal blocks are $\{i\}$ and $\{1, 2, 3\} \setminus \{i\}$, and b_i for the image of the rank-2 projection whose transversal blocks are the two singletons $\{j\}$, $\{k\}$ ($\{i, j, k\} = \{1, 2, 3\}$) and whose nontransversal block is $\{i\}$. In Q (and already in $\mathcal{P}_3/\mu$) the hypotheses (H1)–(H3) hold with E the set of all six pairs (so in particular for the triangle): w is the class of the partition of $\{1, 2, 3\}$ into one block, z that of the discrete partition, m_i that of the partition $\{i\} \mid \{j, k\}$, and the products a_ib_j ($i \neq j$) have rank 2; all this is a direct verification in $\mathcal{P}_3$ (also done by computer). But $w \neq z$, since

the two partitions are different. By Theorem 4.2 no $*$-homomorphism $Q \to \mathcal{P}_X$ is injective on $\{w, z\}$. □

Remark 4.20. (1) The obstruction lives in the projection algebra: the eleven-element projection algebra of Q_0 (six projections of rank 2 and the five "partition" projections w, z, m_1, m_2, m_3) satisfies the hypotheses of Theorem 4.2 with $w \neq z$, so it is not a subalgebra of the projection algebra of any $\mathcal{P}_X$. Both known obstructions, (Q1) and the hexagon condition, are therefore statements about projection algebras.

(2) Concretely, $Q_0 = D \cup (\Pi_3 \times \Pi_3)$ where Π_3 is the set of five partitions of $\{1, 2, 3\}$, $\Pi_3 \times \Pi_3$ is a 5×5 square band (the image of the ideal of rank ≤ 1, via $\alpha \mapsto (\ker \alpha, \operatorname{coker} \alpha)$), and D is the rank-2 $\mathcal{D}$-class of $\mathcal{P}_3$ with its $\mathcal{H}$-classes collapsed, a 6×6 "trace" with trivial groups whose zero pattern is the 6-cycle $a_1 b_2 a_3 b_1 a_2 b_3$ of matched pairs. The elements of D act on Π_3 by $\varepsilon \mapsto \ker(x\alpha)$ for any α with $\ker \alpha = \varepsilon$. Every proper $*$-subsemigroup of Q_0 generated by five of the six rank-2 projections embeds in $\mathcal{P}_3$; all six are needed.

(3) Computer searches confirm that $\mathcal{P}_3/\mu$ has no $*$-embedding into $\mathcal{P}_n$ for $n \leq 7$, that Q_0 has none for $n \leq 5$, and that the hypotheses of Theorem 4.2 have no solution with $w \neq z$ among the projections of $\mathcal{P}_n$ for $n \leq 4$ (a check of the theorem, not needed for it).

(4) The congruence μ is the $*$-invariant congruence of $\mathcal{P}_3$ that collapses the ideal of rank ≤ 1 onto the square band $\Pi_3 \times \Pi_3$; in the classification of the congruences of partition monoids (East–Ruškuc; East–Mitchell–Ruškuc–Torpey) it is one of the congruences below the Rees congruence of that ideal, and ρ is its join with the "$\mathcal{H}$-relation on rank 2". For $n = 4$ the analogous quotients contain those of $\mathcal{P}_3$ and give nothing new (Section 6); the further obstruction of Section 5 was found instead by enumerating small projection algebras.

5. The conditions are not sufficient: a wiring obstruction

In this section we show that (Q1) together with the triangle condition, and indeed together with all the matching-pattern conditions of Section 4, is not sufficient for embeddability, even for finite regular $*$-semigroups. The obstruction is a quasi-identity of a new kind: its hypotheses involve matched pairs and commuting pairs at the same time, and its validity in $\mathcal{P}_X$ rests on the fact that commuting projections share their N-blocks (Lemma 1.1(b)), so that matched pairs to a commuting pair are "wired" through the same nontransversal blocks.

5.1. **Graph semigroups.** Let $G = (V, E)$ be a graph, B a set and $\perp\colon V \to B$ a map; extend $\perp$ to $V \sqcup B$ by $\perp(m) = m$ for $m \in B$, and write $v \bowtie v'$ for $v = v'$ or $vv' \in E$. On the set

$$Q(G, \perp) = (V \times V) \sqcup (B \times B)$$

define $(x, y)^* = (y, x)$ and

$$(p, q)(r, s) = \begin{cases} (p, s) & \text{if } q, r \in V \text{ and } q \bowtie r, \\ (\perp p, \perp s) & \text{otherwise.} \end{cases}$$

Lemma 5.1. *$Q(G, \perp)$ is a regular $*$-semigroup with $|V|^2 + |B|^2$ elements. Its projections are the pairs (v, v), $v \in V$, and (m, m), $m \in B$; identifying them with $V \sqcup B$, the projection algebra is*

$$\theta_v(v') = \begin{cases} v & v \bowtie v' \\ \perp v & \text{otherwise,} \end{cases} \qquad \theta_v(m) = \perp v, \qquad \theta_m(x) = m \qquad (v, v' \in V,\ m \in B,\ x \in V \sqcup B).$$

Its matched pairs of top projections are the edges of G, and it satisfies (Q1) iff $\perp$ is a proper colouring of G (adjacent vertices have distinct values).

Proof. For $x = (p, q)$, $y = (r, s)$, $z = (t, u)$ one checks the four cases according to whether $q \bowtie r$ and $s \bowtie t$: if both hold, $(xy)z = (p, u) = x(yz)$; in the other three cases both products equal $(\bot p, \bot u)$, because a pair with a coordinate in B is never matched. The identities $(xy)^* = y^*x^*$ and $xx^*x = x$ are immediate ($q \bowtie q$ for all q). The projections are the symmetric pairs, and $\theta_{(v,v)}(v', v') = (v, v')(v, v) = (v, v)$ if $v \bowtie v'$, while otherwise it is $(\bot v, \bot v')(v, v) = (\bot v, \bot v)$; the other values are computed in the same way. The projections below (v, v) are (v, v) and $(\bot v, \bot v)$, and (m, m) has no other projection below it; so two distinct top projections v, v' have a common lower bound iff $\bot v = \bot v'$, and then $(v, v)(v', v') = (v, v') \neq (v', v) = (v', v')(v, v)$ if $vv' \in E$, while the two products are both $(\bot v, \bot v)$ if $vv' \notin E$. □

Thus $Q(G, \bot)$ is the Rees matrix semigroup over the trivial group with sandwich matrix "adjacency plus identity" of G, in which the zero has been replaced by the rectangular band $B \times B$ via $\bot$. For $|V| \leq 3$ and every proper colouring $\bot$ the semigroup $Q(G, \bot)$ embeds in $\mathcal{P}_4$ (computer search); our counterexample is the smallest graph for which this fails, and Theorem 5.8 in Section 5.4 determines exactly which $Q(G, \bot)$ embed.

Definition 5.2. Let $\mathfrak{Q} = Q(G, \bot)$ where G is the path $a{-}c{-}b{-}d$ on $V = \{a, b, c, d\}$, $B = \{w, z\}$, $\bot(a) = \bot(b) = w$ and $\bot(c) = \bot(d) = z$; so $|\mathfrak{Q}| = 20$. Its projection algebra $\mathfrak{P} = \{a, b, c, d, w, z\}$ has the matched pairs

$$a \bowtie c, \qquad b \bowtie c, \qquad b \bowtie d, \qquad w \bowtie z,$$

and every other value of θ is "the bottom of the subscript": $\theta_a(b) = \theta_b(a) = w$, $\theta_c(d) = \theta_d(c) = z$, $\theta_a(d) = w$, $\theta_d(a) = z$, $\theta_a(z) = \theta_b(z) = w$, $\theta_c(w) = \theta_d(w) = z$, and $\theta_w \equiv w$, $\theta_z \equiv z$.

The colouring is proper, so $\mathfrak{Q}$ satisfies (Q1). The triangle condition holds in $\mathfrak{P}$ (a direct search over the 6^6 assignments), hence by Theorem 4.14 so do all the non-star matching-pattern conditions; in fact the same search shows that the hexagon, cyclic octagon, directed 4-cycle and double 2-cycle conditions hold in $\mathfrak{P}$, and so do the 2-index square and 3-index star conditions, which fail in $\mathcal{P}_2$.

5.2. **The obstruction.**

Theorem 5.3. *Let a, b, c, d be projections of a partition monoid $\mathcal{P}_X$ such that*

$$a \bowtie c, \qquad b \bowtie c, \qquad b \bowtie d, \qquad cd = dc =: z, \qquad \theta_a(z) = \theta_b(z) = \theta_a(d).$$

Then $z = c$ and $a = b$.

This is a quasi-identity of projection algebras: $cd = dc$ is equivalent to $\theta_c(d) = \theta_d(c)$ in every regular $*$-semigroup (if $cdc = dcd =: e$ then $cd \in \mathcal{H}_e$ and $(cd)^2 = cdcd = ed = e$, so $dc = (cd)^* = (cd)^{-1} = cd$), and $p \bowtie q$ means $\theta_p(q) = p$, $\theta_q(p) = q$. In the binary notation $x * y = \theta_y(x)$ of Lemma 4.7, the theorem reads

$$c{*}a = a,\ a{*}c = c,\ c{*}b = b,\ b{*}c = c,\ d{*}b = b,\ b{*}d = d,\ d{*}c = c{*}d,\ (d{*}c){*}a = (d{*}c){*}b = d{*}a \implies a = b,$$

and it is independent of the conditions found so far: it is not implied by (Q1) and the triangle condition ($\mathfrak{P}$), and together with (Q1) it does not imply the triangle condition, since it holds in the projection algebra $\mathcal{P}(Q_0)$ of Theorem 4.19 and in the Mace4 models of Remark 4.15 (a direct check of all quadruples). It does not imply (Q1) either (it holds in the 5-element $Q(K_2, \bot)$ with constant $\bot$).

Before the proof we note the consequence.

Corollary 5.4. *The projection algebra $\mathfrak{P}$ does not embed in $\mathcal{P}(\mathcal{P}_X)$, and the 20-element regular $*$-semigroup $\mathfrak{Q}$ does not $*$-embed in $\mathcal{P}_X$, for any set X. Consequently (Q1) together with the triangle condition (or with all the matching-pattern conditions of Section 4) is not sufficient for*

a finite regular $*$-semigroup to embed in a partition monoid; the smallest counterexample has order between 9 *and* 20 *(between* 9 *and* 19 *by Theorem 5.16).*

Proof. In $\mathfrak{P}$ the hypotheses of Theorem 5.3 hold, but $a \neq b$. The bound 9 is Section 6. □

The rest of the section proves Theorem 5.3. Recall from Lemma 1.1(c) that a matched pair $p \bowtie q$ comes with a bijection $\mathsf{T}(p) \to \mathsf{T}(q)$, sending a $\mathsf{T}(p)$-block to the unique $\mathsf{T}(q)$-block in the same class of $\varepsilon_p \vee \varepsilon_q$; we write $\varphi\colon \mathsf{T}(a) \to \mathsf{T}(c)$, $\psi\colon \mathsf{T}(b) \to \mathsf{T}(c)$, $\chi\colon \mathsf{T}(b) \to \mathsf{T}(d)$ for the three bijections, and $w = \theta_a(z)$. For projections p, q let $\Gamma(p, q)$ be the bipartite graph whose vertices are the blocks of p and the blocks of q (a set which is a block of both appears twice) and whose edges join a p-block to a q-block meeting it; the classes of $\varepsilon_p \vee \varepsilon_q$ are the unions of the connected components of $\Gamma(p, q)$, and Lemma 1.1(c) says that for $p \bowtie q$ every component contains at most one $\mathsf{T}(p)$-block and at most one $\mathsf{T}(q)$-block, one of each if any. We say that a subgraph *touches* a block if one of its vertices meets that block.

Step 1: the structure of z. By Lemma 1.1(b) applied to $cd = dc$, every class K of $\varepsilon_c \vee \varepsilon_d$ is of one of the types (i) K is a union of $\mathsf{T}(c)$-blocks and a union of $\mathsf{T}(d)$-blocks; (ii) K is a union of $\mathsf{T}(c)$-blocks and a single $\mathsf{N}(d)$-block; (iii) K is a union of $\mathsf{T}(d)$-blocks and a single $\mathsf{N}(c)$-block; (iv) K is a single block of both. By Lemma 1.1(a) the blocks of $z = \theta_c(d)$ are exactly these classes, the T-blocks being those of type (i). In particular $\mathsf{N}(z)$ consists of the $\mathsf{N}(c)$-blocks (types (iii), (iv)) and the type (ii) classes, and symmetrically of the $\mathsf{N}(d)$-blocks and the type (iii) classes.

Step 2: transport. Let $p \bowtie q$ with bijection $\varphi\colon \mathsf{T}(p) \to \mathsf{T}(q)$, and let $x \leq q$. For a set Y which is a union of $\mathsf{T}(q)$-blocks put $\varphi^{-1}[Y] = \bigcup\{\varphi^{-1}(C) : C \in \mathsf{T}(q),\ C \subseteq Y\}$. We claim: the blocks of $\theta_p(x)$ are the $\mathsf{N}(p)$-blocks and the sets $\varphi^{-1}[Y]$ for the blocks $Y \subseteq \operatorname{supp}(q)$ of x, with the label of Y; and the class of $\varepsilon_p \vee \varepsilon_x$ containing $\varphi^{-1}[Y]$ is

$$\Omega_Y = \varphi^{-1}[Y] \cup \bigcup\{\mathsf{N}(p)\text{-blocks}\} = Y \cup \bigcup\{\mathsf{N}(q)\text{-blocks}\},$$

these being all the classes meeting $\operatorname{supp}(p)$. Indeed, by Lemma 1.1(a) the blocks of x outside $\operatorname{supp}(q)$ are $\mathsf{N}(q)$-blocks and those inside are unions of $\mathsf{T}(q)$-blocks. The classes of $\varepsilon_p \vee \varepsilon_q$ meeting $\operatorname{supp}(p)$ are the sets $A \cup (\mathsf{N}(p)\text{-blocks}) = \varphi(A) \cup (\mathsf{N}(q)\text{-blocks})$, $A \in \mathsf{T}(p)$, each containing exactly one $\mathsf{T}(q)$-block. Passing to $\varepsilon_p \vee \varepsilon_x \geq \varepsilon_p \vee \varepsilon_q$, a block $Y \subseteq \operatorname{supp}(q)$ of x merges the classes of the $\mathsf{T}(q)$-blocks it contains, and the other blocks of x lie inside single classes; since each class contains one $\mathsf{T}(q)$-block, which lies in one Y, the classes of $\varepsilon_p \vee \varepsilon_x$ meeting $\operatorname{supp}(p)$ are the Ω_Y. Now Lemma 1.1(a) gives the blocks $\Omega_Y \cap \operatorname{supp}(p) = \varphi^{-1}[Y]$, transversal iff Ω_Y meets $\operatorname{supp}(x)$, i.e. (as $\Omega_Y \cap \operatorname{supp}(q) = Y$) iff Y is a T-block of x.

Applied to (a, c, z), (b, c, z) and (b, d, z), this describes w in three ways: its blocks are the $\mathsf{N}(a)$-blocks and the sets $\varphi^{-1}[K]$, K of type (i) or (ii); the $\mathsf{N}(b)$-blocks and the sets $\psi^{-1}[K]$, K of type (i) or (ii); the $\mathsf{N}(b)$-blocks and the sets $\chi^{-1}[K]$, K of type (i) or (iii); in each case the block is transversal iff K is of type (i).

Step 3: the common join. Since $w \leq a$, every block of w is a union of a-blocks, and by Step 2 every block of w lies in a class of $\varepsilon_a \vee \varepsilon_z$; hence $\varepsilon_a \vee \varepsilon_z = \varepsilon_w \vee \varepsilon_z$, and likewise for b. Write $\Pi = \varepsilon_a \vee \varepsilon_z = \varepsilon_b \vee \varepsilon_z$. By Step 2 the classes of Π meeting $\operatorname{supp}(a)$, equivalently meeting $\operatorname{supp}(c)$ (a class containing $\varphi(A)$ contains A), are the sets $\Omega_K = K \cup (\mathsf{N}(c)\text{-blocks}) = \varphi^{-1}[K] \cup (\mathsf{N}(a)\text{-blocks}) = \psi^{-1}[K] \cup (\mathsf{N}(b)\text{-blocks})$ for K of type (i), (ii); and the classes meeting $\operatorname{supp}(d)$ are $\Omega'_K = K \cup (\mathsf{N}(d)\text{-blocks}) = \chi^{-1}[K] \cup (\mathsf{N}(b)\text{-blocks})$ for K of type (i), (iii). Fix a class Ω of Π meeting one of the four supports and write $S_p = \Omega \cap \operatorname{supp}(p)$ and $\mathsf{N}(p)|_\Omega$ for the set of $\mathsf{N}(p)$-blocks contained in Ω. All classes of $\varepsilon_a \vee \varepsilon_c$, $\varepsilon_b \vee \varepsilon_c$, $\varepsilon_b \vee \varepsilon_d$, $\varepsilon_a \vee \varepsilon_d$ are contained in classes of Π, so all the joins may be computed inside Ω, using only the blocks contained in Ω.

(F1) If $\Omega = \Omega_K$ with K of type (i), then $\Omega = \Omega'_K$, so $S_c = S_d = K$ and $S_a = S_b = \varphi^{-1}[K] = \psi^{-1}[K] = \chi^{-1}[K] =: W$; moreover $\mathsf{N}(a)|_\Omega = \mathsf{N}(b)|_\Omega$ and $\mathsf{N}(c)|_\Omega = \mathsf{N}(d)|_\Omega$. Indeed, the blocks

of w inside Ω are, by the first description, the $\mathsf{N}(a)$-blocks in Ω and the single T-block $\varphi^{-1}[K]$ (the sets $\varphi^{-1}[K']$, $K' \neq K$, lie in other classes); by the second and third descriptions they are the $\mathsf{N}(b)$-blocks in Ω and $\psi^{-1}[K]$, resp. $\chi^{-1}[K]$. Comparing gives the claims about a and b. An $\mathsf{N}(c)$-block $E \subseteq \Omega$ is a block of z of type (iii) or (iv); type (iii) is impossible, since $E \subseteq \Omega \setminus K$, which is a union of $\mathsf{N}(d)$-blocks, whereas a type (iii) class is a union of $\mathsf{T}(d)$-blocks. So E is of type (iv), an $\mathsf{N}(d)$-block; symmetrically.

(F2) If $\Omega = \Omega_K$ with K of type (ii), then Ω contains exactly one class K' of type (iii), and $\Omega = \Omega'_{K'}$; $S_c = K$ is an $\mathsf{N}(d)$-block, $S_d = K'$ is an $\mathsf{N}(c)$-block, and the remaining $\mathsf{N}(c)$-blocks in Ω are exactly the remaining $\mathsf{N}(d)$-blocks in Ω (the type (iv) classes inside Ω). The sets $S_a = \varphi^{-1}[K]$ and $S_b = \psi^{-1}[K] = \chi^{-1}[K']$ are N-blocks of w, and either $S_a = S_b$ and $\mathsf{N}(a)|_\Omega = \mathsf{N}(b)|_\Omega$, or $S_a \cap S_b = \emptyset$, $S_a \in \mathsf{N}(b)$, $S_b \in \mathsf{N}(a)$ and $\mathsf{N}(a)|_\Omega \setminus \{S_b\} = \mathsf{N}(b)|_\Omega \setminus \{S_a\}$. Indeed, Ω contains a $\mathsf{T}(b)$-block B and hence $\chi(B)$, so it meets $\operatorname{supp}(d)$ and $\Omega = \Omega'_{K'}$ for some K' of type (i) or (iii); as $K' \subseteq \Omega \setminus K \subseteq \bigcup \mathsf{N}(c)$, K' is of type (iii), and it is the only type (iii) class in Ω because $\Omega \setminus K'$ is a union of $\mathsf{N}(d)$-blocks; $\Omega \cap \operatorname{supp}(c) = K$ excludes further classes of types (i), (ii). The $\mathsf{N}(c)$-blocks in Ω are the blocks of z of types (iii), (iv) in Ω, i.e. K' and the type (iv) classes, and symmetrically for d. Finally the blocks of w inside Ω are the $\mathsf{N}(a)$-blocks in Ω together with $\varphi^{-1}[K]$, and also the $\mathsf{N}(b)$-blocks together with $\psi^{-1}[K]$, and also the $\mathsf{N}(b)$-blocks together with $\chi^{-1}[K']$; a union of $\mathsf{T}(b)$-blocks is not an $\mathsf{N}(b)$-block, so $\psi^{-1}[K] = \chi^{-1}[K']$, and comparing the first two descriptions gives the alternative.

Step 4: the hypothesis $\theta_a(d) = w$. By Lemma 1.1(a), the blocks of $\theta_a(d)$ inside $\operatorname{supp}(a)$ are the sets $L \cap \operatorname{supp}(a)$, L a class of $\varepsilon_a \vee \varepsilon_d$, transversal iff L meets $\operatorname{supp}(d)$; and each L lies in a class of Π. Since $\theta_a(d) = w$, we get from Step 2: in a class Ω of type (F1) all $\mathsf{T}(a)$-blocks contained in W lie in one class of $\varepsilon_a \vee \varepsilon_d$; in a class of type (F2) all $\mathsf{T}(a)$-blocks contained in S_a lie in one class L of $\varepsilon_a \vee \varepsilon_d$, and L does not meet $\operatorname{supp}(d)$. We show that both are impossible unless they are vacuous, i.e. that $|\mathsf{T}(c) \cap K| = 1$ for K of type (i) (Step 5) and that there is no K of type (ii) (Step 6). Then $\mathsf{T}(z) = \mathsf{T}(c)$ and $\mathsf{N}(z) = \mathsf{N}(c)$ by Step 1, i.e. $z = c$; and then $w = \theta_a(c) = a$ and $w = \theta_b(c) = b$, so $a = b$, proving the theorem.

Step 5: in a class Ω of type (F1)*, distinct $\mathsf{T}(a)$-blocks lie in distinct classes of $\varepsilon_a \vee \varepsilon_d$.* Let $\mathcal{N}' = \mathsf{N}(a)|_\Omega = \mathsf{N}(b)|_\Omega$ and $\mathcal{N} = \mathsf{N}(c)|_\Omega = \mathsf{N}(d)|_\Omega$, and let $\mathcal{G}$ be the graph on $\mathcal{N}' \sqcup \mathcal{N}$ joining blocks that meet. Since the blocks of $\mathcal{N}'$ are a-blocks and b-blocks and those of $\mathcal{N}$ are c-blocks and d-blocks, $\mathcal{G}$ is a subgraph of each of $\Gamma(a,c)$, $\Gamma(b,c)$, $\Gamma(b,d)$, $\Gamma(a,d)$, so each component κ of $\mathcal{G}$ lies in a single component of each of them. By Lemma 1.1(c) for the three matched pairs, κ touches at most one $\mathsf{T}(a)$-block, at most one $\mathsf{T}(b)$-block, at most one $\mathsf{T}(c)$-block and at most one $\mathsf{T}(d)$-block, and if it touches A and C then $C = \varphi(A)$; if B and C then $C = \psi(B)$; if B and D then $D = \chi(B)$. Moreover, if κ touches a $\mathsf{T}(a)$-block A then the point of contact lies in $W = \bigcup \mathsf{T}(b)$, so κ touches a $\mathsf{T}(b)$-block, and conversely; likewise for c and d, as $K = \bigcup \mathsf{T}(c) = \bigcup \mathsf{T}(d)$. Consequently, if κ touches A and D then $D = \sigma(A)$, where $\sigma = \chi\psi^{-1}\varphi$: κ touches some B and some C, and $C = \varphi(A) = \psi(B)$, $D = \chi(B)$. The same holds for a point $x \in W \cap K$ and the blocks A, B, C, D containing it, since meeting blocks lie in the same component.

Now let $A_1 \neq A_2$ be $\mathsf{T}(a)$-blocks in the same class of $\varepsilon_a \vee \varepsilon_d$, and take a path in $\Gamma(a,d)$ from A_1 to A_2 with no $\mathsf{T}(a)$-block in its interior. The interior vertices are blocks of $\mathcal{N}'$, of $\mathcal{N}$ and $\mathsf{T}(d)$-blocks. A maximal interior segment without $\mathsf{T}(d)$-blocks is a connected subgraph of $\mathcal{G}$, hence lies in a component κ. If the path contains no $\mathsf{T}(d)$-block, a single κ touches A_1 and A_2, which is impossible. Otherwise let $D_1, \dots, D_r$ be the $\mathsf{T}(d)$-blocks on the path in order. The segment from A_1 to D_1 is empty (A_1 meets D_1) or a κ touching A_1 and D_1; either way $D_1 = \sigma(A_1)$, and likewise $D_r = \sigma(A_2)$. The segment between D_i and D_{i+1} is a κ touching both, so $D_i = D_{i+1}$. Hence $\sigma(A_1) = \sigma(A_2)$ and $A_1 = A_2$, a contradiction.

Step 6: in a class Ω of type (F2)*, some $\mathsf{T}(a)$-block lies in a class of $\varepsilon_a \vee \varepsilon_d$ meeting* $\mathrm{supp}(d)$. Write $\sim$ for "in the same class of $\varepsilon_a \vee \varepsilon_d$" (for blocks of a and of d inside Ω), $\mathcal{N}$ for the common $\mathsf{N}(c)/\mathsf{N}(d)$-blocks in Ω, and recall that the c-blocks in Ω are the $\mathsf{T}(c)$-blocks, K' and $\mathcal{N}$, the d-blocks are the $\mathsf{T}(d)$-blocks, K and $\mathcal{N}$, every $\mathsf{N}(b)$-block in Ω is an $\mathsf{N}(a)$-block or S_a, and every $\mathsf{N}(a)$-block in Ω is an $\mathsf{N}(b)$-block or S_b. Suppose, for a contradiction, that no $\mathsf{T}(a)$-block is $\sim$ to a $\mathsf{T}(d)$-block.

(α) *Every $\mathsf{T}(a)$-block A satisfies $A \sim K$.* Take a path in $\Gamma(a,c)$ from A to $\varphi(A)$ with no T-block in its interior; its interior c-blocks are blocks of $\mathcal{N}$ or K', and its interior a-blocks are $\mathsf{N}(a)$-blocks. If K' occurs, consider its first occurrence and the a-block u preceding it ($u = A$ if K' comes first): u meets $K' = \bigcup \mathsf{T}(d)$, hence some $\mathsf{T}(d)$-block D, so $u \sim D$; the path from A to u is a path in $\Gamma(a,d)$ (its c-blocks are in $\mathcal{N}$), so $A \sim D$, contrary to assumption. So K' does not occur, and the a-block u preceding $\varphi(A)$ ($u = A$ if the path has length one) meets $\varphi(A) \subseteq K$; as K is a d-block, $u \sim K$ and $A \sim K$.

(β) Hence $K \not\sim D$ for every $\mathsf{T}(d)$-block D.

(γ) *Every $\mathsf{T}(d)$-block D satisfies $S_b \sim D$, and $S_a \neq S_b$.* Let $D = \chi(B)$ and take a path in $\Gamma(b,d)$ from B to D with no T-block in its interior; its interior d-blocks are in $\mathcal{N} \cup \{K\}$ and its interior b-blocks are $\mathsf{N}(a)$-blocks or S_a. If S_a occurs, look at its last occurrence and the d-block v following it: v meets $S_a = \bigcup \mathsf{T}(a)$, hence some $\mathsf{T}(a)$-block A, and the path from v to D is a path in $\Gamma(a,d)$, so $A \sim D$, contrary to assumption. If K occurs, the path from K to D is a path in $\Gamma(a,d)$, contradicting (β). So the interior consists of blocks of $\mathcal{N}$ and $\mathsf{N}(a)$-blocks, and the first d-block v_1 meets B. If $S_a = S_b$ then $B \cap v_1$ contains a point of some $\mathsf{T}(a)$-block A, and $A \sim D$ along the path, a contradiction; so $S_a \neq S_b$, S_b is an a-block meeting v_1, and $S_b \sim D$ along the path (if the path is B, D alone, S_b meets D). As χ is a bijection onto the $\mathsf{T}(d)$-blocks in Ω, this holds for every D.

(δ) Now take any $\mathsf{T}(b)$-block B and a path $B = u_0, v_1, u_1, v_2, \ldots, v_m = \psi(B)$ in $\Gamma(b,c)$ with no T-block in its interior: the c-blocks v_j ($j < m$) are in $\mathcal{N} \cup \{K'\}$ and the b-blocks u_j ($1 \leq j < m$) are $\mathsf{N}(a)$-blocks or S_a. We show by induction on j that u_j is an $\mathsf{N}(a)$-block with $u_j \sim S_b$ (where $u_0 = S_b \supseteq B$). For u_{j-1} (an a-block $\sim S_b$) and v_j: if $v_j \in \mathcal{N}$ then v_j is a d-block meeting u_{j-1}, so $v_j \sim S_b$, and if $v_j = K'$ then u_{j-1} meets some $\mathsf{T}(d)$-block $D \sim S_b$ (by (γ)). If now $u_j = S_a$: in the first case v_j meets S_a in a point of some $\mathsf{T}(a)$-block A, so $A \sim v_j \sim S_b \sim D$ for any D; in the second case a point of $S_a \cap K'$ lies in some $\mathsf{T}(a)$-block A and some $\mathsf{T}(d)$-block D, so $A \sim D$; both contradict the assumption. So u_j is an $\mathsf{N}(a)$-block, and $u_j \sim v_j \sim S_b$ resp. $u_j \sim D \sim S_b$ (it meets K' in a point of some D). Finally u_{m-1} (or S_b itself if $m = 1$) meets $\psi(B) \subseteq K$, so $K \sim S_b \sim D$, contradicting (β). This contradiction proves Step 6 and the theorem. □

Remark 5.5. (1) The hypothesis $\theta_a(d) = \theta_a(z)$ was used only in Step 4; Theorem 5.6 below removes it. (2) The mechanism is different from that of the matching-pattern conditions. There the matchings alone carry the information (Theorem 4.14); here the commuting pair c, d forces c and d to share their N-blocks (Lemma 1.1(b)), and the equality $\theta_a(z) = \theta_b(z)$ forces a and b to share theirs (Step 3); through these shared blocks the matchings $a \bowtie c$, $b \bowtie c$, $b \bowtie d$ transport the pairing of T-blocks to a pairing between a and d. The hubs K, K' of Step 6 (a union of $\mathsf{T}(c)$-blocks which is a single $\mathsf{N}(d)$-block, and vice versa) are the only place where this is delicate.

5.3. **The full wiring condition.**

Theorem 5.6. *In every partition monoid $\mathcal{P}_X$, for projections a, b, c, d,*

$$\text{(S)} \qquad a \bowtie c, \quad b \bowtie c, \quad b \bowtie d, \quad cd = dc, \quad \theta_a(cd) = \theta_b(cd) \implies a \bowtie d.$$

In the binary notation: $c * a = a$, $a * c = c$, $c * b = b$, $b * c = c$, $d * b = b$, $b * d = d$, $d * c = c * d$, $(d * c) * a = (d * c) * b$ imply $d * a = a$ and $a * d = d$.

Condition (S) implies Theorem 5.3 in every projection algebra (Prover9 finds the derivation in a fraction of a second): $a \bowtie d$ gives $\theta_a(d) = a$, so $\theta_a(z) = a = \theta_a(c)$ with $z \leq c$, and θ_a is injective on $c^{\downarrow}$ (its inverse there is θ_c), so $z = c$; then $w = \theta_a(c) = a$ and $w = \theta_b(c) = b$. The converse fails, even in the presence of (Q1) and the triangle condition: see Example 5.7.

Proof. We keep the notation of the proof of Theorem 5.3; Steps 1–3 there did not use the extra hypothesis. Since $\varepsilon_a \vee \varepsilon_d \leq \Pi$, it suffices to show that inside each class Ω of Π meeting $\mathrm{supp}(a) \cup \mathrm{supp}(d)$ every class of $\varepsilon_a \vee \varepsilon_d$ contains exactly one $\mathsf{T}(a)$-block and one $\mathsf{T}(d)$-block. Classes of type (F1) are handled by Step 5 and its mirror image: Step 5 shows that distinct $\mathsf{T}(a)$-blocks are not connected in $\Gamma(a, d)$; the same argument with the roles of a and d exchanged (a path between distinct $\mathsf{T}(d)$-blocks with no interior $\mathsf{T}(d)$-block; the segments between consecutive $\mathsf{T}(a)$-blocks are components κ touching two $\mathsf{T}(a)$-blocks, which is impossible, so there is at most one $\mathsf{T}(a)$-block on the path, and it is σ^{-1} of both ends) shows that distinct $\mathsf{T}(d)$-blocks are not connected; and every $\mathsf{T}(a)$-block A is connected to a $\mathsf{T}(d)$-block, because a path in $\Gamma(a, c)$ from A to $\varphi(A)$ ends with an a-block meeting $\varphi(A) \subseteq K = \bigcup \mathsf{T}(d)$, hence meeting some $\mathsf{T}(d)$-block, and its other vertices are $\Gamma(a, d)$-vertices; dually for $\mathsf{T}(d)$-blocks. It remains to treat a class Ω of type (F2), which we do in three steps: a bound on $p = |\mathsf{T}(a) \cap \Omega|$, a reduction to a finite combinatorial statement, and a verification of that statement by computer.

The bound. We use the notation of Step 6: $K = S_c$, $K' = S_d$, $\mathcal{N}$ the common $\mathsf{N}(c)/\mathsf{N}(d)$-blocks in Ω, and the alternative (B1) $S_a = S_b$ with $\mathsf{N}(a)|_\Omega = \mathsf{N}(b)|_\Omega$, or (B2) $S_a \cap S_b = \emptyset$ with $S_a \in \mathsf{N}(b)$, $S_b \in \mathsf{N}(a)$ and the remaining $\mathsf{N}(a)$- and $\mathsf{N}(b)$-blocks in Ω equal; in both cases write $\mathcal{N}'$ for the common ones. Let $\mathcal{G}$ be the graph on $\mathcal{N}' \sqcup \mathcal{N}$ joining blocks that meet, with components κ; as in Step 5, every κ lies in a single component of each of the four graphs $\Gamma(a, c)$, $\Gamma(b, c)$, $\Gamma(b, d)$, $\Gamma(a, d)$, and (by the three hypotheses) touches at most one $\mathsf{T}(a)$-, one $\mathsf{T}(b)$-, one $\mathsf{T}(c)$- and one $\mathsf{T}(d)$-block. Now let $A \in \mathsf{T}(a)$ and take a path in $\Gamma(a, c)$ from A to $\varphi(A)$ with no T-block in its interior. Its interior vertices are plain blocks, K' and (in case (B2)) S_b. If it contains no hub, it is A followed by a single κ touching A and $\varphi(A)$, or A meets $\varphi(A)$; in case (B1) the point of contact of κ (or of A) with A lies in some $B \in \mathsf{T}(b)$ and κ (or that point) meets K, so B lies in the component of K in $\Gamma(b, d)$ and in the component of $\varphi(A)$ in $\Gamma(b, c)$, whence $A = \varphi^{-1}\psi(B_K)$ where B_K is the unique $\mathsf{T}(b)$-block in the $\Gamma(b, d)$-component of K; in case (B2) the point of contact lies in S_a, so $\varphi(A)$ lies in the component of S_a in $\Gamma(b, c)$, whence $A = \varphi^{-1}(C_a)$ where C_a is the unique $\mathsf{T}(c)$-block of that component. If the path contains K' then A lies in the component of K' in $\Gamma(a, c)$, which contains a unique $\mathsf{T}(a)$-block; and if it contains S_b then A lies in the component of S_b in $\Gamma(a, c)$, again with a unique $\mathsf{T}(a)$-block. Hence $p \leq 2$ in case (B1) and $p \leq 3$ in case (B2).

The reduction. Every point of Ω lies in one block of each of a, b, c, d. On the a-side its blocks are either a pair (A, B) of T-blocks (case (B1)), or (A, S_a) or (S_b, B) (case (B2)), or a common block $F \in \mathcal{N}'$; on the c-side they are (C, K), (K', D) or a common $E \in \mathcal{N}$. The four graphs $\Gamma(\cdot, \cdot)$, and hence the truth of the hypotheses and of the conclusion, depend only on which pairs (an a-side datum, a c-side datum) occur as points. Points whose two data are both plain lie in a component κ of $\mathcal{G}$, and in all four graphs the blocks of κ are connected to each other and to exactly the T-blocks (and, through them, hubs) that κ touches. Consequently the component structure of the four graphs is determined by (i) the set of *direct overlaps*, i.e. pairs of a special a-side datum and a special c-side datum occurring as a point, and (ii) the set of *types* of the components κ, the type of κ being the set of T-blocks it touches (at most one of each of the four kinds, and in case (B1) a $\mathsf{T}(a)$-block iff a $\mathsf{T}(b)$-block, since the contact points lie in cells $A \cap B$, which must be nonempty). Two components of the same type can be merged without changing anything, and a component touching no T-block is irrelevant. So a class of type (F2) is described, up to the equivalence that matters, by a subset of the at most $(p + 1)^4$ types and

a set of direct overlaps, subject to: every T-block occurs in some type or overlap (blocks are nonempty), and in case (B1) a type or overlap involving the cell (A, B) requires that cell to be nonempty, i.e. to occur.

The verification. For each $p \leq 6$ and each of the cases (B1), (B2) we encoded the finitely many such descriptions as a propositional formula (presence variables for types, overlaps and cells; reachability in each of the four finite graphs by iterated squaring), added the fineness of $\Gamma(a, c)$, $\Gamma(b, c)$, $\Gamma(b, d)$ and the negation of the fineness of $\Gamma(a, d)$, and found the formula unsatisfiable (three different SAT solvers, well under a second each; `hubtypes.py`). Together with the bound $p \leq 3$ this proves the theorem in classes of type (F2). (The same encoding shows that classes of type (F2) with $p \geq 3$ do not exist at all, so the actual bound is $p \leq 2$; and a randomised test with an independent implementation of the reachability found no counterexample either, `hubcheck_rand.py`.) (A computer-free proof of the hub case is a special case of Theorem 5.29, see Corollary 5.30 and Remark 5.31(2).) □

Example 5.7. Let $\mathfrak{P}_9 = \{a, b, c, d, m, n, k, w, z\}$ with θ given by the table (rows p, columns q, entry $\theta_p(q)$):

	a	b	c	d	m	n	k	w	z
a	a	m	a	m	m	m	m	w	w
b	m	b	b	b	m	m	m	w	w
c	c	c	c	z	n	n	z	z	z
d	k	d	z	d	k	z	k	z	z
m	m	m	m	m	m	m	m	w	w
n	n	n	n	z	n	n	z	z	z
k	k	k	z	k	k	z	k	z	z
w	w	w	w	w	w	w	w	w	w
z	z	z	z	z	z	z	z	z	z

It is $\mathfrak{P}$ with a middle layer inserted: the matched pairs are $a \bowtie c$, $b \bowtie c$, $b \bowtie d$, $m \bowtie n$, $m \bowtie k$, $w \bowtie z$; $m \leq a, b$, $n \leq c$, $k \leq d$, $w \leq m$, $z \leq n, k$; and $\theta_a(d) = \theta_a(b) = \theta_b(a) = m$, $\theta_d(a) = k$, $\theta_c(d) = \theta_d(c) = z$, $\theta_a(z) = \theta_b(z) = w$. Mace4 produced it as a model of the projection algebra axioms, (Q1), the triangle clause and the quasi-identity of Theorem 5.3, with the hypotheses of (S) as ground equations and $\theta_a(d) \neq a$ (there is none of order ≤ 8; the run needs the memory limit raised because of the six-variable triangle clause). So $\mathfrak{P}_9$ satisfies (Q1), the triangle condition and Theorem 5.3, but violates (S): the pair a, d is only "half wired", $\theta_a(d) = m$ lying strictly between w and a. By Theorem 5.6 it does not embed in any $\mathcal{P}(\mathcal{P}_X)$ (the SAT search confirms this for $\mathcal{P}(\mathcal{P}_m)$, $m \leq 8$), and any regular $*$-semigroup with projection algebra $\mathfrak{P}_9$ (it has at least $16 + 9 + 4 = 29$ elements) is a further finite counterexample to sufficiency which Theorem 5.3 does not detect. Thus (S) is the condition to keep.

5.4. **Graph semigroups: the complete answer.** For the family of Section 5.1 the two conditions (Q1) and (S) are also sufficient, and both have a transparent graph-theoretic meaning. For a colouring $\perp\colon V \to B$ write $V_m = \perp^{-1}(m)$, and for $m \neq n$ let $G[V_m, V_n]$ be the bipartite graph of edges between V_m and V_n.

Theorem 5.8. *Let $G = (V, E)$ be a graph, B a set and $\perp\colon V \to B$ a map. The following are equivalent:*

(i) *$Q(G, \perp)$ $*$-embeds in some partition monoid;*
(ii) *the projection algebra $V \sqcup B$ of $Q(G, \perp)$ embeds in some $\mathcal{P}(\mathcal{P}_X)$;*
(iii) *$V \sqcup B$ satisfies (Q1) and (S);*

(iv) $\perp$ *is a proper colouring, and for any two colours* $m \neq n$ *the bipartite graph* $G[V_m, V_n]$ *contains no induced path* $a-c-b-d$ *with* $\perp a = \perp b = m$ *and* $\perp c = \perp d = n$*; equivalently, every* $G[V_m, V_n]$ *is a disjoint union of complete bipartite graphs and isolated vertices.*

When they hold, $Q(G, \perp)$ *embeds in* $\mathcal{P}_X$ *with* $|X| = |V| + \beta + 2|B|$*, where* β *is the number of connected components with at least one edge of the graphs* $G[V_m, V_n]$*,* $m \neq n$*, and the image consists of elements of rank* ≤ 1*.*

Proof. (i)⇒(ii) is clear and (ii)⇒(iii) is (Q1) together with Theorem 5.6.

(iii)⇒(iv). By Lemma 5.1, (Q1) says that $\perp$ is proper. We work out what (S) says in $V \sqcup B$. A vertex and a bottom element are never matched ($\theta_v(m) = \perp v \neq v$), so in an instance of (S) either all of a, b, c, d lie in B, where everything is matched and the conclusion holds, or all lie in V, with $a \bowtie c$, $b \bowtie c$, $b \bowtie d$ meaning equality or adjacency. Two distinct vertices c, d commute iff $\theta_c(d) = \theta_d(c)$, i.e. iff they are non-adjacent with $\perp c = \perp d$ (if $c \bowtie d$ the two values are $c \neq d$). If $c = d$ then $cd = c$, and $\theta_a(c) = a$, $\theta_b(c) = b$, so the last hypothesis forces $a = b$ and the conclusion $a \bowtie d = b \bowtie d$ holds. If $c \neq d$ then $cd = \perp c \in B$, $\theta_a(cd) = \perp a$ and $\theta_b(cd) = \perp b$. Hence, for a proper colouring, (S) is exactly the statement

$$a \bowtie c, \quad b \bowtie c, \quad b \bowtie d, \quad c \neq d, \quad \perp c = \perp d, \quad \perp a = \perp b \implies a \bowtie d \qquad (a, b, c, d \in V).$$

An induced path $a-c-b-d$ in some $G[V_m, V_n]$ with $a, b \in V_m$, $c, d \in V_n$ satisfies the hypotheses (c, d are distinct and, having the same colour, non-adjacent), so (S) gives $a \bowtie d$, i.e. $ad \in E$, contradicting "induced". Finally, a bipartite graph without induced 4-vertex paths is a disjoint union of complete bipartite graphs: if a connected component is not complete bipartite, choose non-adjacent $p \in V_m$, $q \in V_n$ in it at minimal distance; the distance is odd and ≥ 3, and the first four vertices of a shortest path from p to q form an induced path $p-c-b-d$ with $b \in V_m$, $c, d \in V_n$ ($p \not\sim d$ by minimality). The converse is obvious.

(iv)⇒(i). Let $\mathcal{C}$ be the set of connected components with at least one edge of the graphs $G[V_m, V_n]$, $m \neq n$; by (iv) each $C \in \mathcal{C}$ is complete bipartite between its two nonempty sides $C \cap V_m$ and $C \cap V_n$. Let

$$X = \{x_v : v \in V\} \sqcup \{y_C : C \in \mathcal{C}\} \sqcup \{z_m, z'_m : m \in B\},$$

and for $m \in B$ let ε_m be the partition of X whose non-singleton blocks are $\{z_m, z'_m\}$ and, for every $C \in \mathcal{C}$ one of whose colours is m, the block $\{y_C\} \cup \{x_u : u \in C \setminus V_m\}$ (the point y_C together with the points of the side of C *not* coloured m). These blocks are pairwise disjoint: a vertex $u \in V_n$ lies in at most one component of $G[V_m, V_n]$, and components with different second colours involve points x_u of different colours. Note that x_v is a singleton of $\varepsilon_{\perp v}$.

Claim. Let $v \in V_m$ and $v' \in V_n$. Then x_v and $x_{v'}$ lie in the same class of $\varepsilon_m \vee \varepsilon_n$ iff $v \bowtie v'$ (i.e. $v = v'$ or $vv' \in E$). Indeed, if $m = n$ the join is ε_m, in which both points are singletons, and $v \bowtie v'$ iff $v = v'$ since the colouring is proper. If $m \neq n$ and v lies in a component $C \in \mathcal{C}$ of $G[V_m, V_n]$, then the set $\{y_C\} \cup \{x_u : u \in C\}$ is a union of blocks of ε_m (the block of y_C, which is $\{y_C\} \cup \{x_u : u \in C \cap V_n\}$, and the singletons x_u, $u \in C \cap V_m$) and likewise of ε_n, so it is a class of the join; if v lies in no such component, $\{x_v\}$ is a class. So $x_v \sim x_{v'}$ iff v, v' lie in a common component of $G[V_m, V_n]$ iff $vv' \in E$, by completeness of the components.

Now define $\varphi\colon Q(G, \perp) \to \mathcal{P}_X$: $\varphi(v, v')$ is the partition of $X \sqcup X'$ whose blocks are the blocks of $\varepsilon_{\perp v}$ on X and of $\varepsilon_{\perp v'}$ on X', except that the two singletons $\{x_v\}$ and $\{x'_{v'}\}$ are merged into one transversal block; and $\varphi(m, n)$ consists of the blocks of ε_m on X and of ε_n on X' (rank 0). Clearly $\varphi(x^*) = \varphi(x)^*$, and φ is injective: $\varphi(x)$ determines its rank, its upper and lower partitions (which determine the colours, since $\{z_m, z'_m\}$ is a block of ε_m only) and, in rank 1, the transversal pair $(x_v, x'_{v'})$. For the products, consider $\varphi(v, q)\varphi(r, s)$ with $v, q, r, s \in V$. The only point of the upper row of $\varphi(v, q)$ connected to the middle row is x_v, and its partner there is x_q; the middle row is glued by $\varepsilon_{\perp q} \vee \varepsilon_{\perp r}$, so by the Claim x_q reaches the point x_r, and through

it the lower point x'_s, iff $q \bowtie r$. Hence the product has the upper blocks of $\varepsilon_{\perp v}$, the lower blocks of $\varepsilon_{\perp s}$, and the transversal block $\{x_v, x'_s\}$ if $q \bowtie r$ and no transversal block otherwise: it is $\varphi(v,s)$ or $\varphi(\perp v, \perp s)$, as required. If one of the factors has rank 0 there is no transversal block in the product, whose upper and lower partitions are those of the first and the last factor: $\varphi(m,n)\varphi(r,s) = \varphi(m,\perp s)$, $\varphi(v,q)\varphi(m,n) = \varphi(\perp v, n)$, $\varphi(m,n)\varphi(m',n') = \varphi(m,n')$. So φ is an injective $*$-homomorphism. □

Remark 5.9. (1) For an induced subgraph $G' = G[V']$ the semigroup $Q(G', \perp|_{V'})$ is a $*$-subsemigroup of $Q(G,\perp)$ (products of pairs over $V' \sqcup \perp(V')$ stay there). So a non-embeddable $Q(G,\perp)$ with $\perp$ proper contains a copy of $\mathfrak{Q}$, and $\mathfrak{Q}$ is the smallest non-embeddable graph semigroup satisfying (Q1); the "finite list of forbidden coloured subgraphs" consists of the monochromatic edge and the bichromatic path $a - c - b - d$.

(2) The theorem was found by a computer survey (`graphsg_survey.py`): for the 108 coloured graphs with at most 4 vertices (up to isomorphism and renaming of colours) the projection algebra embeds in some $\mathcal{P}(\mathcal{P}_m)$, $m \le 5$, exactly when (iv) holds (the search was run for the 48 proper colourings; the others fail (Q1)), and then already for $m \le 4$; the triangle condition holds in all of them. The explicit embedding of the proof was then verified by computer for all 788 coloured graphs with 5 vertices (190 embeddable) and all 13 712 with 6 vertices (1 824 embeddable) (`graphsg_thm.py`). The construction is not economical: for $\mathfrak{Q}$ with $\perp$ injective it uses $|X| = 4+3+8 = 15$ while $\mathcal{P}_4$ suffices.

(3) The triangle condition holds in every $Q(G,\perp)$ with $\perp$ proper, and for a trivial reason. Three pairwise commuting projections with a common pairwise product w are either all equal or three distinct pairwise non-adjacent vertices of one colour $w \in B$ (or elements of B, all equal to w); a triangle configuration with $w \ne z$ therefore has $a_i \in V_w$ and $b_i \in V_z$ with $w \ne z$, and then a_i and b_i cannot commute. Thus the family $Q(G,\perp)$ sees only (Q1) and (S), and by the theorem it can witness no further obstruction: an embedding, when it exists, uses only projections of rank ≤ 1, whose T-blocks are single points. The layered graph semigroups of Section 5.5, which contain $\mathfrak{P}_9$, allow the commuting products to lie several levels below the matched vertices; they see no further obstruction either (Theorem 5.12).

5.5. **Layered graph semigroups.** Theorem 5.8 shows that graph semigroups cannot witness obstructions beyond (Q1) and (S). The algebra $\mathfrak{P}_9$ of Example 5.7 suggests a generalisation in which the bottom band of $Q(G,\perp)$ is replaced by another graph semigroup, and so on, so that the product of two unmatched projections may lie several levels below them. It turns out that this family, too, sees only (Q1) and (S).

Definition 5.10. A *layered graph* is a finite set $V = V_1 \sqcup \cdots \sqcup V_L$ ($L \ge 1$) with a reflexive symmetric relation $\bowtie$ on each level V_i, complete on the bottom level V_L, and maps $\perp\colon V_i \to V_{i+1}$ ($1 \le i < L$) which preserve $\bowtie$. For $p \in V_i$ and $\ell \ge i$ write $p_\ell = \perp^{\ell-i} p$ (p descended to level ℓ), call p_L the *root* of p and $T_r = \{p : p_L = r\}$ the *tree* of the root $r \in V_L$; the trees partition V. For $q, r \in V$ let

$$f(q,r) = \min\{\ell \ge \max(\operatorname{lv} q, \operatorname{lv} r) : q_\ell \bowtie r_\ell\},$$

which exists because the bottom level is complete. The *layered graph semigroup* $Q(V)$ is the set of pairs (p,q) with p, q on a common level, with $(p,q)^* = (q,p)$ and

$$(p,q)(r,s) = (p_\ell, s_\ell), \qquad \ell = f(q,r).$$

For $L = 2$ this is $Q(G,\perp)$ ($V_1 = V$, $V_2 = B$); every map into a complete graph preserves $\bowtie$. The layered graph with $V_1 = \{a,b,c,d\}$ ($a \bowtie c$, $b \bowtie c$, $b \bowtie d$), $V_2 = \{m,n,k\}$ ($m \bowtie n$, $m \bowtie k$), $V_3 = \{w,z\}$, $\perp\colon a,b \mapsto m$, $c \mapsto n$, $d \mapsto k$, $m \mapsto w$, $n,k \mapsto z$ gives a 29-element regular $*$-semigroup whose projection algebra is $\mathfrak{P}_9$ (`layered.py`).

Lemma 5.11. *$Q(V)$ is a regular $*$-semigroup; its projections are the pairs (p,p), identified with V, and*

$$\theta_p(q) = p_{f(p,q)}, \qquad q \le p \iff q = p_\ell \text{ for some } \ell.$$

(If $\perp$ does not preserve $\bowtie$, the multiplication is not associative.) Moreover:

(a) *(Q1) holds in $Q(V)$ iff distinct matched elements have different roots;*

(b) *two elements p, q commute iff $p_f = q_f$ for $f = f(p,q)$, and then $pq = p_f$; under (Q1) this says that p, q lie in a common tree, and p_f is their least common ancestor;*

(c) *assuming (Q1), (S) holds in $Q(V)$ iff for every level i and every two roots $r \neq r'$ the bipartite graph of matched pairs between $T_r \cap V_i$ and $T_{r'} \cap V_i$ contains no induced path on four vertices, i.e. is a disjoint union of complete bipartite graphs and isolated vertices;*

(d) *assuming (Q1), the triangle condition holds in $Q(V)$.*

Proof. Preservation of $\bowtie$ gives $q_\ell \bowtie r_\ell \Rightarrow q_{\ell'} \bowtie r_{\ell'}$ for $\ell' \ge \ell$, hence $f(q_k, r) = \max(k, f(q,r))$ for $k \ge \operatorname{lv} q$. So $((p,q)(r,s))(t,u) = (p_{\ell_2}, u_{\ell_2})$ with $\ell_2 = f(s_{\ell_1}, t) = \max(f(q,r), f(s,t))$, $\ell_1 = f(q,r)$, and symmetrically $(p,q)((r,s)(t,u)) = (p_{\ell_4}, u_{\ell_4})$ with $\ell_4 = f(q, r_{\ell_3}) = \max(f(s,t), f(q,r))$, $\ell_3 = f(s,t)$. The identities $(xy)^* = y^*x^*$ and $xx^*x = x$ are clear (f is symmetric and $f(q,q) = \operatorname{lv} q$). If $p \bowtie q$ but $\perp p \not\bowtie \perp q$ (so $\operatorname{lv} p + 1 < L$), then $((p,p)(q,q))(\perp q, \perp q) = (p,q)(\perp q, \perp q) = (\perp p, \perp q)$, while $(p,p)((q,q)(\perp q, \perp q)) = (p,p)(\perp q, \perp q) = (p_\ell, q_\ell)$ with $\ell \ge \operatorname{lv} p + 2$. The projections are the symmetric pairs, and $\theta_p(q) = (p,p)(q,q)(p,p) = (p_f, q_f)(p,p) = (p_f, p_f)$ with $f = f(p,q)$, since $f(q_f, p) = f$; so $\theta_p(q) = q$ iff $q = p_f$.

(b) is the formula for θ together with the remark after Theorem 4.2. Note that within a tree, $p_f \bowtie q_f$ with $p_f \neq q_f$ would be two distinct matched elements with a common descendant, so under (Q1) "matched" means "equal" inside a tree and $f(p,q)$ is the level of the least common ancestor of p and q.

(a) Matched elements lie on a common level, so "no common descendant" means "different roots". If $p \bowtie q$ are distinct with a common root then $r = p_L = q_L$ satisfies $r \le p, q$ but $\theta_p(q) = p \neq q = \theta_q(p)$. Conversely let $r \le p$, $r \le q$ with $p \neq q$, say $r = p_\ell = q_\ell$; at the level $f = f(p,q) \le \ell$ the matched elements p_f, q_f have the common descendant r, so $p_f = q_f$ and $\theta_p(q) = p_f = q_f = \theta_q(p)$.

(c) Matched elements lie on a common level, so in an instance of (S) all of a, b, c, d lie on one level i. If $c = d$ then $cd = c$, $\theta_a(c) = a$, $\theta_b(c) = b$ and the hypothesis $\theta_a(cd) = \theta_b(cd)$ gives $a = b$, so the instance is trivial. If $c \neq d$ then $cd = dc$ says by (b) that c, d lie in a common tree, with $z := c_f = d_f$, $f = f(c,d)$; since $a_f \bowtie c_f = z$ we get $f(a,z) = f$ and $\theta_a(z) = a_f$, similarly $\theta_b(z) = b_f$, so the last hypothesis says that a, b lie in a common tree and their least common ancestor is at level $\le f$. By (Q1) the tree of a, b differs from that of c, d. Thus, writing $\mu(p,q)$ for the level of the least common ancestor of two elements of a tree, (S) says: for every path $a - c - b - d$ of matchings with $a, b \in T_r$, $c, d \in T_{r'}$, $r \neq r'$, if $\mu(a,b) \le \mu(c,d)$ then $a \bowtie d$. Applied to the reversed path $d - b - c - a$ (with c, d in the role of a, b) it says the same under $\mu(c,d) \le \mu(a,b)$; one of the two always holds. So (S) is equivalent to: every path $a - c - b - d$ with $a, b \in T_r \cap V_i$ and $c, d \in T_{r'} \cap V_i$ closes up, $a \bowtie d$; that is, the bipartite graph in (c) has no induced 4-vertex path, which (as in the proof of Theorem 5.8) means that it is a disjoint union of complete bipartite graphs.

(d) Let $a_1, a_2, a_3, b_1, b_2, b_3$ be a triangle configuration. By (b) the a_i pairwise commute, so they lie in one tree; likewise the b_i lie in one tree; and a_i commutes with b_i, so all six lie in one tree. Then $a_1 \bowtie b_2$ forces $a_1 = b_2$ by (a), and similarly $a_2 = b_3$, $a_3 = b_1$. Hence $\{a_1, a_2, a_3\} = \{b_1, b_2, b_3\}$ and $w = a_1a_2 = b_2b_3 = z$. □

Theorem 5.12. *For a layered graph V the following are equivalent:*

(i) *$Q(V)$ $*$-embeds in some partition monoid;*

(ii) *the projection algebra V of $Q(V)$ embeds in some $\mathcal{P}(\mathcal{P}_X)$;*

(iii) *V satisfies (Q1) and (S);*
(iv) *distinct matched elements have different roots, and for every level i and every two roots $r \neq r'$ the bipartite graph of matched pairs between $T_r \cap V_i$ and $T_{r'} \cap V_i$ is a disjoint union of complete bipartite graphs and isolated vertices.*

When they hold, $Q(V)$ embeds in $\mathcal{P}_X$ with $|X| = |V| + |V_L| + \beta$, where β is the number of complete bipartite components in (iv), and every T-block of the image is a single point.

Proof. (i)⇒(ii) is clear, (ii)⇒(iii) is (Q1) and Theorem 5.6, and (iii)⇔(iv) is Lemma 5.11(a),(c). For (iv)⇒(i) we generalise the construction of Theorem 5.8, with trees in the role of colours and the non-root elements in the role of vertices. Let $\mathcal{C}$ be the set of complete bipartite components in (iv), over all levels $i < L$ and pairs of roots; each $C \in \mathcal{C}$ lies on one level and meets exactly two trees. Let

$$X = \{x_v : v \in V \setminus V_L\} \sqcup \{y_C : C \in \mathcal{C}\} \sqcup \{z_r, z'_r : r \in V_L\},$$

and for a root r let ε_r be the partition of X whose non-singleton blocks are $\{z_r, z'_r\}$ and, for each $C \in \mathcal{C}$ meeting T_r, the block $\{y_C\} \cup \{x_u : u \in C \setminus T_r\}$. These blocks are disjoint (an element lies in at most one component per level and pair of trees, and for a fixed r the elements x_u in the blocks belong to the other tree of C). As in the Claim in the proof of Theorem 5.8, for $u \in T_r$ and $v \in T_{r'}$, $r \neq r'$, the points x_u, x_v lie in a common class of $\varepsilon_r \vee \varepsilon_{r'}$ iff u, v lie in a common component $C \in \mathcal{C}$, iff $u \bowtie v$; the class of x_u is $\{y_C\} \cup \{x_w : w \in C\}$ or $\{x_u\}$, and in particular contains no other point $x_{u'}$ with $u' \in T_r$ on a different level. Note also that every x_v, $v \in T_r$, is a singleton of ε_r.

For $p \in T_r$ let $\hat{p}$ be the projection with partition ε_r whose T-blocks are the singletons $\{x_{p_j}\}$, $\operatorname{lv} p \leq j < L$ (the chain from p down to, but excluding, the root); so $\hat{r}$ has rank 0, $\operatorname{rank} \hat{p} = L - \operatorname{lv} p$, and $\operatorname{supp} \hat{p}$ determines p. We check $\theta_{\hat{p}}(\hat{q}) = \widehat{p_f}$, $f = f(p,q)$, using Lemma 1.1(a). If $p, q \in T_r$, the join is ε_r itself, whose classes meeting $\operatorname{supp} \hat{p}$ are the singletons $\{x_{p_j}\}$, transversal in $\theta_{\hat{p}}(\hat{q})$ iff $x_{p_j} \in \operatorname{supp} \hat{q}$, i.e. iff $p_j = q_k$ for some $k \geq \operatorname{lv} q$, i.e. iff $j \geq f$ (the level of the least common ancestor); so $\theta_{\hat{p}}(\hat{q})$ is ε_r with the T-blocks $\{x_{p_j}\}$, $j \geq f$, which is $\widehat{p_f}$. If $p \in T_r$, $q \in T_{r'}$ with $r \neq r'$, the classes of $\varepsilon_r \vee \varepsilon_{r'}$ meeting $\operatorname{supp} \hat{p}$ are the classes K_j of the points x_{p_j}, each containing only this one T-block of $\hat{p}$; K_j meets $\operatorname{supp} \hat{q} = \{x_{q_k} : k \geq \operatorname{lv} q\}$ iff $p_j \bowtie q_k$ for some $k \geq \operatorname{lv} q$, which forces $k = j$, iff $j \geq \operatorname{lv} q$ and $p_j \bowtie q_j$, iff $j \geq f$ (by preservation of $\bowtie$ and the definition of f). So again $\theta_{\hat{p}}(\hat{q}) = \widehat{p_f}$. (For $q = r'$ a root, $\operatorname{supp} \hat{q} = \emptyset$ and $\theta_{\hat{p}}(\hat{q}) = \hat{r} = \widehat{p_L}$, as required; and $\theta_{\hat{r}}(\hat{q}) = \hat{r}$ for every q since $\hat{r}$ has rank 0.) Injectivity: ε_r determines r (the block $\{z_r, z'_r\}$), and the T-blocks determine p. So $p \mapsto \hat{p}$ embeds the projection algebra.

For the semigroup, send (p,q) (with p, q on level i) to the element $\varphi(p,q)$ whose upper and lower partitions are those of $\hat{p}$ and $\hat{q}$, with the transversal blocks $\{x_{p_j}, x'_{q_j}\}$, $i \leq j < L$. This is injective and $*$-compatible, and $\varphi(p,q)\varphi(r,s) = \varphi(p_\ell, s_\ell)$, $\ell = f(q,r)$: in the product, the upper point x_{p_j} is joined to the middle point x_{q_j}, whose class in the middle join $\varepsilon_{T(q)} \vee \varepsilon_{T(r)}$ contains a point x_{r_k}, $k \geq \operatorname{lv} r$ (necessarily $k = j$), iff $j \geq f(q,r)$, by the two computations above, and then x_{p_j} is joined to x'_{s_j}; for $j < \ell$ the block of x_{p_j} in the product is the nontransversal singleton $\{x_{p_j}\}$, and the other blocks are unchanged. This is $\varphi(p_\ell, s_\ell)$. □

Remark 5.13. (1) The search behind the theorem (`layered_search.py`, `layered_search2.py`) enumerated the layered graphs with surjective descent maps and level sizes (n_1, n_2, n_3), $n_1 \leq 4$, $n_2 \leq 3$, $n_3 \leq 2$ (the largest shape $(4,3,2)$ only partly, the run having been interrupted), and the four-level shapes $(2,2,2,1)$, $(3,2,2,1)$, $(2,2,2,2)$, $(3,2,2,2)$; it computed their projection algebras, kept those satisfying (Q1) (up to isomorphism) and tested the triangle condition, (S) and embeddability into $\mathcal{P}(\mathcal{P}_m)$, $m \leq 8$ (backtracking, then SAT). The triangle condition never failed, and every algebra satisfying (S) embedded, with $m \leq 7$; the trees (no matchings above the bottom level) need the largest m. The construction of the theorem was then verified as an

injective $*$-homomorphism on all these structures and on several hundred random layered graphs with up to four levels, and the predicate (iv) was checked against (Q1) and (S) computed from the table (`layered_thm.py`).

(2) In a layered graph every principal down-set $p^{\downarrow}$ is a chain, every value $\theta_p(q)$ lies in that chain, and (under (Q1)) the commuting pairs are the pairs in a common tree. This is the reason why the triangle condition is invisible here: it needs two commuting triples with a common meet, which in a tree forces the triples to coincide. Any semigroup separating (Q1)+(S) from embeddability must therefore have projections with non-chain down-sets (as $\mathcal{P}(Q_0)$ does, where the triangle condition fails), or must force T-blocks of size > 1; both constructions of this section use only singleton T-blocks.

5.6. **A fourth obstruction.** Theorems 5.8 and 5.12 say where *not* to look for further obstructions. A census of the projection algebras of order 7 finds one.

The census. Mace4 enumerated the projection algebras of order 7 satisfying (Q1) and (S) (the axioms, the clause for (Q1) and the two four-variable clauses for (S); the six-variable triangle clause was omitted and the triangle condition tested afterwards). To make the run resumable it was split into 49 chunks by the values of $0 * 1$ and $1 * 0$, of which 21 are redundant by the symmetry exchanging 0 and 1 (`pa7_driver.sh`); isofilter within the chunks and a canonical form across them (`pa7_post.py`) leave 1 663 algebras up to isomorphism (there are 295 of order 6). All of them satisfy the triangle condition. For 1 662 of them the SAT search finds an embedding into $\mathcal{P}(\mathcal{P}_m)$ with $m \le 6$. The remaining one has no embedding into $\mathcal{P}(\mathcal{P}_m)$ for $m \le 8$ (the case $m = 8$ takes a quarter of an hour). With the elements renamed it is the following algebra.

Definition 5.14. $\mathfrak{P}_7 = \{a, c, b, x, u, v, w\}$ with θ given by (rows p, columns q, entry $\theta_p(q)$)

	a	c	b	x	u	v	w
a	a	a	u	u	u	u	u
c	c	c	c	v	v	v	v
b	w	b	b	w	w	w	w
x	u	v	v	x	u	v	v
u	u	u	u	u	u	u	u
v	v	v	v	v	v	v	v
w	w	w	w	w	w	w	w

In words: $a \bowtie c \bowtie b$ is a path; u, v, w form a bottom rectangular band ($\theta_u \equiv u$ etc., all pairs matched); $u \le a$, $v \le c$, $w \le b$, and every value not listed so far is the bottom of the subscript—so far this is the projection algebra of the graph semigroup $Q(a - c - b, \bot)$ with the injective colouring $\bot a = u$, $\bot c = v$, $\bot b = w$, which embeds in $\mathcal{P}_4$. The new element is x, with $u, v \le x$: it commutes with a and with c, $xa = u$, $xc = v$, and it sees b as it sees c, $\theta_x(b) = v$, while $\theta_b(x) = w$. Deleting any element leaves a subalgebra (of order 6, hence embeddable), so $\mathfrak{P}_7$ is a minimal non-embeddable algebra. There is a 19-element regular $*$-semigroup $\mathfrak{T}_{19}$ with projection algebra $\mathfrak{P}_7$: adjoin to $Q(a - c - b, \bot)$ an element x with $x \cdot (p, q) = (\lambda p, \bot q)$, $(p, q) \cdot x = (\bot p, \lambda q)$ for the pairs (p, q) of top elements, $x \cdot (m, n) = (\lambda m, n)$, $(m, n) \cdot x = (m, \lambda n)$ for the pairs of bottom elements, $x^2 = x$, $x^* = x$, where $\lambda a = \lambda u = u$ and $\lambda c = \lambda b = \lambda v = \lambda w = v$ (`fourth.py` checks associativity and computes the projection algebra).

$\mathfrak{P}_7$ satisfies (Q1), the triangle condition and (S) (by the census, and directly by computer), hence all the matching-pattern conditions. The obstruction is the following quasi-identity.

Proposition 5.15. *Let*

$$\text{(R)} \qquad a \bowtie c, \quad b \bowtie c, \quad xa = ax, \quad xc = cx, \quad \theta_x(b) = \theta_x(c) \implies a \bowtie b.$$

(a) *(R) holds in $\mathcal{P}(\mathcal{P}_m)$ for $m \le 5$ (all 580 902 instances of the hypotheses in $\mathcal{P}(\mathcal{P}_5)$ were checked), and for $m \le 6$ in the following stronger sense: a SAT search for four projections of $\mathcal{P}_6$ satisfying the hypotheses with $\theta_a(b) \ne a$, or with $\theta_b(a) \ne b$, is unsatisfiable (`fourth.py`, `pa_sat.py` with a partial table).*

(b) *(R) fails in $\mathfrak{P}_7$, with (a, b, c, x) as named; it is the only instance. Theorem 5.16 below proves (R) in every $\mathcal{P}_X$; the computations in (a) were the evidence that led to it.*

(c) *(R) holds in every projection algebra of order ≤ 7 satisfying (Q1) and (S) other than $\mathfrak{P}_7$, and in every projection algebra of order ≤ 6 satisfying (Q1) and the triangle condition, in particular in $\mathfrak{P}$; and it holds in $\mathfrak{P}_9$. Hence (R) is independent of (Q1), the triangle condition and (S): $\mathfrak{P}$ satisfies (R) but not (S), $\mathfrak{P}_7$ satisfies (S) but not (R), and $\mathcal{P}(Q_0)$ and $Q(K_2, const)$ show the rest as before.*

(d) *In $\mathcal{P}(\mathcal{P}_5)$ the hypothesis $\theta_a(b) = \theta_a(x)$ (which also holds in $\mathfrak{P}_7$) does not strengthen (R): adding it changes nothing, and each of the five hypotheses listed is needed.*

Theorem 5.16. *(R) holds in every partition monoid $\mathcal{P}_X$. Consequently $\mathfrak{P}_7$ embeds in no $\mathcal{P}(\mathcal{P}_X)$, $\mathfrak{T}_{19}$ embeds in no $\mathcal{P}_X$, and the smallest regular $*$-semigroup satisfying (Q1), the triangle condition and (S) that embeds in no partition monoid has order between 9 and 19.*

Proof. Let a, b, c, x be projections of $\mathcal{P}_X$ with $a \bowtie c$, $b \bowtie c$, $xa = ax =: u$, $xc = cx =: v$ and $\theta_x(b) = v$. We use the notation of Lemma 1.1; "block" means a block of the relevant partition ε_p of X, and two blocks of different partitions are *adjacent* if they meet. A *chain* in a join $\varepsilon_p \vee \varepsilon_q$ is a sequence of blocks, alternately of p and of q, in which consecutive blocks are adjacent; two blocks lie in a common class of the join iff they are joined by a chain. Write $A_1, \dots, A_s$ for the $\mathsf{T}(a)$-blocks, C_i for the $\mathsf{T}(c)$-block matched with A_i by $a \bowtie c$ and B_i for the $\mathsf{T}(b)$-block matched with C_i by $b \bowtie c$ (Lemma 1.1(c)); Γ_i and Δ_i denote the classes of $\varepsilon_a \vee \varepsilon_c$ and $\varepsilon_b \vee \varepsilon_c$ containing C_i, so that Γ_i contains exactly the T-blocks A_i, C_i and Δ_i exactly B_i, C_i. Write H for a generic $\mathsf{T}(x)$-block and $Y = X \setminus \operatorname{supp} x$.

Step 1: the commuting pairs. By Lemma 1.1(b) the classes of $\varepsilon_x \vee \varepsilon_a$ are of four types: the $\mathsf{T}(u)$-blocks $U_1, \dots, U_t$ (each a union of A's and a union of H's), the $\mathsf{N}(a)$-blocks inside $\operatorname{supp} x$ (unions of H's), the $\mathsf{N}(x)$-blocks inside $\operatorname{supp} a$ (unions of A's), and the common N-blocks; and $\varepsilon_u = \varepsilon_x \vee \varepsilon_a$. Likewise for c, with the $\mathsf{T}(v)$-blocks $V_1, \dots, V_{t'}$. In particular every block of a and every block of c lies inside $\operatorname{supp} x$ or inside Y; every A_i lies in some U_l or in an $\mathsf{N}(x)$-block, and every C_i in some V_k or in an $\mathsf{N}(x)$-block. An $\mathsf{N}(x)$-block M containing a $\mathsf{T}(c)$-block C_i is a union of C's; it is not an $\mathsf{N}(a)$-block (an ε_a-block which is a union of ε_c-blocks would be a class of $\varepsilon_a \vee \varepsilon_c$ containing C_i and no A), so it is a union of A's as well, and the class of C_i in $\varepsilon_a \vee \varepsilon_c$, being made of the A's and C's inside M, is $A_i = C_i$. Hence, with $I = \{i : A_i \subseteq \operatorname{supp} u\}$:

$$A_i \subseteq \operatorname{supp} u \iff C_i \subseteq \operatorname{supp} v, \qquad \text{and } A_i = C_i \subseteq Y \text{ for } i \notin I$$

(if $A_i \subseteq \operatorname{supp} u \subseteq \operatorname{supp} x$ but $C_i \subseteq Y$, then $A_i = C_i \subseteq Y$, a contradiction). Similarly an $\mathsf{N}(x)$-block is an $\mathsf{N}(a)$-block iff it is an $\mathsf{N}(c)$-block. So *a and c have the same blocks inside Y*, with the same labels. For $i \in I$ let $U_{l(i)} \supseteq A_i$ and $V_{k(i)} \supseteq C_i$.

Step 2: two elementary facts. (a) An $\mathsf{N}(a)$-block meets at most one C, an $\mathsf{N}(c)$-block meets at most one A and at most one B, an $\mathsf{N}(b)$-block meets at most one C, a C meets at most one A and at most one B: for instance an $\mathsf{N}(a)$-block meeting C_i and C_j lies in $\Gamma_i \cap \Gamma_j$. (b) $u \bowtie v$, and $\theta_u(b) = u$, $\theta_v(b) = v$. Indeed $g = ac$ satisfies $gg^* = a$, $g^*g = c$, and $xg = xac = uc = uxc = uv = axc = acx = gx$; so $u = xgg^*x = (uv)(uv)^* = uvu$ and $v = xg^*gx = (uv)^*(uv) = vuv$. Since $u \le x$, $\theta_u(y) = \theta_u(\theta_x(y))$ for every projection y (as $u = ux = xu$); so $\theta_u(b) = \theta_u(v) = u$ and $\theta_v(b) = \theta_v(v) = v$. By $u \bowtie v$ every class of $\varepsilon_u \vee \varepsilon_v = \varepsilon_x \vee \varepsilon_a \vee \varepsilon_c$ meeting $\operatorname{supp} u \cup \operatorname{supp} v$ contains exactly one U and one V; since Γ_i lies in such a class, $l(i) = l(j)$ iff $k(i) = k(j)$ for $i, j \in I$, i.e. the A's inside U_l are matched with the C's inside one $V_{k(l)}$, and U_l and $V_{k(l)}$ lie in

one class of $\varepsilon_x \vee \varepsilon_a \vee \varepsilon_c$; two $\mathsf{T}(x)$-blocks lying in U_l and V_k respectively are connected in that join only if $k = k(l)$.

Step 3: the hypothesis $\theta_x(b) = v$. By Lemma 1.1(a) the blocks of $\theta_x(b)$ are the $\mathsf{N}(x)$-blocks and the sets $U_K = K \cap \operatorname{supp} x$ (a union of H's) for the classes K of $\varepsilon_x \vee \varepsilon_b$ meeting $\operatorname{supp} x$, U_K being transversal iff K meets $\operatorname{supp} b$; and the blocks of v are the $\mathsf{N}(x)$-blocks, the T-blocks V_k and the $\mathsf{N}(c)$-blocks inside $\operatorname{supp} x$. So the partition $\mathcal{Q}$ of $\operatorname{supp} x$ into the V_k and the $\mathsf{N}(c)$-blocks inside $\operatorname{supp} x$ is also the partition induced by $\varepsilon_x \vee \varepsilon_b$: there is a class K_k with $K_k \cap \operatorname{supp} x = V_k$, which meets $\operatorname{supp} b$, and for each $\mathsf{N}(c)$-block $\tilde{N} \subseteq \operatorname{supp} x$ a class $K_{\tilde{N}}$ with $K_{\tilde{N}} \cap \operatorname{supp} x = \tilde{N}$, which does not meet $\operatorname{supp} b$. Consequently every ε_b-block Z meets $\operatorname{supp} x$ inside a single part of $\mathcal{Q}$.

Step 4: the classes Δ_i avoid $\operatorname{supp} x$ except through C_i. We claim that no $\mathsf{N}(c)$-block inside $\operatorname{supp} x$ lies in a Δ_i. Suppose $\tilde{N} \subseteq \Delta_i$ and take a shortest chain $\tilde{N} = E_0, Z_1, E_1, Z_2, \ldots, B_i$ in Δ_i (Z's blocks of b, E's blocks of c). Z_1 meets an $H \subseteq \tilde{N}$, so $Z_1 \subseteq K_{\tilde{N}}$. Inductively all blocks of the chain lie in $K_{\tilde{N}}$: an ε_b-block meeting a block of $K_{\tilde{N}}$ lies in $K_{\tilde{N}}$; an $\mathsf{N}(c)$-block E meeting such a Z is an $\mathsf{N}(x)$-block (then $E \subseteq K_{\tilde{N}}$) or lies inside $\operatorname{supp} x$ (then the $H \subseteq E$ meeting Z lies in $K_{\tilde{N}} \cap \operatorname{supp} x = \tilde{N}$, so $E = \tilde{N}$, excluded by minimality); and if $E = C_i$ then either $C_i \subseteq V_{k(i)}$, and the $H \subseteq V_{k(i)}$ meeting Z would lie in $\tilde{N}$, or C_i lies in an $\mathsf{N}(x)$-block, which is then in $K_{\tilde{N}}$. So $B_i \subseteq K_{\tilde{N}}$, contradicting $K_{\tilde{N}} \cap \operatorname{supp} b = \emptyset$. Consequently: every $\mathsf{N}(c)$-block in Δ_i lies in Y and is an $\mathsf{N}(x)$-block and an $\mathsf{N}(a)$-block (Step 1); an ε_b-block Z meeting $\operatorname{supp} x$ inside some V_k meets a $C_p \subseteq V_k$, so $Z \subseteq \Delta_p$ and $Z \cap V_k \subseteq C_p$ by Step 2(a); and an ε_b-block meeting $\operatorname{supp} x$ inside an $\mathsf{N}(c)$-block $\tilde{N}$ lies in a class of $\varepsilon_b \vee \varepsilon_c$ containing $\tilde{N}$, hence containing no B and no C.

Step 5: the indices outside I. Let $i \notin I$, so $A_i = C_i \subseteq Y$, and let D be the class of A_i in $\varepsilon_a \vee \varepsilon_b$. We claim that every ε_a-block in D lies in Y. Otherwise take a shortest chain $A_i, Z_1, Y_1, \ldots, Z_t, Y_t$ in D from A_i to a block $Y_t \subseteq \operatorname{supp} x$ of a; then $Y_1, \ldots, Y_{t-1} \subseteq Y$ are blocks of c (Step 1), so the chain up to Z_t lies in $\varepsilon_b \vee \varepsilon_c$ and $Z_t \subseteq \Delta_i$. Now Z_t meets $Y_t \subseteq \operatorname{supp} x$ inside one part of $\mathcal{Q}$ (Step 3): inside a V_k is impossible, since then $Z_t \subseteq \Delta_p$ for a $C_p \subseteq V_k$, so $p = i$ and $C_i \subseteq \operatorname{supp} v$; and inside an $\mathsf{N}(c)$-block $\tilde{N}$ is impossible since then $\tilde{N} \subseteq \Delta_i$, contradicting Step 4. So all blocks of a in D are blocks of c, D is contained in Δ_i, and D contains no A or B other than A_i and B_i. Moreover $B_i \in D$: a chain from $C_i = A_i$ to B_i in Δ_i has interior ε_c-blocks which are $\mathsf{N}(a)$-blocks (Step 4), so it is a chain in $\varepsilon_a \vee \varepsilon_b$. Thus the classes of $\varepsilon_a \vee \varepsilon_b$ containing A_i or B_i with $i \notin I$ contain exactly one $\mathsf{T}(a)$-block and one $\mathsf{T}(b)$-block. In particular every class containing a block A_i or B_i with $i \in I$ contains no A_j, B_j with $j \notin I$.

Step 6: consecutive blocks of a inside $\operatorname{supp} x$. Let $Y_1, Z_1, \ldots, Z_r, Y_2$ be a chain in $\varepsilon_a \vee \varepsilon_b$ whose end blocks Y_1, Y_2 are blocks of a inside $\operatorname{supp} x$ and whose interior blocks of a lie in Y. The interior blocks of a are $\mathsf{N}(x)$-blocks and $\mathsf{N}(c)$-blocks (Step 1, Step 5: no A_j with $j \notin I$ occurs), so all Z's lie in one class K of $\varepsilon_x \vee \varepsilon_b$ and in one class of $\varepsilon_b \vee \varepsilon_c$; K meets $\operatorname{supp} x$ (as Z_1 meets Y_1), and $Z_1 \cap Y_1$, $Z_r \cap Y_2$ lie in the part $K \cap \operatorname{supp} x$ of $\mathcal{Q}$. If this part is V_k, then $Z_1 \subseteq \Delta_p$, $Z_r \subseteq \Delta_{p'}$ with $Z_1 \cap V_k \subseteq C_p$, $Z_r \cap V_k \subseteq C_{p'}$ (Step 4), $p = p'$ because Z_1, Z_r lie in one class of $\varepsilon_b \vee \varepsilon_c$, and Y_1, Y_2 both meet C_p. If the part is an $\mathsf{N}(c)$-block $\tilde{N}$, then Y_1, Y_2 both meet $\tilde{N}$. In both cases Y_1 *and* Y_2 *lie in a common class of* $\varepsilon_a \vee \varepsilon_c$.

Step 7: no class of $\varepsilon_a \vee \varepsilon_b$ contains two $\mathsf{T}(a)$-blocks or two $\mathsf{T}(b)$-blocks. Let A_i, A_j lie in a class D, $i \neq j$. By Step 5, $i, j \in I$, and D contains no A_p with $p \notin I$. Take a chain from A_i to A_j in D and list its blocks of a lying inside $\operatorname{supp} x$: the first is A_i, the last is A_j, and by Step 6 consecutive ones lie in a common class of $\varepsilon_a \vee \varepsilon_c$. Hence $\Gamma_i = \Gamma_j$, so $i = j$. Now let B_i, B_j lie in a class D, $i \neq j$; again $i, j \in I$ by Step 5. Note that B_i lies in the class $K_{k(i)}$ of $\varepsilon_x \vee \varepsilon_b$: a chain from C_i to B_i in Δ_i has interior ε_c-blocks which are $\mathsf{N}(x)$-blocks (Step 4), and its first ε_b-block meets $C_i \subseteq V_{k(i)}$. Take a chain from B_i to B_j in D. If it contains no block of a inside $\operatorname{supp} x$, its blocks of a are blocks of c and $\Delta_i = \Delta_j$. Otherwise let Y_1 be the first block of a inside $\operatorname{supp} x$ on

the chain; the blocks between B_i and Y_1 are ε_b-blocks and $\mathsf{N}(x)$-blocks, so they lie in $K_{k(i)}$ and in Δ_i; the last of them, Z, meets Y_1 inside $K_{k(i)} \cap \operatorname{supp} x = V_{k(i)}$, and $Z \cap V_{k(i)} \subseteq C_i$ (Step 4, as $Z \subseteq \Delta_i$); so Y_1 meets C_i and lies in Γ_i. Symmetrically the last block of a inside $\operatorname{supp} x$ lies in Γ_j, and by Step 6 all of them lie in one class of $\varepsilon_a \vee \varepsilon_c$; so $\Gamma_i = \Gamma_j$ and $i = j$.

Step 8: $xb = xcb$. For a class K of $\varepsilon_x \vee \varepsilon_b$ let $B(K) = \{p : B_p \subseteq K\}$, and let $C(K)$ be the set of p with $C_p \subseteq V_k = K \cap \operatorname{supp} x$ or $C_p \subseteq M \subseteq K$ for an $\mathsf{N}(x)$-block M. We claim $B(K) = C(K)$, that no class K meeting $\operatorname{supp} x$ contains an $\mathsf{N}(x)$-block containing a C, and that a class not meeting $\operatorname{supp} x$ contains at most one such $\mathsf{N}(x)$-block. Take a shortest chain from B_p to C_p in Δ_p; its interior ε_c-blocks are $\mathsf{N}(x)$-blocks (Step 4), so all its blocks except C_p lie in the class K of B_p, and the last ε_b-block Z meets C_p. If $C_p \subseteq V_k$, the $\mathsf{T}(x)$-block containing a point of $Z \cap C_p$ lies in K, so $K = K_k$ and $p \in C(K)$; if $C_p \subseteq M$ then $M \subseteq K$ and again $p \in C(K)$. Reading the chain backwards gives the converse, so $B(K) = C(K)$. Next let $M \subseteq K$ be an $\mathsf{N}(x)$-block containing C's (a union of C's and of A's), let W be any block of x in K other than M, and take a shortest chain $M, Z_1, X_1, Z_2, \ldots, Z_t, W$ in K. Let $C_p \subseteq M$ be a $\mathsf{T}(c)$-block met by Z_1; then $Z_1 \subseteq \Delta_p$. We show inductively that all Z_j lie in Δ_p and all X_j are $\mathsf{N}(x)$-blocks which are $\mathsf{N}(c)$-blocks. If X_j is a $\mathsf{T}(x)$-block H, then Z_j meets H, which lies in a V_k (then Z_j meets a $C_q \subseteq V_k$, $q = p$, contradicting $C_p \subseteq M \subseteq Y$) or in an $\mathsf{N}(c)$-block $\tilde{N}$ (then $\tilde{N} \subseteq \Delta_p$, contradicting Step 4). If X_j is an $\mathsf{N}(x)$-block containing C's, it meets $Z_j \subseteq \Delta_p$ in a point of some C_q, so $q = p$ and $X_j \supseteq C_p$, i.e. $X_j = M$, contradicting minimality. So X_j is an $\mathsf{N}(c)$-block, $X_j \subseteq \Delta_p$, and $Z_{j+1} \subseteq \Delta_p$. The same alternatives apply to W: W is not a $\mathsf{T}(x)$-block, so K does not meet $\operatorname{supp} x$; and if W contains C's then $W = M$. This proves the claim. Now compare the elements xb and xcb of $\mathcal{P}_X$. The upper blocks of xb are the $\mathsf{N}(x)$-blocks and the sets U_K, transversal iff K meets $\operatorname{supp} b$; the lower blocks are the $\mathsf{N}(b)$-blocks and, for each class K, $\bigcup_{p \in B(K)} B'_p$, attached to U_K if K meets $\operatorname{supp} x$. For $xcb = x(cb)$ the middle join is $\varepsilon_x \vee \varepsilon_c = \varepsilon_v$: the upper blocks are the $\mathsf{N}(x)$-blocks, the V_k (transversal) and the $\mathsf{N}(c)$-blocks inside $\operatorname{supp} x$; the lower blocks are the $\mathsf{N}(b)$-blocks, $\bigcup_{C_p \subseteq V_k} B'_p$ attached to V_k, and $\bigcup_{C_p \subseteq M} B'_p$ for the $\mathsf{N}(x)$-blocks M containing C's. By Step 3 and the claim these coincide: $B(K_k) = C(K_k) = \{p : C_p \subseteq V_k\}$, and the classes K not meeting $\operatorname{supp} x$ with $B(K) \neq \emptyset$ correspond bijectively to the $\mathsf{N}(x)$-blocks M containing C's, with $B(K) = \{p : C_p \subseteq M\}$. So $xb = xcb$, and multiplying by a on the left, $ub = axb = axcb = ucb$.

Step 9: every $\mathsf{T}(a)$-block shares its class with a $\mathsf{T}(b)$-block. By Step 7 the upper blocks of the element ab are the $\mathsf{N}(a)$-blocks and the single blocks A_p, and A_p is transversal in ab iff its class $D(A_p)$ in $\varepsilon_a \vee \varepsilon_b$ contains a $\mathsf{T}(b)$-block, in which case the lower part of that block is the union of the B's in $D(A_p)$. Multiply by x on the left: the middle join is $\varepsilon_x \vee \varepsilon_a$, so the transversal block of $x(ab)$ at U_l has lower part the union of the $\mathsf{T}(b)$-blocks lying in the classes $D(A_p)$, $p \in I_l = \{p : A_p \subseteq U_l\}$; by Step 7 each $D(A_p)$ contains at most one $\mathsf{T}(b)$-block, and distinct p give distinct classes, so this is a union of at most $|I_l|$ distinct $\mathsf{T}(b)$-blocks, with equality iff every $D(A_p)$, $p \in I_l$, contains one. On the other hand $x(ab) = (xa)b = ub = ucb$ by Step 8, and in $u(cb)$ the middle join is $\varepsilon_u \vee \varepsilon_c = \varepsilon_x \vee \varepsilon_a \vee \varepsilon_c$, whose class containing U_l contains the C's of $V_{k(l)}$ and no other C (Step 2(b); an $\mathsf{N}(x)$-block which is a union of A's and C's is a class by itself); so the transversal block of ucb at U_l has lower part $\bigcup_{p \in I_l} B_p$, a union of $|I_l|$ distinct $\mathsf{T}(b)$-blocks. Comparing the two lower parts, which are unions of pairwise disjoint nonempty sets, $D(A_p)$ contains a $\mathsf{T}(b)$-block for every $p \in I_l$ and every l, i.e. for every $p \in I$; for $p \notin I$ this is Step 5.

Conclusion. The s blocks $A_1, \ldots, A_s$ lie in s distinct classes of $\varepsilon_a \vee \varepsilon_b$ (Step 7), each of which contains at least one of the s blocks $B_1, \ldots, B_s$ (Step 9); so each contains exactly one, and no other class meets $\operatorname{supp} a \cup \operatorname{supp} b$. By Lemma 1.1(c) (and its proof, which shows that this property characterises matched pairs), $a \bowtie b$. □

Remark 5.17. The proof uses the hypothesis $\theta_x(b) = \theta_x(c)$ only through Step 3, i.e. through the partition $\mathcal{Q}$ of $\operatorname{supp} x$ common to $\varepsilon_x \vee \varepsilon_b$ and $\varepsilon_x \vee \varepsilon_c$; the commutation of x with a and with

c enters through the four types of Lemma 1.1(b), which make every block of a and of c lie inside $\operatorname{supp} x$ or outside it and make a and c coincide outside $\operatorname{supp} x$ (Step 1). The mechanism is thus different from that of (S): there the commuting pair c, d shares its nontransversal blocks and the matchings transport the transversal blocks across them; here the single projection x splits X into a part where a and c agree and a part where b and c induce the same coarse structure, and the two parts communicate only through the $\mathsf{T}(c)$-blocks (Step 4). The identity $xb = xcb$ of Step 8 holds in $\mathfrak{P}_7$ as well (for the elements of $\mathfrak{T}_{19}$), so it is not by itself an obstruction; what $\mathfrak{T}_{19}$ violates is Step 9, i.e. the way ab and ucb are compared through their transversal blocks.

Like (Q1), the triangle condition and (S), the condition (R) defines a proper quasivariety, i.e. the class it defines is not closed under $*$-homomorphic images: $\mathfrak{T}_{19}$ is a $*$-homomorphic image of a $*$-subsemigroup of $\mathcal{P}_5$. Indeed, let $\mathfrak{T}_{26}$ be defined like $\mathfrak{T}_{19}$ but with the bottom band $\{u, v, v', w\} \times \{u, v, v', w\}$ and $\lambda b = \lambda v' = \lambda w = v'$, $\lambda c = \lambda v = v$, $\lambda a = \lambda u = u$ (so $xbx = v'$ is no longer equal to $xcx = v$). Then $\mathfrak{T}_{26}$ is a regular $*$-semigroup, identifying v' with v is a $*$-homomorphism onto $\mathfrak{T}_{19}$, and $\mathfrak{T}_{26}$ $*$-embeds in $\mathcal{P}_5$ (its 8-element projection algebra embeds in $\mathcal{P}(\mathcal{P}_5)$ by the SAT search, and since $\mathfrak{T}_{26}$ is generated by its projections the embedding extends; `fourth_pre.py`). The same example shows that the class of regular $*$-semigroups embeddable in partition monoids is not closed under $*$-homomorphic images with a 19-element image, the previous witnesses being the 32-element $Q(a - c - b - d, \perp)$ with $\perp$ injective and its image $\mathfrak{Q}$.

5.7. **A uniform view: path-closing conditions.** Both (S) and (R) have the same shape: a *matched path* $p_0 \bowtie p_1 \bowtie \ldots \bowtie p_n$, an auxiliary projection x tied to some vertices of the path by commutation or order, two vertices which are "indistinguishable through x", and the conclusion that the path *closes*, $p_0 \bowtie p_n$:

$$\text{(R)}: \quad p_0 \bowtie p_1 \bowtie p_2,\ xp_0 = p_0x,\ xp_1 = p_1x,\ \theta_x(p_1) = \theta_x(p_2) \implies p_0 \bowtie p_2,$$
$$\text{(S)}: \quad p_0 \bowtie p_1 \bowtie p_2 \bowtie p_3,\ x = p_1p_3 = p_3p_1,\ \theta_{p_0}(x) = \theta_{p_2}(x) \implies p_0 \bowtie p_3.$$

In (R) the auxiliary projection *sees* two consecutive vertices alike; in (S) it *is seen* alike by the two even vertices, and it lies below the two odd ones. To find out whether this scheme contains further obstructions we searched it systematically (`pathclose_np.py`): for a path of length n and the atomic relations $xp_i = p_ix$, $x \le p_i$, $p_i \le x$, $\theta_x(p_i) = \theta_x(p_j)$, $\theta_{p_i}(x) = \theta_{p_j}(x)$, $x = \theta_{p_i}(p_j)$, we listed all sets of at most three atoms H such that "path + $H \Rightarrow p_0 \bowtie p_n$" holds for all instances in $\mathcal{P}(\mathcal{P}_4)$ with distinct vertices and x not a vertex, is not vacuous (some closed instance satisfies H), and is minimal; each survivor was then tested on more than a million random instances in $\mathcal{P}(\mathcal{P}_5)$ (`pathclose5.py`; none failed) and, by Mace4, for countermodels among the projection algebras of order ≤ 9 satisfying (Q1) alone, (Q1)+(S), and (Q1)+(S)+(R) (`pc_mace.sh`). The outcome:

- $n = 2$: ten minimal conditions. Two of them are (R) and its mirror image; the other eight have no countermodel of order ≤ 9 with (Q1), so they are, at least up to that order, consequences of (Q1).
- $n = 3$: 42 minimal conditions. Fourteen are consequences of (Q1) in the same sense; the other 28 fail in $\mathfrak{P}$ (a countermodel of order 6 with (Q1)) and have no countermodel of order ≤ 9 with (Q1)+(S). Among them are (S), its mirror image, the variants in which $x = p_1p_3$ is weakened to $x \le p_1$, $x \le p_3$ or to $x \le p_1$, $xp_3 = p_3x$, and the symmetric form

$$\text{(S}'\text{)}: \quad p_0 \bowtie p_1 \bowtie p_2 \bowtie p_3,\ \theta_{p_0}(x) = \theta_{p_2}(x),\ \theta_{p_1}(x) = \theta_{p_3}(x),\ x \le p_i \text{ or } xp_i = p_ix \text{ for some } i \implies p_0 \bowtie p_3,$$

  in which x is seen alike by the two even and by the two odd vertices. (R) plays no role for $n = 3$: adding it to (Q1) produces no further consequences here.
- $n = 4$: 18 minimal conditions, all without countermodels of order ≤ 9 with (Q1) alone.

- Two auxiliary projections x, y (`pathclose2.py`; the atoms above for each of x, y, together with $xy = yx$, $x \le y$, $y \le x$, $\theta_x(p_i) = \theta_y(p_i)$, $\theta_{p_i}(x) = \theta_{p_i}(y)$, $x = \theta_y(p_i)$, $y = \theta_x(p_i)$, $x = \theta_{p_i}(y)$, $y = \theta_{p_i}(x)$; x, y distinct and not vertices). Length 2: 60 minimal conditions, 40 consequences of (Q1) and 20 of (Q1)+(R) (up to order 9), all surviving the random test in $\mathcal{P}(\mathcal{P}_5)$; among the latter is the cleaner form

$$p_0 \bowtie p_1 \bowtie p_2, \quad xp_0 = p_0x, \quad y \le p_1, \quad \theta_x(p_2) = \theta_y(p_2) \implies p_0 \bowtie p_2$$

  (in (R) one has $y = xp_1$). Length 3: 184 minimal conditions in $\mathcal{P}(\mathcal{P}_4)$, of which 16 fail in $\mathcal{P}(\mathcal{P}_5)$ (all of the form "x is seen alike by p_0, p_2 and by p_1, p_3, and $y = \theta_x(p_i)$"); of the remaining 168, 52 are consequences of (Q1) and 116 of (Q1)+(S) up to order 9, none needing (R). Typical members of the second group are

$$p_0 \bowtie p_1 \bowtie p_2 \bowtie p_3, \quad xp_0 = p_0x, \quad \theta_{p_1}(x) = \theta_{p_3}(x), \quad \theta_{p_0}(y) = \theta_{p_2}(y) \implies p_0 \bowtie p_3,$$

  with two different auxiliary projections seen alike by the odd and by the even vertices. No condition with a countermodel of order ≤ 9 satisfying (Q1), (S) and (R) was found.

So within this scheme (one or two auxiliary projections, at most three atoms, paths of length ≤ 4 resp. ≤ 3) the only obstructions beyond (Q1) are (R), at length 2, and (S), at length 3, each with a cluster of equivalent forms; and, curiously, (R) is of "sees" type and (S) of "seen" type, the "seen" conditions of length 2 and the "sees" conditions of length 3 being consequences of (Q1). The triangle condition is of a different shape (two commuting triples, an equality as conclusion) and needs six variables; whether it, too, is the first member of a family, and whether two auxiliary projections or longer paths produce further path-closing obstructions, is open.

5.8. **The census of order 8: seven more candidate obstructions.** The chunked Mace4 enumeration of Section 5.6 does not scale to order 8 (the first of 4096 chunks alone has millions of labelled models). The following observation replaces it by a census "by extension".

Lemma 5.18. *Let P be a finite projection algebra with $|P| \ge 2$ and let e be a maximal element of $(P, \le)$. Then $P \setminus \{e\}$ is a subalgebra. Consequently, if $\mathcal{K}$ is a class of projection algebras defined by quasi-identities, the members of $\mathcal{K}$ of order $n+1$ are exactly the one-point extensions, by a new maximal element, of the members of $\mathcal{K}$ of order n.*

Proof. If $\theta_q(p) = e$ then $e \le q$, so $q = e$. Subalgebras of members of $\mathcal{K}$ are in $\mathcal{K}$. □

For each algebra B of order n Mace4 is given the table of B as ground equations and finds all values $\theta_e(q)$, $\theta_q(e)$ ($q \in B$) with e maximal, subject to the axioms, (Q1), (S) and (R); the results are reduced by a canonical form (`pa8/extend.py`, a fraction of a second per base algebra). Starting from the one-element algebra, the numbers of projection algebras satisfying (Q1), (S) and (R) are

$$1,\ 2,\ 5,\ 16,\ 62,\ 295,\ 1\,662,\ 10\,828,\ 80\,039,\ 658\,727 \qquad (\text{orders } 1, \dots, 10),$$

in agreement with the direct enumerations of orders 6 and 7 ($1\,662 = 1\,663 - 1$, the missing algebra being $\mathfrak{P}_7$, which violates (R)); the step from order 8 to order 9 takes about 40 minutes on two cores, the step from order 9 to order 10 (1 849 833 labelled extensions) 8.2 hours on one core; the analysis of the 658 727 algebras of order 10 has not been done. Every algebra of order 8 in the list satisfies the triangle condition, and 10 821 of them embed in $\mathcal{P}(\mathcal{P}_m)$ with $m \le 7$ (`pa8/post8.py`: backtracking for $m \le 4$, SAT for $5 \le m \le 7$). The remaining seven do not embed in $\mathcal{P}(\mathcal{P}_m)$ for $m \le 8$ (SAT; the case $m = 8$ takes from 25 seconds for $\mathfrak{P}_8^2$ to 2.6 hours for $\mathfrak{P}_8^7$; `pa8/sat8.py`). Since every algebra of order 7 satisfying (Q1), (S) and (R) embeds, all

proper subalgebras of the seven embed; so each of them, if it embeds in no $\mathcal{P}(\mathcal{P}_X)$ at all, is a minimal non-embeddable algebra. Rows p, columns q, entries $\theta_p(q)$:

$\mathfrak{P}_8^1$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	6	6
2	0	0	2	0	0	2	0	2
3	0	0	0	3	0	3	0	3
4	4	4	4	4	4	4	4	4
5	4	4	5	5	4	5	4	5
6	6	6	6	6	6	6	6	6
7	6	6	7	7	6	7	6	7

$\mathfrak{P}_8^2$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	0	6	6	6	6
2	0	0	2	0	0	0	0	2
3	0	0	0	3	0	3	0	3
4	4	4	4	4	4	4	4	4
5	4	4	4	5	4	5	4	4
6	6	6	6	6	6	6	6	6
7	6	6	7	7	6	6	6	7

$\mathfrak{P}_8^3$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	1	0	1
2	0	0	2	2	0	0	0	2
3	0	0	2	3	4	4	0	2
4	4	4	4	4	4	4	4	4
5	4	5	4	4	4	5	4	4
6	6	6	6	6	6	6	6	6
7	6	7	7	7	6	6	6	7

$\mathfrak{P}_8^4$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	6	6	6	6	6
2	0	0	2	0	0	2	0	0
3	3	3	3	3	3	3	3	3
4	3	3	3	3	4	3	3	4
5	3	3	5	3	3	5	3	5
6	6	6	6	6	6	6	6	6
7	6	6	6	6	7	7	6	7

$\mathfrak{P}_8^5$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0
2	0	0	2	6	7	7	6	7
3	3	3	3	3	3	3	3	3
4	3	3	4	3	4	3	3	4
5	3	5	5	3	3	5	3	5
6	6	6	6	6	6	6	6	6
7	6	6	7	6	7	7	6	7

$\mathfrak{P}_8^6$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	1
2	0	0	2	0	2	2	0	2
3	3	3	3	3	3	3	3	3
4	3	3	4	3	4	4	3	3
5	3	3	4	3	4	5	6	6
6	6	6	6	6	6	6	6	6
7	6	7	7	6	6	6	6	7

$\mathfrak{P}_8^7$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	3	3	0	3	0
2	0	0	2	0	2	5	5	5
3	3	3	3	3	3	3	3	3
4	3	3	4	3	4	6	6	3
5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7

Six of the seven are one-point extensions of projection algebras of graph semigroups. Deleting the element x indicated below leaves the projection algebra of $Q(G, \perp)$ with $|V| = 4$ and three colours (`pa8/cand8b.py`):

- $\mathfrak{P}_8^1$, $x = 1$: $V = \{2, 3, 5, 7\}$, $E = \{25, 27, 35, 37, 57\}$ (K_4 minus the edge 23), $\perp$: $2, 3 \mapsto 0$, $5 \mapsto 4$, $7 \mapsto 6$; x lies above 0 and 6, commutes with $2, 3$ (product 0) and with 7 (product 6), and $\theta_x(5) = 0$, $\theta_5(x) = 4$.
- $\mathfrak{P}_8^2$, $x = 1$: same V and $\perp$, $E = \{27, 35, 37\}$ (a path $2 - 7 - 3 - 5$); x as in $\mathfrak{P}_8^1$ except $\theta_x(5) = 6$ (and $\theta_x(4) = 6$).

- $\mathfrak{P}_8^3$, $x=3$: $V=\{1,2,5,7\}$, $E=\{15,17,27\}$ (a path $5-1-7-2$), $\bot\colon 1,2\mapsto 0$, $5\mapsto 4$, $7\mapsto 6$; x lies above 0, 4 and the vertex 2, commutes with 1 (product 0) and with 5 (product 4), and $\theta_x(7)=2$, $\theta_7(x)=7$.
- $\mathfrak{P}_8^4$, $x=1$: $V=\{2,4,5,7\}$, $E=\{25,47,57\}$ (a path $2-5-7-4$), $\bot\colon 2\mapsto 0$, $4,5\mapsto 3$, $7\mapsto 6$; x lies above 0 and 6, commutes with 2 (product 0) and with 7 (product 6), and $\theta_x(4)=\theta_x(5)=6$, $\theta_4(x)=\theta_5(x)=3$.
- $\mathfrak{P}_8^5$, $x=2$: $V=\{1,4,5,7\}$, $E=\{15,47,57\}$ (a path $1-5-7-4$), $\bot\colon 1\mapsto 0$, $4,5\mapsto 3$, $7\mapsto 6$; x lies above 0, 6 and the vertex 7, commutes with 1 (product 0), and $\theta_x(4)=\theta_x(5)=7$, $\theta_4(x)=4$, $\theta_5(x)=5$.
- $\mathfrak{P}_8^6$, $x=5$: $V=\{1,2,4,7\}$, $E=\{17,24,27\}$ (a path $1-7-2-4$), $\bot\colon 1,2\mapsto 0$, $4\mapsto 3$, $7\mapsto 6$; x lies above 3, 6 and the vertex 4, commutes with 7 (product 6), and $\theta_x(1)=3$, $\theta_1(x)=0$, $\theta_x(2)=4$, $\theta_2(x)=2$.

The seventh, $\mathfrak{P}_8^7$, is different: its bottom band has five elements $0,3,5,6,7$, and above it lie $1\geq 0,3$, $2\geq 0,5$, $4\geq 3,6$ with $2\bowtie 4$, 1 commuting with 2 (product 0) and with 4 (product 3); the element 7 lies below no other element, and $\theta_1(7)=0$, $\theta_2(7)=5$, $\theta_4(7)=3$.

Each of the seven violates a quasi-identity valid in $\mathcal{P}(\mathcal{P}_6)$, found by the methods of Section 5.7 (`pa8/pc3_hit.py`: for an instance in $\mathfrak{P}_8^k$ of a matched path with auxiliaries, the minimal subsets of the atoms true there that are contained in no counterexample signature of $\mathcal{P}(\mathcal{P}_4)$, then a SAT check of validity without distinctness assumptions, `pa8/pc3_verify.py`) or by greedy minimisation of the diagram of $\mathfrak{P}_8^k$ under a SAT validity test (`pa8/qi_extract.py`). Since $\mathfrak{P}_8^k$ satisfies (Q1), the triangle condition, (S) and (R), each of these quasi-identities is independent of those four.

Caveat. Apart from $\mathfrak{P}_8^7$, $\mathfrak{P}_8^3$, $\mathfrak{P}_8^2$, $\mathfrak{P}_8^4$ and $\mathfrak{P}_8^5$, whose conditions (V), (R′), (P_1), (P_2) and (Q) are proved in Theorems 5.20, 5.23, 5.25 and 5.29 below, none of the seven has been *proved* non-embeddable, and none of the other quasi-identities below has been proved for all $\mathcal{P}_X$: the evidence is the absence of embeddings into $\mathcal{P}(\mathcal{P}_m)$ for $m\leq 8$ and the validity of the quasi-identities in $\mathcal{P}(\mathcal{P}_m)$ for $m\leq 7$. That a finite algebra embeds in some $\mathcal{P}(\mathcal{P}_X)$ but in no $\mathcal{P}(\mathcal{P}_m)$ with m small does happen: a semilattice with k join-irreducible elements embeds in $\mathcal{P}(\mathcal{P}_k)$ (partial identities) and, e.g. for the 9-element semilattice consisting of chains of lengths 1, 4 and 3 glued at the bottom, in no $\mathcal{P}(\mathcal{P}_m)$ with $m\leq 7$ (SAT); so in the census of order 9 an SAT search with $m\leq 7$ is not conclusive, and the SAT results below for the two algebras $\mathfrak{P}_8^1,\mathfrak{P}_8^6$ are evidence, not proof.

Proposition 5.19. *The following quasi-identities hold in $\mathcal{P}(\mathcal{P}_m)$ for $m\leq 7$ (SAT, variables not assumed distinct; `pa8/verify6.py`, the case $m=6$ taking between 20 seconds and 12 minutes each; `pa8/verify7b.py` for $m=7$ on M.K.'s machine: (Q), and hence (P_2), 11.3 hours, (E_6) 21 minutes, (E_1) 3 minutes; (V), (R′), (P_1), (Q) and hence (P_2) are proved below, in Theorems 5.20, 5.23, 5.25, 5.29).*

(a) (R′)*: $a\bowtie c$, $b\bowtie c$, $xc=cx$, $\theta_x(a)=\theta_x(c)$, $ya=ay$, $\theta_x(b)=\theta_y(b)\implies a\bowtie b$. It fails in $\mathfrak{P}_8^3$ ($(a,c,b,x,y)=(5,1,7,2,3)$) and in $\mathfrak{P}_7$ ($(a,c,b,x,y)=(a,c,b,v,x)$). (R′) holds in every $\mathcal{P}_X$ (Theorem 5.23). In every projection algebra (R′) implies (R): given the hypotheses of (R), put $x'=xc=cx$ and $y'=x$. Then $x'\leq c$, so $x'c=cx'$; $\theta_{x'}(a)=cx\,a\,xc=c(xax)c=c(xa)c=x(cac)=xc=\theta_{x'}(c)$, using $xa=ax$, $xc=cx$ and $cac=c$; $y'=x$ commutes with a; and $\theta_{x'}(b)=c\,\theta_x(b)\,c=c\,\theta_x(c)\,c=cxc=xc=\theta_x(c)=\theta_x(b)=\theta_{y'}(b)$. So (R′) gives $a\bowtie b$. (This substitution was found by Prover9.) Each of the four hypotheses on x,y is needed, and with $y=x$ the hypotheses force $a=c$ (since then $xa=xc\leq a,c$ and matched projections with a common lower bound are equal by (Q1)).*

(b) (V): $bcb = b,\ ata \le b,\ ctc \le a \implies ata \le btb$. *It fails in* $\mathfrak{P}_8^7$ $((a,b,c,t) = (1,2,4,7)$: $ata = 0,\ btb = 5,\ ctc = 3)$. *(V) holds in every* $\mathcal{P}_X$, *with* $ata = btb$ *(Theorem 5.20). In* $\mathcal{P}(\mathcal{P}_m)$, $m \le 6$, *the hypotheses of (V) also force* $tat = tbt$ *(SAT), and each of the three hypotheses is needed for* $tat = tbt$ *already in* $\mathcal{P}(\mathcal{P}_4)$; *but* $tat = tbt$ *together with* $ata \le b$ *does not give* $ata \le btb$, *so this is not a decomposition of (V) into two steps.*

(c) (P_1): $p_0 \bowtie p_1 \bowtie p_2 \bowtie p_3,\ xp_0 = p_0x,\ xp_1 = p_1x,\ \theta_{p_0}(x) = \theta_{p_2}(x),\ \theta_x(p_1) = \theta_x(p_3) \implies p_0 \bowtie p_3$, *and* ($P_2$): $p_0 \bowtie p_1 \bowtie p_2 \bowtie p_3,\ xp_0 = p_0x,\ xp_2 = p_2x,\ \theta_x(p_1) = \theta_x(p_2),\ \theta_{p_1}(x) = \theta_{p_3}(x) \implies p_0 \bowtie p_3$. *($P_1$) fails in* $\mathfrak{P}_8^2$ $((p_0,\dots,p_3;x) = (2,7,3,5;1))$ *and (P_2) in* $\mathfrak{P}_8^4$ $((2,5,7,4;1))$; *(P_1) holds in every* $\mathcal{P}_X$ *(Theorem 5.25). More precisely (`pa8/pc1_classify.py`): with one auxiliary projection, a path of length 3, the atoms of Section 5.7 and the chords* $p_0 \bowtie p_2$, $p_1 \bowtie p_3$ *as further atoms, there are 76 minimal valid hypothesis sets of size* ≤ 4 *and 8 of size 5, all valid in* $\mathcal{P}(\mathcal{P}_5)$; *exactly 8 of size 4 and 2 of size 5 have a countermodel among the 12 870 algebras of order* $2 \le n \le 8$ *satisfying* (Q1), *(S) and (R), always a single one,* $\mathfrak{P}_8^2$ *or* $\mathfrak{P}_8^4$. *Up to reversal of the path the eight of size 4 are (P_1), (P_2), and*

$$xp_0 = p_0x,\ xp_2 = p_2x,\ \theta_{p_1}(x) = \theta_{p_3}(x),\ \theta_x(p_2) = \theta_x(p_3) \quad (\mathfrak{P}_8^4),$$
$$xp_1 = p_1x,\ xp_2 = p_2x,\ \theta_x(p_0) = \theta_x(p_2),\ \theta_{p_1}(x) = \theta_{p_3}(x) \quad (\mathfrak{P}_8^2);$$

the two of size 5 are $xp_0 = p_0x,\ xp_1 = p_1x,\ xp_2 = p_2x,\ \theta_x(p_0) = \theta_x(p_2),\ \theta_x(p_1) = \theta_x(p_3)$ *and its reversal* ($\mathfrak{P}_8^2$). *All of them combine two commutation atoms, one "seen alike" atom for two vertices of the same parity and one "sees alike" atom.*

(d) (Q): $p_0 \bowtie p_1 \bowtie p_2 \bowtie p_3,\ xp_0 = p_0x,\ \theta_{p_1}(x) = \theta_{p_3}(x),\ yp_0 = p_0y,\ yp_2 = p_2y,\ \theta_y(p_1) = \theta_y(p_2) \implies p_0 \bowtie p_3$, *which fails in* $\mathfrak{P}_8^5$ $((p_0,\dots,p_3;\,x,y) = (1,5,7,4;\,0,2))$ *and also in* $\mathfrak{P}_8^4$ $((2,5,7,4;\,0,1))$. *No condition of the scheme of Section 5.7 with two auxiliaries, a path of length* ≤ 3, *chords and at most 5 atoms fails in* $\mathfrak{P}_8^1$ *or* $\mathfrak{P}_8^6$.

(e) *For* $\mathfrak{P}_8^6$: $dbd = d,\ aca = a,\ ada = a,\ u \le c,\ u \le d,\ bcb = aua \implies aua = a$ *(fails with* $(a,b,c,d,u) = (2,1,5,7,6)$; *the conclusion was found in the form* $aua \bowtie a$, *which is equivalent because* $\theta_a(aua) = aua$*). For* $\mathfrak{P}_8^1$: $z \le b,\ aba = z,\ dbd = d,\ xcx = z,\ aca = a,\ bcb = b,\ e \le x,\ e \le d,\ ada = a \implies aba = a$ *(fails with* $(a,b,c,d,x,e,z) = (2,3,5,7,1,6,0)$; *here e only serves, via (Q1), to make x and d commute). Both are greedy-minimal subsets of the diagram of the algebra, and need not be the most natural formulations.*

Theorem 5.20. *(V) holds in every partition monoid* $\mathcal{P}_X$: *if* b,c,a,t *are projections with* $bcb = b$, $ata \le b$ *and* $ctc \le a$, *then* $ata = btb$; *in particular* $ata \le btb$. *Consequently* $\mathfrak{P}_8^7$ *embeds in no* $\mathcal{P}(\mathcal{P}_X)$, *and it is a minimal non-embeddable projection algebra.*

Proof. Write $z = ata$, $w = ctc$, $s = btb$. We first reduce the claim, by computations valid in every projection algebra satisfying (Q1), to a single statement about $\mathcal{P}_X$.

Step 1: three identities. (i) *If* $k \le a$ *and* $ab = ba$ *then* $bkb \le a$: indeed $a(bkb)a = (ab)k(ba) = (ba)k(ab) = b(aka)b = bkb$. (ii) *If* $ata \le b$ *then* $ab = ba$ *and* $ata = ete$, *where* $e = ab$: $z = ata$ lies below a and below b, so a and b commute by (Q1); $e = ab$ is a projection with $e \le a$, $e \le b$, and $eze = a(bzb)a = aza = z$, so $z \le e$ and $z = eze = e(ata)e = (ea)t(ae) = ete$ because $ea = ae = e$. (iii) *If* $bcb = b$ *then* $s = b(csc)b$: $s = btb = (bcb)t(bcb) = b\,c(btb)c\,b$.

Step 2: reduction. Suppose we know that

$$csc = w, \qquad \text{i.e.} \qquad c(btb)c = ctc. \tag{4}$$

Then $s = b(csc)b = bwb$ by (iii), and $bwb \le a$ by (i), since $w \le a$ and $ab = ba$ (by (ii), as $z \le b$). So $s \le a$ and $s \le b$, hence $s \le e = ab$ and $s = ese = e(btb)e = (eb)t(be) = ete$; and $z = ete$ by

(ii). Thus $s = z$. It remains to prove (4) in $\mathcal{P}_X$; in fact we show that $\theta_c(s)$ and $\theta_c(t)$ have the same blocks.

Step 3: the setting in $\mathcal{P}_X$. We use Lemma 1.1(a) and the notation of the proof of Theorem 5.16: a chain in a join $\varepsilon_p \vee \varepsilon_q$ is a sequence of blocks, alternately of p and of q, consecutive ones meeting; two blocks lie in a common class iff they are joined by a chain. Recall also that for projections p, q:

- (α) $q \leq p$ iff every block of p lies in a block of q and every $\mathsf{N}(p)$-block is a block of q; then every block of q is an $\mathsf{N}(p)$-block or a union of $\mathsf{T}(p)$-blocks, and the $\mathsf{N}(p)$-blocks are $\mathsf{N}(q)$-blocks. (If $q = pqp$, this is Lemma 1.1(a); conversely, if the two conditions hold then $\varepsilon_p \vee \varepsilon_q = \varepsilon_q$ and every ε_q-class meeting $\operatorname{supp} p$ is a union of $\mathsf{T}(p)$-blocks, so $pqp = q$.)
- (β) $pqp = p$ iff every class of $\varepsilon_p \vee \varepsilon_q$ meeting $\operatorname{supp} p$ contains exactly one $\mathsf{T}(p)$-block and meets $\operatorname{supp} q$ (Lemma 1.1(a)). In particular every block of q meets at most one $\mathsf{T}(p)$-block.

Put $J_a = \varepsilon_a \vee \varepsilon_t$, $J_b = \varepsilon_b \vee \varepsilon_t$, $J_c = \varepsilon_c \vee \varepsilon_t$. By Lemma 1.1(a) the blocks of z are the $\mathsf{N}(a)$-blocks and the sets $Z_K = K \cap \operatorname{supp} a$ for the classes K of J_a meeting $\operatorname{supp} a$, Z_K being transversal iff K meets $\operatorname{supp} t$; similarly the blocks of w are the $\mathsf{N}(c)$-blocks and the sets $W_L = L \cap \operatorname{supp} c$ ($L \in J_c$), and those of s are the $\mathsf{N}(b)$-blocks and the sets $S_H = H \cap \operatorname{supp} b$ ($H \in J_b$). From $w \leq a$ and (α): every block of a lies in a block of w; every block of w is an $\mathsf{N}(a)$-block or a union of $\mathsf{T}(a)$-blocks; and every $\mathsf{N}(a)$-block is an $\mathsf{N}(w)$-block, i.e. an $\mathsf{N}(c)$-block or a set W_L with $L \cap \operatorname{supp} t = \emptyset$. From $z \leq b$: every block of b lies in a block of z; every block of z is an $\mathsf{N}(b)$-block or a union of $\mathsf{T}(b)$-blocks; and every $\mathsf{N}(b)$-block is a block of z, non-transversal. From $bcb = b$ and (β): every block of c meets at most one $\mathsf{T}(b)$-block, and for every $\mathsf{T}(b)$-block B the class Γ_B of $\varepsilon_b \vee \varepsilon_c$ containing B meets $\operatorname{supp} c$. Since blocks of a lie in blocks of w, which lie in classes of J_c, we have $J_a \leq J_c$ (every class of J_a lies in a class of J_c); likewise $J_b \leq J_a$. For a class K of J_a let $\lambda(K)$ be the class of J_c containing it. Call a block of c *inner* if it meets $\operatorname{supp} a$ and *outer* otherwise. An inner $\mathsf{T}(c)$-block lies in a set W_L which is a union of $\mathsf{T}(a)$-blocks; an inner $\mathsf{N}(c)$-block is a union of $\mathsf{T}(a)$-blocks (being a block of w that is not an $\mathsf{N}(a)$-block); an outer block of c inside a class L of J_c with $W_L \neq \emptyset$ is either a $\mathsf{T}(c)$-block inside W_L, W_L being then an $\mathsf{N}(a)$-block, or an $\mathsf{N}(c)$-block which is an $\mathsf{N}(a)$-block.

Step 4: two facts. (1) *No $\mathsf{T}(b)$-block lies in a block M which is both an $\mathsf{N}(a)$-block and an $\mathsf{N}(c)$-block.* Otherwise let $B \subseteq M$ be one. The only block of c meeting B is M, and the blocks of b meeting M lie inside M (M is a block of z, and blocks of b lie in blocks of z); so $\Gamma_B = M$ does not meet $\operatorname{supp} c$, contradicting (β). Consequently every block which is both an $\mathsf{N}(a)$-block and an $\mathsf{N}(c)$-block is an $\mathsf{N}(b)$-block (a block of z containing no $\mathsf{T}(b)$-block).

(2) *Let K be a class of J_a and $M \subseteq K$ an $\mathsf{N}(a)$-block containing a $\mathsf{T}(b)$-block. Then M is the only block of z inside K that meets* $\operatorname{supp} b$*; in particular Z_K, if nonempty, is an $\mathsf{N}(b)$-block.* By (1) M is not an $\mathsf{N}(c)$-block, so $M = W_L$ with $L = \lambda(K)$, $L \cap \operatorname{supp} t = \emptyset$; this already shows that no other $\mathsf{N}(a)$-block in K contains a $\mathsf{T}(b)$-block (it would also equal $W_{\lambda(K)}$). Suppose Z_K contains a $\mathsf{T}(b)$-block B. Every $\mathsf{T}(a)$-block in K lies in L but not in $W_L = M$ (which is disjoint from $\operatorname{supp} a$), hence in an $\mathsf{N}(c)$-block, which is then inner, a union of $\mathsf{T}(a)$-blocks, and contained in L. Take a chain $B = B_0, C_1, B_1, C_2, \ldots$ from B to a $\mathsf{T}(c)$-block in Γ_B (it exists by (β)). C_1 meets $B \subseteq Z_K \subseteq \operatorname{supp} a$ in a point of a $\mathsf{T}(a)$-block of K, so C_1 is an inner $\mathsf{N}(c)$-block, $C_1 \subseteq L \cap \operatorname{supp} a$; then B_1 meets $\operatorname{supp} a$ in a point of a $\mathsf{T}(a)$-block $A_1 \subseteq C_1 \subseteq L$, whose class K_1 satisfies $\lambda(K_1) = L$, and $B_1 \subseteq Z_{K_1}$; so C_2 meets a $\mathsf{T}(a)$-block of K_1, which lies in L but not in W_L, and C_2 is again an inner $\mathsf{N}(c)$-block. Inductively no block of the chain is a $\mathsf{T}(c)$-block, a contradiction.

Step 5: the blocks of $\theta_c(s)$ and $\theta_c(t)$ inside $\operatorname{supp} c$. By Lemma 1.1(a) the blocks of $\theta_c(s)$ are the $\mathsf{N}(c)$-blocks and the sets $\Theta \cap \operatorname{supp} c$ for the classes Θ of $\varepsilon_c \vee \varepsilon_s$ meeting $\operatorname{supp} c$, transversal iff Θ meets $\operatorname{supp} s$; the blocks of $\theta_c(t) = w$ are the $\mathsf{N}(c)$-blocks and the W_L, transversal iff L meets

$\operatorname{supp} t$. Every block of s lies in a class of J_c: an $\mathsf{N}(b)$-block is a block of z, hence lies in a class of J_a, and $S_H \subseteq H$, a class of $J_b \leq J_a \leq J_c$. Hence $\varepsilon_c \vee \varepsilon_s \leq J_c$: every class Θ lies in a class L of J_c, $\Theta \cap \operatorname{supp} c \subseteq W_L$, and if Θ meets $\operatorname{supp} s$ then L meets $\operatorname{supp} t$ (a block $S_H \subseteq \Theta$ is transversal only if $H \subseteq L$ meets $\operatorname{supp} t$). So (4) follows once we show, for every class L of J_c with $W_L \neq \emptyset$: *(A) any two $\mathsf{T}(c)$-blocks in W_L lie in a common class of $\varepsilon_c \vee \varepsilon_s$, and (B) if L meets* $\operatorname{supp} t$ *then this class meets* $\operatorname{supp} s$. Fix L; then W_L is a union of $\mathsf{T}(a)$-blocks (*type T*) or an $\mathsf{N}(a)$-block M (*type N*, and then $L \cap \operatorname{supp} t = \emptyset$, so (B) is vacuous).

Type T. Let C, C' be $\mathsf{T}(c)$-blocks in W_L and take a chain $C = C_0, T_1, C_1, \ldots, T_r, C_r = C'$ in J_c (blocks C_i of c, blocks T_i of t, all inside L). We claim that consecutive inner blocks of c on the chain, say C_i and C_j with $C_{i+1}, \ldots, C_{j-1}$ outer, lie in a common class of $\varepsilon_c \vee \varepsilon_s$; since C_0 and C_r are inner ($W_L \subseteq \operatorname{supp} a$), (A) follows. The outer blocks $C_{i+1}, \ldots, C_{j-1}$ are $\mathsf{N}(c)$-blocks which are $\mathsf{N}(a)$-blocks (the $\mathsf{T}(c)$-blocks of L lie in $W_L \subseteq \operatorname{supp} a$), hence $\mathsf{N}(b)$-blocks by (1); the blocks $T_{i+1}, C_{i+1}, \ldots, T_j$ lie in a single class K'' of J_a (each lies in a class, and consecutive ones meet). Let $x \in T_{i+1} \cap C_i$ and $y \in T_j \cap C_j$; both lie in $\operatorname{supp} a$ (C_i, C_j are inner: a $\mathsf{T}(c)$-block inside W_L, or an $\mathsf{N}(c)$-block which is a union of $\mathsf{T}(a)$-blocks), so $x, y \in Z_{K''}$. If $Z_{K''}$ is an $\mathsf{N}(b)$-block, it is a block of s meeting C_i and C_j. Otherwise $Z_{K''}$ is a union of $\mathsf{T}(b)$-blocks; let $B_x \ni x$, $B_y \ni y$ be the $\mathsf{T}(b)$-blocks; then $B_x, T_{i+1}, C_{i+1}, T_{i+2}, \ldots, T_j, B_y$ is a chain in J_b, so $B_x, B_y \subseteq H$ for one class H of J_b, and the block $S_H \supseteq B_x \cup B_y$ of s meets C_i and C_j. This proves the claim. For (B), let L meet $\operatorname{supp} t$, and take a chain $C = C_0, T_1, C_1, \ldots, C_q, T_{q+1}$ in L from C to a $\mathsf{T}(t)$-block T_{q+1}. Let C_j ($j \leq q$) be the last inner block of c on it; then $C_{j+1}, \ldots, C_q$ are $\mathsf{N}(b)$-blocks as above, and $T_{j+1}, C_{j+1}, \ldots, T_{q+1}$ lie in one class K' of J_a, which meets $\operatorname{supp} t$. The point $x \in T_{j+1} \cap C_j$ lies in $Z_{K'}$, a transversal block of z, hence a union of $\mathsf{T}(b)$-blocks (an $\mathsf{N}(b)$-block is a non-transversal block of z); the $\mathsf{T}(b)$-block $B_x \ni x$ is joined to T_{q+1} by the chain $B_x, T_{j+1}, C_{j+1}, \ldots, C_q, T_{q+1}$ in J_b, so its class H meets $\operatorname{supp} t$, S_H is a transversal block of s, and S_H meets C_j, which lies in the class of C by the claim.

Type N. Here $W_L = M$ is an $\mathsf{N}(a)$-block and a block of z. If M is an $\mathsf{N}(b)$-block, it is a block of s containing every $\mathsf{T}(c)$-block of W_L, and (A) holds. Otherwise M is a union of $\mathsf{T}(b)$-blocks $B_1, \ldots, B_k$; each $\mathsf{T}(c)$-block $C \subseteq M$ lies inside one B_i (it is covered by them and meets at most one), so (A) follows if we show that $B_1, \ldots, B_k$ lie in a single class H of J_b (then $S_H \supseteq M$). Let K_0 be the class of J_a containing M. By (2), M is the only block of z inside K_0 meeting $\operatorname{supp} b$; so every $\mathsf{N}(a)$-block in K_0 other than M is an $\mathsf{N}(b)$-block, and Z_{K_0} is an $\mathsf{N}(b)$-block if nonempty. Hence every block of b inside K_0 is one of: a $\mathsf{T}(b)$-block B_i; an $\mathsf{N}(a)$-block $M' \neq M$ in K_0; or Z_{K_0}; and for a point $p \in K_0$ we write $\beta(p)$ for the block of b containing it, which is Z_{K_0} whenever $p \in \operatorname{supp} a$. Let $C \subseteq B_1$ and $C' \subseteq B_2$ be $\mathsf{T}(c)$-blocks and take a chain $C = C_0, T_1, C_1, \ldots, T_r, C_r = C'$ in L. Every block of t meeting M, Z_{K_0} or an $\mathsf{N}(a)$-block in K_0 lies in K_0; in particular $T_1, T_r \subseteq K_0$. Consider a block C_l of the chain with $T_l \subseteq K_0$, and let $u \in T_l \cap C_l$. If C_l is a $\mathsf{T}(c)$-block, then $C_l \subseteq M$ lies inside one B_i, and $\beta(u) = \beta(v) = B_i$ for the point $v \in T_{l+1} \cap C_l$. If C_l is an outer $\mathsf{N}(c)$-block, it is an $\mathsf{N}(a)$-block inside K_0 and an $\mathsf{N}(b)$-block by (1), and $\beta(u) = \beta(v) = C_l$. If C_l is an inner $\mathsf{N}(c)$-block, then $u \in \operatorname{supp} a \cap K_0$ and $\beta(u) = Z_{K_0}$; if also $T_{l+1} \subseteq K_0$ then $\beta(v) = Z_{K_0}$; if not, let $T_{l'}$ ($l' > l+1$) be the next block of t on the chain contained in K_0 (it exists, as $T_r \subseteq K_0$): the block $C_{l'-1}$ meets $T_{l'-1} \not\subseteq K_0$ and $T_{l'} \subseteq K_0$, so it meets two classes of J_a and is therefore an inner $\mathsf{N}(c)$-block, and the point $v' \in T_{l'} \cap C_{l'-1}$ lies in $\operatorname{supp} a \cap K_0$, so $\beta(v') = Z_{K_0} = \beta(u)$. Since consecutive points of the chain on a common block $T_l \subseteq K_0$ have blocks β joined by T_l in J_b, we obtain, walking along the chain and skipping the excursions outside K_0, a chain in J_b from $\beta(x_0) = B_1$ ($x_0 \in T_1 \cap C_0$) to $\beta(y_r) = B_2$ ($y_r \in T_r \cap C_r$). So B_1, B_2 lie in one class of J_b, as required.

Conclusion. By Step 5 the blocks of $\theta_c(s)$ inside $\operatorname{supp} c$ are the sets W_L, transversal iff L meets $\operatorname{supp} t$, and its other blocks are the $\mathsf{N}(c)$-blocks; so $\theta_c(s) = \theta_c(t) = w$, which is (4), and $ata = btb$

by Step 2. The statements about $\mathfrak{P}_8^7$ follow from Proposition 5.19(b) and the minimality noted above. □

Remark 5.21. (1) The hypothesis $bcb = b$ enters the proof of (4) only through the facts (1) and (2) of Step 4; SAT finds no counterexample in $\mathcal{P}(\mathcal{P}_m)$, $m \le 6$, to the quasi-identity $ata \le b,\ ctc \le a \Rightarrow c(btb)c = ctc$ without it. (2) The algebraic part (Steps 1–2) shows more generally: in any projection algebra satisfying (Q1), if $bcb = b$, $ata \le b$, $ctc \le a$ and $c(btb)c = ctc$, then $ata = btb$; and $\theta_b \circ \theta_c$ is the identity on $\{q : q \le b\}$ whenever $bcb = b$ (as $q = bqb = bcb\,q\,bcb = b(cqc)b$), so θ_c separates the elements below b. (3) Prover9 (M.K.) finds that (V) is equivalent, in every projection algebra (without (Q1)), to the three-atom clause $\theta_a(t) \le \theta_d(c),\ \theta_c(t) \le a \Rightarrow \theta_a(t) \le \theta_{\theta_d(c)}(t)$ (in the binary notation: $(A*B)*(C*D) \ne A*B \vee (A*C)*B \ne A*C \vee (A*B)*(A*(C*D)) = A*B$, with $(A,B,C,D) = (t,a,c,d)$). This is (V) with b replaced by $b' = \theta_d(c)$: the fixing hypothesis $bcb = b$ has been absorbed, because $\theta_d(c)\,c\,\theta_d(c) = \theta_d(c)$ is an identity of projection algebras ($c*(c*d) = c*d$ follows from the axioms), and conversely every b with $bcb = b$ is of this form ($b = \theta_b(c)$). So (V) says: the statement $\theta_a(t) \le q,\ \theta_c(t) \le a \Rightarrow \theta_a(t) \le \theta_q(t)$, which for $q = c$ is a triviality under (Q1) ($\theta_a(t)$ is then a common lower bound of a and c, so $ac = ca =: e$ and $\theta_a(t) = \theta_e(t) = \theta_c(t)$), remains true for every restriction $q = \theta_d(c)$ of c; and its conclusion reads $\theta_a(t) \le \theta_b\theta_c\theta_b(t)$, which is the form $s = b(csc)b$ in which it enters Step 1(iii) of the proof. The case $q = c$ itself (the clause with $d = c$) is thus a consequence of (Q1), and $\mathfrak{P}_8^7$ satisfies it while violating (V).

Remark 5.22. (1) The path-closing scheme now contains, besides (R) and (S), the one-auxiliary conditions of size 4 and 5 of (c) and the two-auxiliary conditions (R′) (length 2) and (Q) (length 3); the searches of Section 5.7 stopped at three atoms, which is why these were missed. Whether the conditions of (c) are the complete list of one-auxiliary conditions of length 3 (no bound on the size) is a finite question for $\mathcal{P}(\mathcal{P}_4)$, but its answer must then be checked in larger $\mathcal{P}(\mathcal{P}_m)$, as all conditions above were only verified for $m \le 7$. (2) (V) is of a new shape: four variables, one "fixing" hypothesis $bcb = b$ (half of a matched pair), two order hypotheses on products, and an order conclusion; Theorem 5.20 shows that its content is the statement $\theta_c(\theta_b(t)) = \theta_c(t)$ about restrictions, the rest being equational. (3) *Order* 9. The 80 039 algebras of order 9 satisfying (Q1), (S) and (R) were analysed in the same way (`pa8/post8.py` with the quasi-identities of Proposition 5.19 as a first filter, `pa8/newconds.py`; then `pa8/pass2_9.py`). All satisfy the triangle condition. Exactly 163 contain one of $\mathfrak{P}_8^1, \dots, \mathfrak{P}_8^7$ as a subalgebra, and these are exactly the algebras violating one of (R′), (V), (P_1), (P_2), (Q) or the two conditions of (e), with one exception: one algebra violates (R′) and contains none of the seven (so (R′), which is valid by Theorem 5.23, has at least two minimal witnesses, and this algebra is a minimal non-embeddable algebra of order 9). Of the remaining 79 875, 79 783 embed in $\mathcal{P}(\mathcal{P}_m)$ with $m \le 7$, 39 are semilattices (which embed in $\mathcal{P}(\mathcal{P}_8)$ but, by SAT, in no $\mathcal{P}(\mathcal{P}_7)$: they have 8 join-irreducibles), 46 embed in $\mathcal{P}(\mathcal{P}_8)$ but in no $\mathcal{P}(\mathcal{P}_7)$, and 7 embed in no $\mathcal{P}(\mathcal{P}_m)$ with $m \le 8$ (SAT, 1 to 30 minutes each; `pa8/cand9.json`, `pa8/cand9.out`). These seven are the candidates of order 9: all their proper subalgebras embed. Three of them are one-point extensions of graph-semigroup projection algebras ($|V| = 4$ with four colours, and $|V| = 5$ twice), one has a five-element bottom band like $\mathfrak{P}_8^7$, and none violates a two-auxiliary path-closing condition of length ≤ 3 with at most four atoms. One of them (the second in `cand9.json`) violates the five-atom condition $p_0 \bowtie p_1 \bowtie p_2 \bowtie p_3,\ xp_0 = p_0x,\ \theta_{p_0}(x) = \theta_{p_2}(x),\ yp_1 = p_1y,\ \theta_y(p_0) = \theta_y(p_1),\ \theta_x(p_3) = \theta_y(p_3) \Rightarrow p_0 \bowtie p_3$ (valid in $\mathcal{P}(\mathcal{P}_5)$), a length-3 relative of (R′); for all seven, greedy minimisation of the diagram (`pa8/qi_all9.py`, results in `pa8/qi_all9.jsonl`) gives quasi-identities with 9 to 14 atoms, valid in $\mathcal{P}(\mathcal{P}_5)$, which we have not tried to interpret. So the count of candidates for minimal non-embeddable algebras is $1, 7, 8$ for orders $7, 8, 9$ (counting the algebra violating (R′)), of which $1, 5, 1$ are proved (Theorems 5.20, 5.23, 5.25, 5.29), and the number of points needed

by embeddable algebras of order n has reached $n-1$ at $n=9$. (4) *Absorbing the normalising hypotheses.* As with (V) (Remark 5.21(3)), the hypotheses of (e) which merely say that a variable is a restriction of another can be absorbed by substitution: $z \le b$ by $z = \theta_b(z')$, $dbd = d$ by $d = \theta_{d'}(b)$, $e \le x$ by $e = \theta_x(e')$. Prover9 (M.K.) found in this way a seven-literal clause equivalent to (E_1) in every projection algebra, $\theta_B(A) = \theta_A(C),\ \theta_E(D) = \theta_A(C),\ \theta_B(D) = B,\ \theta_A(D) = A,\ \theta_E(F) \le \theta_G(A),\ \theta_B(\theta_G(A)) = B \Rightarrow \theta_B(A) = B$, which is ($\mathrm{E}_1$) under $(a,b,c,d,x,e,z) = (B, A, D, \theta_G(A), E, \theta_E(F), \theta_A(C))$; the converse direction is the instance $(A,B,C,D,E,F,G) = (b,a,z,c,x,e,d)$, which Prover9 has difficulty finding (instantiating a seven-variable clause by paramodulation), but which is immediate by hand. More useful is that under (Q1) the variable e can be eliminated altogether, since it only serves to make x and d commute: (E_1) is equivalent to

$$(\mathrm{E}_1') \qquad aca = a,\ bcb = b,\ dbd = d,\ ada = a,\ xd = dx,\ xcx = aba \le b \implies aba = a$$

(given (E_1), put $e = xd$, which lies below x and d; given (E_1'), x and d commute by (Q1), and Prover9 verifies this direction), and (E_1') fails in $\mathfrak{P}_8^1$ with $(a,b,c,d,x) = (2,3,5,7,1)$. Similarly u can be specialised to cd in (E_6): the consequence

$$(\mathrm{E}_6') \qquad aca = a,\ ada = a,\ dbd = d,\ cd = dc,\ bcb = a(cd)a \implies a(cd)a = a$$

still fails in $\mathfrak{P}_8^6$ (with $(a,b,c,d) = (2,1,5,7)$, $cd = 6$), so it is an obstruction of five variables and five atoms, though possibly weaker than (E_6). The two share the pattern $aca = a$, $ada = a$, $dbd = d$: a is covered by c and by d (in the sense $\theta_a(c) = a$), and d by b. On the 12 046 instances of (E_6') in $\mathcal{P}(\mathcal{P}_4)$ with a,b,c,d distinct (`pa8/e6/`), the conclusion comes with much more: $bcb = aba = bab = a$, so $a \le b$; $a \bowtie cd$; $\theta_d(a) = cd$; and $\theta_c(a) = \theta_c(b)$. Of these only $\theta_{cd}(a) = cd$ is a consequence of the axioms, (Q1) and the hypotheses (Prover9); $\theta_d(a) = cd$ is not a consequence of $aca = a$, $ada = a$, $cd = dc$ alone (it fails in $\mathcal{P}(\mathcal{P}_4)$ without the hypotheses on b), and adding it to the hypotheses does not make (E_6') equational either. So a proof of (E_6') will need a block argument of a new kind, without an auxiliary projection commuting with two of the others; we have not attempted it. (5) *Towards (Q).* Write the path of (Q) as $a \bowtie c \bowtie b \bowtie d$, so that the hypotheses are $xa = ax$, $\theta_c(x) = \theta_d(x)$, $ya = ay$, $yb = by$, $\theta_y(c) = \theta_y(b)$ and the conclusion is $a \bowtie d$. On all 349 343 instances in $\mathcal{P}(\mathcal{P}_4)$ (`pa8/q/`): $cd = dc$ (by (Q1), as $\theta_c(x) = \theta_d(x)$ is a common lower bound), $xc = xd$, $cx = dx$, $\theta_y(c) = \theta_y(b) = \theta_y(d)$, $y\,ad = y\,abd = y\,acbd$ and $x\,ad = x\,acd = x\,acbd$; but $ad = acbd$ fails in 420 instances with distinct path elements, and no join $\varepsilon_a \vee \varepsilon_d \vee \varepsilon_p \vee \varepsilon_q$ with p,q among $c, b, cd, x, y, ya, yb, \theta_c(x), xa$ separates the $\mathsf{T}(a)$-blocks in all instances. So, unlike (R'), the conclusion of (Q) is not the coherence of the three matchings, and a proof cannot go through a separation statement for a coarser join. Note also that a proof of (Q) contains a proof of (S) (Proposition 5.32(a)), hence of the hub case of Theorem 5.6, which until Section 5.11 rested on a computer check. A search for a common generalisation of (R'), (P_1) and (Q) in the two-auxiliary path-closing language (all 627 392 paths of length 3 in $\mathcal{P}(\mathcal{P}_4)$, 90 atoms, all instantiations of the auxiliaries by restrictions of the given ones; `pa8/gen/`) found none, not even for any two of the three. Section 5.11 proves (Q) (Theorem 5.29).

5.9. **Proof of (R').**

Theorem 5.23. *(R') holds in every partition monoid $\mathcal{P}_X$: if a,b,c,x,y are projections with*

$$a \bowtie c, \quad b \bowtie c, \quad xc = cx, \quad \theta_x(a) = \theta_x(c), \quad ya = ay, \quad \theta_x(b) = \theta_y(b),$$

then $a \bowtie b$. Consequently $\mathfrak{P}_8^3$ embeds in no $\mathcal{P}(\mathcal{P}_X)$ and is a minimal non-embeddable projection algebra, and so is the algebra of order 9 of Remark 5.22(3) that violates (R') but contains none of $\mathfrak{P}_8^1, \dots, \mathfrak{P}_8^7$.

The proof has the same shape as that of Theorem 5.20: an algebraic part, valid in every projection algebra satisfying (Q1), reduces the theorem to a combinatorial statement about blocks, and the combinatorial part is a chain argument in $\mathcal{P}_X$. The algebraic part is longer here, and we found most of it with Prover9 (`pa8/rp/`); the combinatorial part was found by testing candidate statements against all instances of the hypotheses in $\mathcal{P}(\mathcal{P}_4)$ (91 172 quintuples, `pa8/rp/harness*.py`) and, for the statements finally used, against random instances in $\mathcal{P}(\mathcal{P}_5)$.

Proof. Throughout, $r = \theta_x(b) = \theta_y(b)$, $v = xc = cx$ and $u = ya = ay$.

Step 1: algebraic consequences. All identities in this step hold in every regular $*$-semigroup whose projection algebra satisfies (Q1), under the hypotheses of the theorem.

(i) $r \le x$ and $r \le y$ (as $r = \theta_x(b)$ and $r = \theta_y(b)$), so $xy = yx =: m$ by (Q1), and $r \le m$.

(ii) $\theta_x(c) = xcx = xc = v$, so the hypothesis $\theta_x(a) = \theta_x(c)$ reads $xax = v$. Hence $xa = xa\,ax\,xa = (xax)a = va$ and, applying $*$, $ax = av$.

(iii) *$v \le y$ and $cy = yc = v$.* First, $\theta_a(r) = a(yby)a = (ay)b(ya) = ubu$. Next, for every element s, $\theta_x(\theta_u(s)) = x\,ya\,s\,ay\,x = m\,asa\,m = \theta_m(\theta_a(s))$, since $xya = ma$ and $ayx = am$. Applying this to $s = b$ and using $\theta_a(r) = ubu$ together with $\theta_a(b) = aba$: $\theta_m(aba) = \theta_x(ubu) = \theta_x(\theta_a(r)) = xa\,xbx\,ax = (xax)b(xax) = vbv$, and $vbv = v$ because $v \le c$ and $cbc = c$ (for $q \le c$, $qbq = qc\,b\,cq = q(cbc)q = q$). So $v = \theta_m(aba) \le m \le y$. Now v is a common lower bound of c and y, so $cy = yc$ by (Q1); the projection $q = cy$ lies below c, so $qbq = q$ by the identity just used, and $qbq = c(yby)c = crc = c(xbx)c = (cx)b(xc) = vbv = v$. Thus $cy = yc = v$.

(iv) *$\theta_a(x) = \theta_a(y) = u$ and $u \bowtie v$.* By (ii) and (iii), $axa = ava = a\,cy\,a = ac\,ya = aca\,y = ay = u$. Put $g = ac$, so that $gg^* = aca = a$ and $g^*g = cac = c$; then $yg = yac = ayc = acy = gy$, hence $u = ya = y\,gg^* = (yg)(yg)^*$ and $v = yc = y\,g^*g = (yg)^*(yg)$, so $u \bowtie v$ (Section 2: $hh^* \bowtie h^*h$ for every h).

(v) *$yax = ax$, $xya = xa$, $axb = yab$ and $xab = xcb$.* We have $cbv = cb\,cx = (cbc)x = cx = v$ and, by (ii), $ax = av = acx$; hence $axbv = avbv = a(vbv) = av = ax$. Since $r \le y$, $y\,xbx = yr = r = xbx$; therefore $yax = y\,axbv = ay\,xbx\,c = a\,xbx\,c = axbv = ax$ (we used $v = xc$ and $ay = ya$). Applying $*$ gives $xay = xa$, i.e. $xya = xa$. Next $xb = (xbx)b = rb = (yby)b = yb$, so $axb = ayb = yab$. Finally $xcb = (xax)b = x(axb) = x(yab) = (xya)b = xab$.

(vi) *$y\,ab = y\,acb$.* Multiplying $xab = xcb$ on the left by a and using $axa = u$ and $axc = avc = av$ (as $vc = v$): $u\,b = av\,b$; and $y\,acb = (ya)(cb) = a(yc)b = avb$. So $ub = y\,ab = y\,acb = ucb$.

(vii) *Reduction to $x \le y$.* The projection $m = xy$ satisfies the hypotheses in place of x: $mc = xyc = xv = v = vy = cxy = cm$ (as $v \le x$, $v \le y$); $\theta_m(c) = mcm = v$ and $\theta_m(a) = xy\,a\,yx = xux = x(ava)x = (xa)v(ax) = (va)v(av) = v(ava)v = vuv = v$ by (ii), (iv); and $\theta_m(b) = \theta_m(\theta_x(b)) = \theta_m(r) = r = \theta_y(b)$, since $m \le x$ and $r \le m$. As the conclusion does not involve x, we may and do assume from now on that $x \le y$.

Step 2: the setting in $\mathcal{P}_X$. We use Lemma 1.1, the facts (α), (β) recalled in the proof of Theorem 5.20, and chains as in the proof of Theorem 5.16. Write $Y = X \setminus \operatorname{supp} y$. Let $A_1, \dots, A_s$ be the $\mathsf{T}(a)$-blocks, C_i the $\mathsf{T}(c)$-block matched with A_i and B_i the $\mathsf{T}(b)$-block matched with C_i (Lemma 1.1(c)); Γ_i and Δ_i denote the classes of $\varepsilon_a \vee \varepsilon_c$ and of $\varepsilon_b \vee \varepsilon_c$ containing C_i, so that Γ_i contains exactly the T-blocks A_i, C_i and Δ_i exactly B_i, C_i; for a block E of c we write $\Gamma(E)$, $\Delta(E)$ for the classes containing it. We shall show that

$$\text{no class of } \varepsilon_a \vee \varepsilon_b \vee \varepsilon_c \text{ contains two } \mathsf{T}(c)\text{-blocks}, \tag{5}$$

and then conclude by a counting argument. Since A_i, B_i, C_i lie in one class of $\varepsilon_a \vee \varepsilon_b \vee \varepsilon_c$, (5) implies that every class of $\varepsilon_a \vee \varepsilon_b$ contains at most one $\mathsf{T}(a)$-block and at most one $\mathsf{T}(b)$-block, and that a class containing A_i contains no $\mathsf{T}(b)$-block other than B_i.

By Lemma 1.1(b) applied to the commuting pairs y, a and y, c and to x, c: every block of a, of c and of x lies inside $\operatorname{supp} y$ or is an $\mathsf{N}(y)$-block; an $\mathsf{N}(y)$-block is an $\mathsf{N}(a)$-block or a union

of $\mathsf{T}(a)$-blocks, and an $\mathsf{N}(c)$-block or a union of $\mathsf{T}(c)$-blocks; every $\mathsf{N}(c)$-block lies inside $\operatorname{supp} x$ (and is then a union of $\mathsf{T}(x)$-blocks) or is an $\mathsf{N}(x)$-block. Since $x \leq y$, every block of x is an $\mathsf{N}(y)$-block or a union of $\mathsf{T}(y)$-blocks ((α)). The classes of $\varepsilon_y \vee \varepsilon_a$ meeting $\operatorname{supp} y$ are the $\mathsf{T}(u)$-blocks $U_1, \dots, U_t$ (unions of $\mathsf{T}(a)$-blocks and of $\mathsf{T}(y)$-blocks) and the $\mathsf{N}(a)$-blocks inside $\operatorname{supp} y$; similarly the classes of $\varepsilon_y \vee \varepsilon_c$ meeting $\operatorname{supp} y$ are the $\mathsf{T}(v)$-blocks $V_1, \dots, V_{t'}$ and the $\mathsf{N}(c)$-blocks inside $\operatorname{supp} y$.

Step 3: a second reduction. Let M be an $\mathsf{N}(y)$-block which is a union of $\mathsf{T}(a)$-blocks. If M were an $\mathsf{N}(c)$-block, it would be a class of $\varepsilon_a \vee \varepsilon_c$ by itself, containing $\mathsf{T}(a)$-blocks but no $\mathsf{T}(c)$-block, contradicting $aca = a$ ((β)); so M is a union of $\mathsf{T}(c)$-blocks, and the class of $\varepsilon_a \vee \varepsilon_c$ of any $A_i \subseteq M$ lies inside M and consists of T-blocks only, whence $\Gamma_i = A_i = C_i$. Symmetrically, an $\mathsf{N}(y)$-block which is a union of $\mathsf{T}(c)$-blocks is a union of $\mathsf{T}(a)$-blocks. Let x', y' be obtained from x, y by declaring all these blocks M transversal. We claim that a, b, c, x', y' still satisfy the hypotheses (with $x' \leq y'$); as the conclusion does not change, we may then assume that *every $\mathsf{N}(y)$-block is an $\mathsf{N}(a)$-block and an $\mathsf{N}(c)$-block*, so that all A_i and all C_i lie inside $\operatorname{supp} y$. Indeed, the partitions $\varepsilon_{x'} = \varepsilon_x$, $\varepsilon_{y'} = \varepsilon_y$ are unchanged, and each M is a class by itself in $\varepsilon_x \vee \varepsilon_a$, in $\varepsilon_x \vee \varepsilon_c$, in $\varepsilon_y \vee \varepsilon_a$ and in $\varepsilon_y \vee \varepsilon_c$, being a block of x and of y which is a union of blocks of a and of c. Hence $x'c = cx'$ and $y'a = ay'$, $y'c = cy'$ by the converse of Lemma 1.1(b) (if every class of $\varepsilon_p \vee \varepsilon_q$ is of one of the four types, the description of pq shows that pq is symmetric, so $pq = qp$): the class M is now of type (i), the others are unchanged; $\theta_{x'}(a) = \theta_{x'}(c)$ because the blocks of these projections are those of $\theta_x(a) = \theta_x(c)$ except that each M becomes a transversal block of both; and $\theta_{x'}(b) = \theta_{y'}(b)$ will follow from Step 4(c) below, whose proof uses only $x \leq y$ and $\theta_x(b) = \theta_y(b)$.

Step 4: the classes of $\varepsilon_y \vee \varepsilon_b$. Let $\sigma = \varepsilon_y \vee \varepsilon_b$. (a) $\varepsilon_x \vee \varepsilon_b = \sigma$. Let W be a block of x which is a union of $\mathsf{T}(y)$-blocks. If W is a $\mathsf{T}(x)$-block, let $\tilde{K}$ be its class in $\varepsilon_x \vee \varepsilon_b$; then $\tilde{K} \cap \operatorname{supp} x$ is a block of $\theta_x(b) = \theta_y(b)$ contained in $\operatorname{supp} y$, hence equal to $K \cap \operatorname{supp} y$ for a class K of σ, and $W \subseteq K$. If W is an $\mathsf{N}(x)$-block, it is itself a block of $\theta_x(b) = \theta_y(b)$ contained in $\operatorname{supp} y$, so again $W = K \cap \operatorname{supp} y$ for a class K of σ. The remaining blocks of x are $\mathsf{N}(y)$-blocks. So every block of x lies in a single class of σ, i.e. $\varepsilon_x \leq \sigma$, and $\varepsilon_x \vee \varepsilon_b = \sigma$. (b) *Two kinds of classes.* By (a) and the two descriptions of r, for every class K of σ meeting $\operatorname{supp} y$ either $K \cap \operatorname{supp} y = \tilde{N}$ is an $\mathsf{N}(x)$-block (a union of $\mathsf{T}(y)$-blocks), and then $K \cap \operatorname{supp} b = \emptyset$ since $\tilde{N}$ is a non-transversal block of $\theta_x(b)$; or $K \cap \operatorname{supp} y = K \cap \operatorname{supp} x$. We call the $\mathsf{N}(x)$-blocks inside $\operatorname{supp} y$ the *$\tilde{N}$-blocks*; each $\tilde{N}$-block is $K \cap \operatorname{supp} y$ for a class K of σ with $K \cap \operatorname{supp} b = \emptyset$, and every other class meeting $\operatorname{supp} y$ has $K \cap \operatorname{supp} y \subseteq \operatorname{supp} x$. (c) In the situation of Step 3, the blocks of $\theta_{x'}(b)$ and $\theta_{y'}(b)$ are indexed by the same classes K of σ (by (a)), and they can differ only for a class $K \supseteq M$, where they are $(K \cap \operatorname{supp} x) \cup M$ and $(K \cap \operatorname{supp} y) \cup M$, both transversal iff K meets $\operatorname{supp} b$. If these differ, then by (b) $K \cap \operatorname{supp} y$ is an $\tilde{N}$-block and $K \cap \operatorname{supp} b = \emptyset$; but then, for $A_i = C_i \subseteq M$, every block of b meeting K lies in K, and every block of c meeting a block Z of b inside K lies in K (a block of c in Y is an $\mathsf{N}(y)$-block, hence lies in a single class of σ; a block of c inside $\operatorname{supp} y$ meets $Z \cap \operatorname{supp} y \subseteq \tilde{N}$, so it is $\tilde{N}$ or contained in it), so a chain from $C_i \subseteq K$ to B_i in Δ_i stays inside K, contradicting $K \cap \operatorname{supp} b = \emptyset$. This completes Step 3. (d) *Every $\tilde{N}$-block is an $\mathsf{N}(c)$-block, and its class $\Delta(\tilde{N})$ contains no $\mathsf{T}(b)$-block and satisfies $\Delta(\tilde{N}) \cap \operatorname{supp} y = \tilde{N}$.* Let K be the class of σ with $K \cap \operatorname{supp} y = \tilde{N}$. As in (c), every block of b or of c reachable by a chain in $\varepsilon_b \vee \varepsilon_c$ from a block of b inside K lies inside K. If $\tilde{N}$ were a union of $\mathsf{T}(c)$-blocks (the alternative, x and c commuting), a block of b meeting some $C_j \subseteq \tilde{N}$ would lie in K, and then $B_j \subseteq \Delta_j \subseteq K$, contradicting $K \cap \operatorname{supp} b = \emptyset$. So $\tilde{N}$ is an $\mathsf{N}(c)$-block, $\Delta(\tilde{N}) \subseteq K$, and the claims follow.

Step 5: the classes of $\varepsilon_x \vee \varepsilon_a$. By Lemma 1.1(a), the blocks of $\theta_x(a)$ inside $\operatorname{supp} x$ are the sets $K \cap \operatorname{supp} x$ for the classes K of $\varepsilon_x \vee \varepsilon_a$ meeting $\operatorname{supp} x$, transversal iff K meets $\operatorname{supp} a$; those of

$\theta_x(c)$ are the $\mathsf{T}(v)$-blocks V_k (transversal) and the $\mathsf{N}(c)$-blocks $\tilde{N}'$ inside $\operatorname{supp} x$ (non-transversal; they are the classes of $\varepsilon_x \vee \varepsilon_c$ of type (ii)). Since $\theta_x(a) = \theta_x(c)$: for every k there is a class K_k of $\varepsilon_x \vee \varepsilon_a$ with $K_k \cap \operatorname{supp} x = V_k$, and for every $\mathsf{N}(c)$-block $\tilde{N}' \subseteq \operatorname{supp} x$ a class $K_{\tilde{N}'}$ with $K_{\tilde{N}'} \cap \operatorname{supp} x = \tilde{N}'$ and $K_{\tilde{N}'} \cap \operatorname{supp} a = \emptyset$. Consequently *every block of c lies inside a single class of $\varepsilon_x \vee \varepsilon_a$*: a $\mathsf{T}(c)$-block lies in some $V_k \subseteq K_k$; an $\mathsf{N}(c)$-block inside $\operatorname{supp} x$ is some $\tilde{N}' \subseteq K_{\tilde{N}'}$; and the other $\mathsf{N}(c)$-blocks are $\mathsf{N}(x)$-blocks. Hence $\varepsilon_a \vee \varepsilon_c \leq \varepsilon_x \vee \varepsilon_a$, and for an $\mathsf{N}(c)$-block $\tilde{N}' \subseteq \operatorname{supp} x$ we get $\Gamma(\tilde{N}') \subseteq K_{\tilde{N}'}$; so $\Gamma(\tilde{N}')$ *contains no $\mathsf{T}(a)$-block, hence no $\mathsf{T}(c)$-block, and $\tilde{N}'$ is the only block of c inside* $\operatorname{supp} x$ *that it contains* (a block of c in $\Gamma(\tilde{N}')$ inside $\operatorname{supp} x$ lies in $K_{\tilde{N}'} \cap \operatorname{supp} x = \tilde{N}'$).

Step 6: the four kinds of blocks of c and their classes. By Steps 2–4, every block of c is of exactly one of the following kinds: a $\mathsf{T}(c)$-block C_i (inside $\operatorname{supp} y$, by Step 3); an $\mathsf{N}(c)$-block inside $\operatorname{supp} x$; an $\tilde{N}$-block; or an $\mathsf{N}(y)$-block inside Y, which is an $\mathsf{N}(a)$-block by Step 3 and is therefore a class of $\varepsilon_a \vee \varepsilon_c$ by itself. Collecting Steps 4(d) and 5:

(1) the blocks of c in Γ_i are C_i and $\tilde{N}$-blocks (an $\mathsf{N}(c)$-block inside $\operatorname{supp} x$ lies in a class without $\mathsf{T}(a)$-blocks, and an $\mathsf{N}(y)$-block in Y is a class by itself);
(2) the blocks of c in Δ_i are C_i, $\mathsf{N}(c)$-blocks inside $\operatorname{supp} x$ and $\mathsf{N}(y)$-blocks in Y (an $\tilde{N}$-block lies in a class without $\mathsf{T}(b)$-blocks);
(3) for an $\mathsf{N}(c)$-block $\tilde{N}' \subseteq \operatorname{supp} x$, the blocks of c in $\Gamma(\tilde{N}')$ are $\tilde{N}'$ and $\tilde{N}$-blocks, and $\Gamma(\tilde{N}')$ contains no $\mathsf{T}(a)$-block;
(4) for an $\tilde{N}$-block $\tilde{N}$, the blocks of c in $\Delta(\tilde{N})$ are $\tilde{N}$ and $\mathsf{N}(y)$-blocks in Y (as $\Delta(\tilde{N}) \cap \operatorname{supp} y = \tilde{N}$), and $\Delta(\tilde{N})$ contains no $\mathsf{T}(b)$-block;
(5) an $\mathsf{N}(y)$-block $E \subseteq Y$ is a class of $\varepsilon_a \vee \varepsilon_c$ by itself.

Step 7: proof of (5). Two blocks E, E' of c lie in one class of $\varepsilon_a \vee \varepsilon_b \vee \varepsilon_c$ iff they are joined by a sequence $E = E_0, E_1, \ldots, E_n = E'$ of blocks of c in which consecutive blocks lie in a common class of $\varepsilon_a \vee \varepsilon_c$ or in a common class of $\varepsilon_b \vee \varepsilon_c$: indeed, if two blocks W, W' (of a, b or c) meet at a point p, and $E(p)$ is the block of c containing p, then W and $E(p)$ lie in a common class of $\varepsilon_a \vee \varepsilon_c$ or of $\varepsilon_b \vee \varepsilon_c$ (according to whether W is a block of a or of b; if W is a block of c then $W = E(p)$), so a chain of blocks meeting consecutively can be replaced by the sequence of the blocks of c at its meeting points. Now start such a sequence at $E_0 = C_p$ and use (1)–(5) to see which blocks of c can occur. From C_p one can move, within Γ_p, only to $\tilde{N}$-blocks of Γ_p, and within Δ_p only to $\mathsf{N}(c)$-blocks $\tilde{N}' \subseteq \operatorname{supp} x$ of Δ_p and to $\mathsf{N}(y)$-blocks $E \subseteq Y$ of Δ_p. From an $\tilde{N}$-block $\tilde{N}$ of Γ_p one can move within $\Gamma(\tilde{N}) = \Gamma_p$ (back) or within $\Delta(\tilde{N})$ to $\mathsf{N}(y)$-blocks $E \subseteq Y$; from such an E one can move within $\Delta(E) = \Delta(\tilde{N})$ only, and $\Gamma(E) = \{E\}$. From $\tilde{N}' \subseteq \operatorname{supp} x$ in Δ_p one can move within $\Delta(\tilde{N}') = \Delta_p$ (back) or within $\Gamma(\tilde{N}')$ to $\tilde{N}$-blocks of $\Gamma(\tilde{N}')$, from which one can move within $\Gamma(\tilde{N}')$ (back) or within their Δ-classes to $\mathsf{N}(y)$-blocks in Y, which are dead ends as before. From an $\mathsf{N}(y)$-block $E \subseteq Y$ in Δ_p one can move within Δ_p only. So the blocks of c reachable from C_p are: C_p; $\tilde{N}$-blocks (of Γ_p, or of the classes $\Gamma(\tilde{N}')$ for $\tilde{N}' \subseteq \operatorname{supp} x$ in Δ_p); $\mathsf{N}(c)$-blocks inside $\operatorname{supp} x$ lying in Δ_p; and $\mathsf{N}(y)$-blocks in Y. None of these is a $\mathsf{T}(c)$-block other than C_p, which proves (5).

Step 8: counting. Let $l \in \{1, \ldots, t\}$ and $I_l = \{p : A_p \subseteq U_l\}$; by Step 3 every A_p lies in some U_l. Consider the class of U_l in $\varepsilon_u \vee \varepsilon_v$ ($\varepsilon_u = \varepsilon_y \vee \varepsilon_a$, $\varepsilon_v = \varepsilon_y \vee \varepsilon_c$); by $u \bowtie v$ (Step 1(iv)) it contains exactly one $\mathsf{T}(v)$-block, $V_{k(l)}$ say. Every Γ_p is contained in a class of $\varepsilon_u \vee \varepsilon_v$ (its blocks of a and of c are contained in blocks of u and of v), so $C_p \subseteq V_{k(l)}$ for $p \in I_l$, and conversely $C_q \subseteq V_{k(l)}$ implies $A_q \subseteq U_l$; thus $I_l = \{q : C_q \subseteq V_{k(l)}\}$. Now compare the two sides of $y\,ab = y\,acb$ (Step 1(vi)). The upper blocks of ab are the $\mathsf{N}(a)$-blocks and the sets $D \cap \operatorname{supp} a$ for the classes D of $\varepsilon_a \vee \varepsilon_b$ meeting $\operatorname{supp} a$, the latter being transversal iff D meets $\operatorname{supp} b$, with lower part $D \cap \operatorname{supp} b$. By (5), $D \cap \operatorname{supp} a$ is a single $\mathsf{T}(a)$-block; write $D(A_p)$ for the class of A_p. In the

product $y \cdot (ab)$ the middle join is $\varepsilon_y \vee \varepsilon_{ab}$, where ε_{ab} is the upper partition of ab; the class of U_l in it is U_l itself, because U_l is a union of $\mathsf{T}(y)$-blocks and of the blocks A_p ($p \in I_l$) of ε_{ab}, and no $\mathsf{N}(a)$-block or $\mathsf{N}(y)$-block meets it. So $y\,ab$ has the upper block U_l, with lower part $\bigcup_{p\in I_l}(D(A_p)\cap \operatorname{supp} b)$ (transversal iff this is nonempty). In $y\,acb = (ya)(cb) = u\,(cb)$ the middle join is $\varepsilon_u \vee \varepsilon_c$, whose class containing U_l contains no other $\mathsf{T}(u)$-block and, being contained in the class of U_l in $\varepsilon_u \vee \varepsilon_v$, contains exactly the $\mathsf{T}(c)$-blocks C_p with $p \in I_l$ (all of which it does contain, as Γ_p is a chain in $\varepsilon_u \vee \varepsilon_c$); so $u\,cb$ has the upper block U_l with lower part $\bigcup_{p\in I_l} B_p$. Hence

$$\bigcup_{p\in I_l}\big(D(A_p)\cap \operatorname{supp} b\big) = \bigcup_{p\in I_l} B_p.$$

By (5), $D(A_p)\cap \operatorname{supp} b \subseteq B_p$ for each p, and the B_p are pairwise disjoint and nonempty; so $D(A_p)\cap \operatorname{supp} b = B_p$, i.e. $B_p \in D(A_p)$, for every $p \in I_l$ and every l, that is, for every p.

Conclusion. Every class of $\varepsilon_a \vee \varepsilon_b$ meeting $\operatorname{supp} a$ is some $D(A_p)$, which contains exactly one $\mathsf{T}(a)$-block and exactly one $\mathsf{T}(b)$-block (B_p) by (5) and Step 8; and every class meeting $\operatorname{supp} b$ contains some B_p, hence is $D(A_p)$. By Lemma 1.1(c) and its proof, $a \bowtie b$. The statements about $\mathfrak{P}_8^3$ and the algebra of order 9 follow from Proposition 5.19(a) and Remark 5.22(3): both violate (R′), and all their proper subalgebras embed (for the algebra of order 9, because its subalgebras of order 8 satisfy (Q1), (S), (R), so they are among the 10 828 algebras of the census, and it contains none of the seven that do not embed). □

Remark 5.24. (1) In the language of Step 1, the hypotheses of (R′) say that $x\,ac = xc$ (x does not distinguish the element ac implementing $a \bowtie c$ from c) and $xb = yb$ (x and y act alike on b); the hypotheses of (R) say that $xb = x\,cb$. The conclusion actually obtained is stronger than $a \bowtie b$: by (5) and Step 8, the classes of $\varepsilon_a \vee \varepsilon_b \vee \varepsilon_c$ meeting $\operatorname{supp} a \cup \operatorname{supp} b \cup \operatorname{supp} c$ are exactly the classes containing the triples A_i, B_i, C_i, so that $ab = acb$; in $\mathcal{P}(\mathcal{P}_4)$ this identity, and its consequences $cab = cb$, $ba = bca$, hold in all instances of the hypotheses, whereas a triangle $a \bowtie b \bowtie c \bowtie a$ alone does not force $ab = acb$ (the three matchings need not compose to the identity). (2) Neither the counting of Step 8 nor its analogue for $x\,ab = x\,cb$ can replace (5): both are blind to the merging of two $\mathsf{T}(a)$-blocks of the same U_l into one class of $\varepsilon_a \vee \varepsilon_b$, which is exactly what happens in $\mathfrak{P}_8^3$ ($\theta_a(b) = u$, $\theta_b(a) = w$ there).

5.10. **Proof of ($\mathbf{P}_1$).** We write the path of (P_1) as $a \bowtie c \bowtie b \bowtie d$ (so $p_0 = a$, $p_1 = c$, $p_2 = b$, $p_3 = d$); the hypotheses are then $xa = ax$, $xc = cx$, $\theta_a(x) = \theta_b(x)$ and $\theta_x(c) = \theta_x(d)$, and the conclusion is $a \bowtie d$.

Theorem 5.25. *(P_1) holds in every partition monoid $\mathcal{P}_X$: if a, b, c, d, x are projections with $a \bowtie c$, $c \bowtie b$, $b \bowtie d$, $xa = ax$, $xc = cx$, $\theta_a(x) = \theta_b(x)$ and $\theta_x(c) = \theta_x(d)$, then $a \bowtie d$. Consequently $\mathfrak{P}_8^2$ embeds in no $\mathcal{P}(\mathcal{P}_X)$ and is a minimal non-embeddable projection algebra.*

Proof. Write $u = xa = ax$ and $v = xc = cx$.

Step 1: x commutes with b, and $xb = u$. We have $\theta_b(x) = \theta_a(x) = axa = u$, so $u \le b$; as also $u \le x$, (Q1) gives $xb = bx$, and then $\theta_b(x) = bxb = xb$, i.e. $xb = bx = u$. (In particular $u \le a, b$, so $ab = ba$ by (Q1); this is not needed below.)

Step 2: the blocks of a, b, c. We use Lemma 1.1 and the notation of the proof of Theorem 5.16: $Y = X \setminus \operatorname{supp} x$; $A_1, \dots, A_s$ are the $\mathsf{T}(a)$-blocks, C_i the $\mathsf{T}(c)$-block matched with A_i, B_i the $\mathsf{T}(b)$-block matched with C_i, D_i the $\mathsf{T}(d)$-block matched with B_i; Γ_i, Δ_i, Λ_i are the classes of $\varepsilon_a \vee \varepsilon_c$, $\varepsilon_c \vee \varepsilon_b$, $\varepsilon_b \vee \varepsilon_d$ containing C_i, C_i, B_i respectively, so that Γ_i contains exactly the T-blocks A_i, C_i; Δ_i exactly C_i, B_i; Λ_i exactly B_i, D_i (Lemma 1.1(c)). Since x commutes with a, b and c (Step 1), every block of a, b or c lies inside $\operatorname{supp} x$ or is an $\mathsf{N}(x)$-block, and Step 1 of the proof of Theorem 5.16, applied to the matched pairs $a \bowtie c$ and $b \bowtie c$, shows that *a, b and c have the same*

blocks inside Y, with the same labels: for $i \notin I := \{i : A_i \subseteq \operatorname{supp} x\}$ we have $A_i = B_i = C_i \subseteq Y$, and an $\mathsf{N}(x)$-block is an $\mathsf{N}(a)$-block iff it is an $\mathsf{N}(b)$-block iff it is an $\mathsf{N}(c)$-block. Inside $\operatorname{supp} x$ we compare the elements xa and xb, which are equal by Step 1: by Lemma 1.1(b) the upper blocks of xa are the $\mathsf{N}(x)$-blocks, the $\mathsf{T}(u)$-blocks $U_1, \dots, U_t$ (the classes of type (i) of $\varepsilon_x \vee \varepsilon_a$, each a union of $\mathsf{T}(a)$-blocks and of $\mathsf{T}(x)$-blocks, with lower part U_l') and the $\mathsf{N}(a)$-blocks inside $\operatorname{supp} x$ (type (ii)); likewise for xb. Hence *the $\mathsf{N}(a)$-blocks inside* $\operatorname{supp} x$ *are exactly the $\mathsf{N}(b)$-blocks inside* $\operatorname{supp} x$*, and each U_l is a union of $\mathsf{T}(a)$-blocks and also a union of $\mathsf{T}(b)$-blocks.* Altogether: a and b have the same N-blocks, and $\operatorname{supp} a \cap \operatorname{supp} x = \operatorname{supp} b \cap \operatorname{supp} x = \operatorname{supp} u = U_1 \cup \dots \cup U_t$. Finally, for $p \in I$ the block C_p meets $\operatorname{supp} u$ only inside $A_p \cap B_p$: a point of $C_p \cap \operatorname{supp} u$ lies in some A_q and some B_r, and C_p meets no $\mathsf{T}(a)$-block other than A_p and no $\mathsf{T}(b)$-block other than B_p.

Step 3: the blocks of d inside $\operatorname{supp} x$*.* By Lemma 1.1(a) the blocks of $\theta_x(d)$ inside $\operatorname{supp} x$ are the sets $K \cap \operatorname{supp} x$ for the classes K of $\varepsilon_x \vee \varepsilon_d$ meeting $\operatorname{supp} x$, transversal iff K meets $\operatorname{supp} d$; the blocks of $\theta_x(c) = v$ inside $\operatorname{supp} x$ are the $\mathsf{T}(v)$-blocks $V_1, \dots, V_{t'}$ (unions of $\mathsf{T}(c)$-blocks; transversal) and the $\mathsf{N}(c)$-blocks inside $\operatorname{supp} x$ (non-transversal). Since $\theta_x(d) = \theta_x(c)$: *every block Z of d meets* $\operatorname{supp} x$ *inside a single V_k, or inside a single $\mathsf{N}(c)$-block $\tilde{N} \subseteq \operatorname{supp} x$, or not at all; and in the second case Z is an $\mathsf{N}(d)$-block*, because the class of $\varepsilon_x \vee \varepsilon_d$ whose trace is $\tilde{N}$ does not meet $\operatorname{supp} d$. We call Z of *V-type*, *$\tilde{N}$-type* or *outer* accordingly.

Now let Z be a block of d lying in Λ_s. A point of Z in $\operatorname{supp} b$ lies in some B_r, and then $r = s$: *a block of Λ_s meets* $\operatorname{supp} b$ *only inside B_s*; in particular, if Z meets a $\mathsf{T}(a)$-block $A_i \subseteq Y$, then Z meets $B_i = A_i$ and $i = s$, and if Z meets $\operatorname{supp} u$ then $s \in I$ and Z meets no $\mathsf{T}(a)$-block inside Y. Suppose Z meets $\operatorname{supp} u$ at a point q; then $q \in B_s$. If Z is of V-type, $q \in V_k \cap \operatorname{supp} u \subseteq \operatorname{supp} c \cap \operatorname{supp} u$ lies in some C_p, hence in $A_p \cap B_p$ by Step 2, and $p = s$: *a V-type block of Λ_s meets* $\operatorname{supp} a$ *only inside A_s*. If Z is of $\tilde{N}$-type, $q \in \tilde{N} \cap B_s$, so $\tilde{N} \in \Delta_s$ (a block of c meeting B_s), and $\tilde{N}$ lies in a class $\Gamma(\tilde{N})$ of $\varepsilon_a \vee \varepsilon_c$ which contains at most one $\mathsf{T}(a)$-block, A_r say; then Z *meets* $\operatorname{supp} a$ *only inside* $\tilde{N} \cap A_r$, and $\tilde{N} \in \Gamma_r \cap \Delta_s$.

Step 4: the key invariant. Call two blocks of d *linked* if they are joined by a chain of blocks of d and $\mathsf{N}(a)$-blocks (consecutive blocks meeting); a linked family lies in a single class of $\varepsilon_b \vee \varepsilon_d$, since $\mathsf{N}(a)$-blocks are $\mathsf{N}(b)$-blocks (Step 2). We claim: *a linked family $\mathcal{F}$ of blocks of d meets at most one $\mathsf{T}(a)$-block; and if $\mathcal{F}$ contains a $\mathsf{T}(d)$-block D_p, it meets* $\operatorname{supp} a$ *only inside A_p.* If no block of $\mathcal{F}$ meets $\operatorname{supp} b$ there is nothing to prove (a $\mathsf{T}(a)$-block met by a block of $\mathcal{F}$ is then $A_i = B_i \subseteq Y$, and $\mathcal{F} \subseteq \Lambda_i$). Otherwise $\mathcal{F} \subseteq \Lambda_s$ for a unique s, and by Step 3 a block of $\mathcal{F}$ meets $\operatorname{supp} a$ only inside A_s (if it is of V-type, or of $\tilde{N}$-type with $\tilde{N} \in \Gamma_s$), or inside A_r for an $\mathsf{N}(c)$-block $\tilde{N} \subseteq \operatorname{supp} x$ with $\tilde{N} \in \Gamma_r \cap \Delta_s$, $r \neq s$ (if it is of $\tilde{N}$-type and meets $\operatorname{supp} u$), or inside a $\mathsf{T}(a)$-block $A_i \subseteq Y$, which forces $i = s$. So the claim follows if we show: *if some $Z_0 \in \mathcal{F}$ is of $\tilde{N}_0$-type with $\tilde{N}_0 \in \Gamma_r \cap \Delta_s$ and $r \neq s$ (or with $\Gamma(\tilde{N}_0)$ containing no $\mathsf{T}(a)$-block), then every block of $\mathcal{F}$ meeting* $\operatorname{supp} x$ *is of $\tilde{N}$-type for some $\tilde{N} \in \Gamma_r \cap \Delta_s$ (resp. $\tilde{N} \in \Gamma(\tilde{N}_0) \cap \Delta_s$).* Indeed, such a family meets $\operatorname{supp} a$ only inside A_r (resp. not at all) by Step 3, and it contains no $\mathsf{T}(d)$-block: a $\mathsf{T}(d)$-block is not of $\tilde{N}$-type, and if it were outer, the chain joining it to Z_0 would have a first block meeting $\operatorname{supp} x$, joined to it through blocks lying in Y, which are $\mathsf{N}(x)$-blocks, so that this first block would lie in a class of $\varepsilon_x \vee \varepsilon_d$ meeting $\operatorname{supp} d$ and could not be of $\tilde{N}$-type. We prove the statement by induction along a chain $Z_0, N_1, Z_1, N_2, \dots$ in $\mathcal{F}$. Let Z_j be of $\tilde{N}_j$-type with $\tilde{N}_j \in \Gamma_r \cap \Delta_s$ and let N_{j+1} be an $\mathsf{N}(a)$-block meeting Z_j. If $N_{j+1} \subseteq Y$, it is an $\mathsf{N}(x)$-block, so Z_{j+1} lies in the same class of $\varepsilon_x \vee \varepsilon_d$ as Z_j, whose trace on $\operatorname{supp} x$ is $\tilde{N}_j$: Z_{j+1} is of $\tilde{N}_j$-type or outer, and in the latter case the next $\mathsf{N}(a)$-block meeting it lies in Y as well (an $\mathsf{N}(a)$-block inside $\operatorname{supp} x$ cannot meet an outer block), so the class of $\varepsilon_x \vee \varepsilon_d$ persists until the next block meeting $\operatorname{supp} x$, which is then of $\tilde{N}_j$-type. If $N_{j+1} \subseteq \operatorname{supp} x$, it meets Z_j inside $\tilde{N}_j$; being a block

of a meeting the block $\tilde{N}_j$ of c, it lies in Γ_r, and being a block of b (Step 2) meeting $\tilde{N}_j$, it lies in Δ_s. Let Z_{j+1} meet N_{j+1} at a point $q \in \operatorname{supp} x$, and let E be the block of c containing q. Then E meets N_{j+1}, so $E \in \Gamma_r \cap \Delta_s$; E is not a $\mathsf{T}(c)$-block, since $C_r \in \Delta_r \neq \Delta_s$, $C_s \in \Gamma_s \neq \Gamma_r$ and $C_p \notin \Gamma_r$ for $p \neq r$ (in the case of a class without $\mathsf{T}(a)$-block, no $\mathsf{T}(c)$-block lies in it); so E is an $\mathsf{N}(c)$-block, inside $\operatorname{supp} x$ as $q \in \operatorname{supp} x$, and by Step 3 $Z_{j+1} \cap \operatorname{supp} x \subseteq E$: Z_{j+1} is of E-type with $E \in \Gamma_r \cap \Delta_s$. This completes the induction and the proof of the claim.

Step 5: the classes of $\varepsilon_a \vee \varepsilon_d$. Let $\rho = \varepsilon_a \vee \varepsilon_d$ and let $D(A_i)$ be the ρ-class of A_i.

(a) $D(A_i)$ *contains no $\mathsf{T}(a)$-block other than A_i.* Let A_i, A_j lie in one class and take a chain in ρ from A_i to A_j; between two consecutive $\mathsf{T}(a)$-blocks on it, the blocks are blocks of d and $\mathsf{N}(a)$-blocks, so the blocks of d there are linked, and by Step 4 the two $\mathsf{T}(a)$-blocks coincide. Hence $A_i = A_j$.

(b) $D_i \in D(A_i)$. Take a chain from B_i to D_i in Λ_i; its blocks of b other than B_i are $\mathsf{N}(b)$-blocks, i.e. $\mathsf{N}(a)$-blocks. If $i \notin I$ then $B_i = A_i$ and the chain is a chain in ρ. If $i \in I$, let Z be the block of d following B_i on the chain; the blocks of d on the chain are linked, and the family contains D_i, so by the last remark of Step 4 Z meets $\operatorname{supp} a$ only inside A_i; but Z meets $B_i \subseteq \operatorname{supp} u \subseteq \operatorname{supp} a$, so Z meets A_i, and replacing B_i by A_i gives a chain in ρ from A_i to D_i.

(c) $D(A_i)$ *contains no $\mathsf{T}(d)$-block other than D_i.* Let $D_p \in D(A_i)$ and take a chain in ρ from D_p to A_i; let A_r be the first $\mathsf{T}(a)$-block on it. The blocks of d before A_r are linked to D_p, so by Step 4 the one meeting A_r meets $\operatorname{supp} a$ only inside A_p, whence $r = p$; and by (a) applied along the rest of the chain, $A_i = A_r = A_p$.

Conclusion. By (a)–(c) the classes $D(A_1), \dots, D(A_s)$ are distinct and $D(A_i)$ contains exactly the T-blocks A_i and D_i; every class of ρ meeting $\operatorname{supp} a \cup \operatorname{supp} d$ contains some A_i or some D_i, hence is one of them. By Lemma 1.1(c) and its proof, $a \bowtie d$. The statement about $\mathfrak{P}_8^2$ follows from Proposition 5.19(c) and the minimality noted in Section 5.8. □

Remark 5.26. (1) The proof is shorter than that of (R′) because the hypotheses make a and b almost equal: they have the same blocks outside $\operatorname{supp} x$ and the same N-blocks, and differ only in how they partition each $\mathsf{T}(u)$-block into T-blocks (where $u = xa = xb$). Off the "diagonal" $A_p \cap B_p$, which is where the $\mathsf{T}(c)$-blocks meet $\operatorname{supp} u$, the overlaps $A_r \cap B_s$ lie in $\mathsf{N}(c)$-blocks, and Step 4 says that the blocks of d reaching such an overlap form a closed family of $\mathsf{N}(d)$-blocks. (2) Unlike (R′), the conclusion is not accompanied by a separation statement for a coarser join: in $\mathcal{P}(\mathcal{P}_4)$ the identity $ad = acbd$ holds in all 9 519 instances of the hypotheses, but the classes of $\varepsilon_a \vee \varepsilon_d \vee \varepsilon_b$ may contain two $\mathsf{T}(a)$-blocks. Step 4 was found by testing candidate statements on those instances and on 41 918 instances in $\mathcal{P}(\mathcal{P}_5)$ (`pa8/p1/`); the statement actually used (a linked family of blocks of d meets at most one $\mathsf{T}(a)$-block) holds in all of them.

5.11. **Proof of (Q).** We write the path of (Q) as $a \bowtie c \bowtie b \bowtie d$, with the bijections $A_i \mapsto C_i \mapsto B_i \mapsto D_i$ of T-blocks and the classes $\Gamma_i \ni A_i, C_i$ of $\varepsilon_a \vee \varepsilon_c$, $\Delta_i \ni C_i, B_i$ of $\varepsilon_c \vee \varepsilon_b$, $\Lambda_i \ni B_i, D_i$ of $\varepsilon_b \vee \varepsilon_d$ as in Section 5.10; the hypotheses are $xa = ax$, $\theta_c(x) = \theta_d(x)$, $ya = ay$, $yb = by$, $\theta_y(c) = \theta_y(b)$, and the conclusion is $a \bowtie d$, i.e. $\theta_a(d) = a$, i.e. $\operatorname{rank}(ad) = \operatorname{rank} a$. This subsection proves (Q) (Theorem 5.29). The statements of Steps 2–7 below were found and checked on all instances of the hypotheses in $\mathcal{P}(\mathcal{P}_4)$ and on 10^5–10^6 sampled instances in $\mathcal{P}(\mathcal{P}_5)$ and $\mathcal{P}(\mathcal{P}_6)$ (`pa8/q2/`) before being proved.

Step 1: normalising the auxiliaries. Put $u = xa = ax$. Then $u \leq a$ and $\theta_c(u) = \theta_d(u)$: in the binary notation $u = a * x = x * a$, and the axioms give $(A * B) * (B * C) = (A * B) * C$, i.e. $\theta_C(q) = \theta_{\theta_C(B)}(q)$ for $q \leq B$, so that $\theta_c(u) = (a*x)*c = (a*x)*(x*c) = (a*x)*(x*d) = \theta_d(u)$ (Prover9, `pa8/q2/p9/`). So we may and do assume $x = u \leq a$. Since $\theta_c(u) = \theta_d(u) =: z$ is a common lower bound of c and d, (Q1) gives $cd = dc$. Further consequences derived by Prover9 in the language of regular $*$-semigroups (`pa8/q2/p9s/`): $yc = ybc$, $ycb = yb$ (from $ya = ay$, $yb = by$, $ycy = yby$ and $b \bowtie c$ only), and $ud = ucd = ucbd$; the first is $yc = (yc)(yc)^*(yc) =$

$(ycy)(c) = (yby)c = byc$, the second follows by multiplying with b on the right, and $ud = ucbd$ is the u-analogue of the identity $x\,ad = x\,acbd$ of Remark 5.22(5). The element u is determined by a partition P of the index set $\{1, \dots, s\}$ into *groups* G and a label for each group: the blocks of u are the $\mathsf{N}(a)$-blocks and the sets $A_G = \bigcup_{i\in G} A_i$, transversal iff G is labelled T. Changing the label of a group from T to N preserves the hypothesis $\theta_c(u) = \theta_d(u)$ (verified; it is also clear from Step 2), and conversely a non-hub group (Step 2) may be labelled T; so the labels carry no information, but it is convenient to keep them. We write $C_G = \bigcup_{i\in G} C_i$, B_G, D_G similarly. Not every $u \le a$ with singleton groups satisfies the hypothesis: in $\mathcal{P}(\mathcal{P}_5)$ there are instances in which the given u has a group of size two and no u' with smaller groups works (`u0.py`).

Step 2: the structure of c and d. By the transport lemma (Step 2 of the proof of Theorem 5.3) $z = \theta_c(u)$ has the blocks $\mathsf{N}(c) \cup \{C_G : G \in P\}$, C_G transversal iff G is. Put $\Pi_1 = \varepsilon_u \vee \varepsilon_c$. Its classes are the sets $\Gamma_G = \bigcup_{i\in G} \Gamma_i$ and the *plain* classes of $\varepsilon_a \vee \varepsilon_c$ (those without T-blocks), and since $z \le c$, $z \le d$ and $z = \theta_d(u)$ we have $\Pi_1 = \varepsilon_u \vee \varepsilon_d = \varepsilon_a \vee \varepsilon_z$, so that $\varepsilon_a \vee \varepsilon_d \le \Pi_1$. More precisely, since $\varepsilon_a \le \varepsilon_u$, $\Pi_1 = \varepsilon_u \vee \varepsilon_d = \rho \vee \varepsilon_u$ with $\rho = \varepsilon_a \vee \varepsilon_d$, and $\rho \vee \varepsilon_u$ is obtained from ρ by identifying the classes that meet the same block A_G; so a class of ρ containing no $\mathsf{T}(a)$-block is a class of Π_1, and hence *every class of ρ contained in some Γ_G contains a $\mathsf{T}(a)$-block A_i, $i \in G$*, and every class of ρ not contained in any Γ_G is a plain class. Comparing the two descriptions of z (blocks $\mathsf{N}(c) \cup \{C_G\}$, and $\mathsf{N}(d)$ together with the traces $L \cap \operatorname{supp} d$ of the classes L of Π_1) gives, for each group G, one of two cases:

(nh) $C_G \subseteq \operatorname{supp} d$ and $C_G = \Gamma_G \cap \operatorname{supp} d$ (a *non-hub* group);
(h) C_G is an $\mathsf{N}(d)$-block, and $\Gamma_G \cap \operatorname{supp} d$ is empty or a single $\mathsf{N}(c)$-block E_G (a *hub*, as in case (F2) of the proof of Theorem 5.6);

every other $\mathsf{N}(c)$-block is an $\mathsf{N}(d)$-block and conversely, and a plain class W meets $\operatorname{supp} d$ in an $\mathsf{N}(c)$-block E_W or not at all. The same construction from the side of b, with $w = \theta_b(z) \le b$ (which satisfies $\theta_c(w) = \theta_d(w) = z$ by transport along $b \bowtie c$, $b \bowtie d$) and *the same partition P* (because $\theta_c(w)$ has the blocks $\mathsf{N}(c) \cup \{C_{G'}\}$ for the groups G' of w), gives $\Delta_G = \Lambda_G$ for every G, where $\Delta_G = \bigcup_{i\in G} \Delta_i$ and $\Lambda_G = \bigcup_{i\in G} \Lambda_i$, with $\Lambda_G \cap \operatorname{supp} d = D_G$ (a block of $z = \theta_d(w)$, by transport along $b \bowtie d$); hence in case (nh) $C_G = D_G$, and in case (h) D_G is a single $\mathsf{N}(c)$-block $E'_G \subseteq \Lambda_G$. Comparing the blocks $\mathsf{N}(c) \cup \{C_G\}$ of $z = \theta_c(u)$ with the blocks $\mathsf{N}(d) \cup \{D_G\}$ of $z = \theta_d(w)$ (transport along $b \bowtie d$), and using $C_G = D_G$ for non-hubs, we see that the $\mathsf{N}(c)$-blocks which are not $\mathsf{N}(d)$-blocks are exactly the sets $D_{G'}$, G' a hub. Each nonempty trace E_G or E_W is an $\mathsf{N}(c)$-block contained in $\operatorname{supp} d$, hence one of these; conversely each $D_{G'}$, G' a hub, is an $\mathsf{N}(c)$-block contained in $\operatorname{supp} d$, hence contained in the trace of its class of Π_1, which is a hub class or a plain class (as $D_{G'} \cap \operatorname{supp} c = \emptyset$), and equals that trace. So the nonempty traces E_G of the hubs and E_W of the plain classes are exactly the sets $D_{G'} = E'_{G'}$, G' a hub, each occurring once; write $E_G = D_{\pi(G)}$, $E_W = D_{\pi(W)}$. In particular the $\mathsf{T}(d)$-blocks contained in a non-hub class Γ_G are exactly D_k, $k \in G$, and those contained in a hub class Γ_G form $D_{\pi(G)}$ for a hub $\pi(G)$. Consequently $\varepsilon_a \vee \varepsilon_d \le \Pi_1$ has, inside a non-hub class, $|G|$ blocks of each kind, and a proof of $a \bowtie d$ has to show that these are matched by $\varepsilon_a \vee \varepsilon_d$, and that $|\pi(G)| = |G|$ for hubs, again with a matching. The hub C_G lies in a single class Λ_j, $j \in G$, and E_G in a single class $\Delta_{k'}$, $k' \in \pi(G)$. Hubs have $|G| \le 2$, and a hub G of size 2 is *cross-wired*: writing $G = \{k, l\}$ with $C_G \subseteq \Lambda_l$, we have $D_G \subseteq \Delta_k$, so that $C_k \subseteq \Delta_k \cap \Lambda_l$, $C_l \subseteq \Delta_l \cap \Lambda_l$, $D_k \subseteq \Delta_k \cap \Lambda_k$, $D_l \subseteq \Delta_k \cap \Lambda_l$, and in particular C_k and D_l do not meet $\operatorname{supp} b$ (the $\mathsf{T}(b)$-block B_m lies in $\Delta_m \cap \Lambda_m$). Indeed, let $C_G \subseteq \Lambda_j$ and $D_G \subseteq \Delta_{j'}$ with $j, j' \in G$, and let $i \in G$. A path in Λ_i from D_i to B_i alternates between blocks of d and blocks of b; its interior blocks of d are $\mathsf{N}(d)$-blocks, i.e. common blocks or hubs $C_{G''}$; every block of b, every common block (an $\mathsf{N}(c)$-block) and $D_i \subseteq D_G$ (contained in the $\mathsf{N}(c)$-block D_G) lies in a single class of $\varepsilon_c \vee \varepsilon_b$, while a hub $C_{G''}$ meets exactly the classes Δ_m, $m \in G''$. So the class of $\varepsilon_c \vee \varepsilon_b$ changes along the path only at hubs $C_{G''}$, and only between

indices in G''. The path starts in $\Delta_{j'}$ and ends in Δ_i; if $i \neq j'$, the first change happens at a hub $C_{G''}$ with $j' \in G''$, i.e. $G'' = G$, and since $C_G \subseteq \Lambda_j$ lies on a path in Λ_i, we get $i = j$. Hence $G \subseteq \{j, j'\}$, and if $|G| = 2$ then $j \neq j'$; put $l = j$, $k = j'$. Further, π need not be the identity: in $\mathcal{P}(\mathcal{P}_4)$, $a = (01|23)$, $c = (0|\mathsf{N}1|2|\mathsf{N}3)$, $b = (03|12)$, $d = (\mathsf{N}0|1|\mathsf{N}2|3)$ (writing only the upper parts, N marking N-blocks) with $u = a$ relabelled N and y the rank 0 projection with the single block X has two singleton hubs exchanged by π; here $ad \neq acbd$.

Step 3: the structure imposed by y. Since y commutes with a and b, Lemma 1.1(b) describes $\varepsilon_y \vee \varepsilon_a$ and $\varepsilon_y \vee \varepsilon_b$; in particular every $\mathsf{N}(y)$-block is an $\mathsf{N}(a)$-block or a union of $\mathsf{T}(a)$-blocks, and an $\mathsf{N}(b)$-block or a union of $\mathsf{T}(b)$-blocks (four kinds: *common* blocks, *bridges* $= \mathsf{N}(b)$-blocks that are unions of $\mathsf{T}(a)$-blocks, *co-bridges*, and unions of $\mathsf{T}(a)$-blocks that are unions of $\mathsf{T}(b)$-blocks), and every block of a or b lies in $\operatorname{supp} y$ or in an $\mathsf{N}(y)$-block. The hypothesis $\theta_y(c) = \theta_y(b)$ says (Lemma 1.1(a),(b)): the classes K of $\varepsilon_y \vee \varepsilon_c$ meeting $\operatorname{supp} y$ have traces $K \cap \operatorname{supp} y$ equal to the classes of $\varepsilon_y \vee \varepsilon_b$ of type (i) (unions Q of $\mathsf{T}(y)$-blocks which are unions of $\mathsf{T}(b)$-blocks; these K meet $\operatorname{supp} c$; we call them classes *of the first kind*) and of type (iii) (single $\mathsf{N}(b)$-blocks $F \subseteq \operatorname{supp} y$; these K do not meet $\operatorname{supp} c$; *of the second kind*). We shall use the following consequence repeatedly. $(*)$ *Let K be a class of $\varepsilon_y \vee \varepsilon_c$ of the second kind, with trace an $\mathsf{N}(b)$-block $F \subseteq \operatorname{supp} y$, and let P be a path in a class Δ_m of $\varepsilon_c \vee \varepsilon_b$ (blocks of c and of b alternately, consecutive blocks meeting, all distinct) or in a class Λ_k of $\varepsilon_b \vee \varepsilon_d$ (blocks of d and of b), whose first block is F or a block of c lying in K. Then every block of c on P or meeting a block of b of P lies in K; in particular P contains no $\mathsf{T}(c)$-block; every block of d on P is a common block (an $\mathsf{N}(c)$-block which is an $\mathsf{N}(d)$-block; it lies in K), except possibly the $\mathsf{T}(d)$-block D_k, which is then not contained in $\operatorname{supp} c$, so that the $\mathsf{N}(c)$-block containing it lies in K.* Call a set *covered* if each of its points lies in a block of y or of c belonging to K; F is covered (its points lie in $\mathsf{T}(y)$-blocks, which are in K as F is the trace), and so is a block of c in K. We show inductively that every block of P is covered, which gives all assertions: let W be covered and let W' be the next block of P, meeting W at q, so that q lies in a block Y of y or of c with $Y \in K$. If W' is a block of c, then $W' = Y$ or W' meets the y-block Y, so $W' \in K$; the same applies to any block of c meeting W if W is a block of b, and shows that a block of c meeting a covered set lies in K. If W' is an $\mathsf{N}(b)$-block, it is an $\mathsf{N}(y)$-block, and then $W' \in K$ ($W' = Y$, or W' meets the block Y of c), or a union of $\mathsf{T}(y)$-blocks, and then the $\mathsf{T}(y)$-block through q lies in K, hence in F, so $W' = F$; in both cases W' is covered. If W' is the $\mathsf{T}(b)$-block B_m (resp. B_k), it is not contained in $\operatorname{supp} y$ (the $\mathsf{T}(y)$-block through q would lie in K, hence in F, but $F \cap W' = \emptyset$), so it lies in an $\mathsf{N}(y)$-block Q', which meets Y and lies in K, and W' is covered. If W' is a block of d, it is a common block (a block of c, covered by the first case), or a hub, or D_k; a hub is a union of $\mathsf{T}(c)$-blocks, and the $\mathsf{T}(c)$-block through q would lie in K, which does not meet $\operatorname{supp} c$, impossible; and for D_k the same argument shows $D_k \not\subseteq \operatorname{supp} c$, so D_k lies in an $\mathsf{N}(c)$-block (Step 2), which lies in K.

From this: (a) an $\mathsf{N}(b)$-block F lying in the class Δ_m of a $\mathsf{T}(c)$-block is an $\mathsf{N}(y)$-block: otherwise it is a union of $\mathsf{T}(y)$-blocks, the class K with trace F is of the second kind, and a path in Δ_m from F to C_m contradicts $(*)$; (a′) an $\mathsf{N}(b)$-block F lying in the class Λ_k of a $\mathsf{T}(d)$-block is an $\mathsf{N}(y)$-block: otherwise, with K as before, a path in Λ_k from F to D_k shows by $(*)$ that D_k lies in an $\mathsf{N}(c)$-block $E \in K$, which is $E_{G'}$ or E_W for a hub G' or a plain class W, hence equals $D_{G''}$ for a hub G'' (Step 2) and lies in a class $\Delta_{k''}$, $k'' \in G''$ (as $D_{G''} \subseteq \Lambda_{G''} = \Delta_{G''}$), and a path in $\Delta_{k''}$ from E to $C_{k''}$ contradicts $(*)$; (b) a $\mathsf{T}(y)$-block meeting $\operatorname{supp} c$ lies in a class K of the first kind, hence in a set Q as above, hence in $\operatorname{supp} b$; so a $\mathsf{T}(y)$-block meets C_i only inside $B_i \cap C_i$ (the only $\mathsf{T}(b)$-block meeting $C_i \subseteq \Delta_i$ is B_i); (b′) a $\mathsf{T}(y)$-block Y meeting D_k lies in $\operatorname{supp} b$, and meets D_k only inside $B_k \cap D_k$: if $D_k \subseteq \operatorname{supp} c$ this is (b), and otherwise D_k lies in an $\mathsf{N}(c)$-block $E \subseteq \Delta_{k''}$ as in (a′), and if the class K of Y were of the second kind, $E \in K$ and

a path in $\Delta_{k''}$ from E to $C_{k''}$ would contradict $(*)$; (c) the $\mathsf{T}(c)$-blocks in the class with trace Q are exactly the C_i with $B_i \subseteq Q$ (this is $ycb = yb$); (d) every $\mathsf{N}(b)$-block meeting $\operatorname{supp} c \cup \operatorname{supp} d$ lies in some Δ_m or Λ_k, hence is an $\mathsf{N}(y)$-block by (a), (a′), hence an $\mathsf{N}(a)$-block or a bridge; (e) every block Z of a satisfies $Z \cap (\operatorname{supp} c \cup \operatorname{supp} d) \subseteq \operatorname{supp} b$, or Z is contained in an $\mathsf{N}(b)$-block: if $Z \subseteq \operatorname{supp} y$ this is (b), (b′), and otherwise Z lies in an $\mathsf{N}(y)$-block, which is an $\mathsf{N}(b)$-block or a union of $\mathsf{T}(b)$-blocks. (The identity $\theta_y(d) = \theta_y(b)$ itself holds in every instance, but it is not a consequence of the hypotheses in projection algebras, not even together with (S): Mace4 finds 6-element countermodels; we do not need it.)

Step 4: plain classes contain no $\mathsf{T}(d)$-block. Suppose $D_j \subseteq E_W \subseteq W$, W plain. Take a path in Λ_j from D_j to B_j. Its $\mathsf{N}(b)$-blocks are $\mathsf{N}(y)$-blocks (Step 3(a′)), hence $\mathsf{N}(a)$-blocks or bridges; a bridge is a union of $\mathsf{T}(a)$-blocks and cannot meet W, and an $\mathsf{N}(a)$-block meeting W lies in W; the $\mathsf{N}(d)$-blocks of the path meet these $\mathsf{N}(a)$-blocks, so they lie in W too, and inductively the whole path lies in W up to B_j. If B_j lies in an $\mathsf{N}(y)$-block Q, then Q is an $\mathsf{N}(a)$-block (Q meets W and is a-good), so $B_j \subseteq W$; then a path in Δ_j from C_j to B_j, read backwards from B_j, lies in W by the same argument, and $C_j \subseteq W$, a contradiction. If $B_j \subseteq \operatorname{supp} y$, a $\mathsf{T}(y)$-block of B_j meets the last $\mathsf{N}(d)$-block of the Λ_j-path if the path has length > 1; that block is an $\mathsf{N}(a)$- and $\mathsf{N}(y)$-block (or the hub of W's class, impossible), and a $\mathsf{T}(y)$-block lies in a union of $\mathsf{T}(a)$-blocks or in an $\mathsf{N}(a)$-block which is a union of $\mathsf{T}(y)$-blocks, so cannot meet an $\mathsf{N}(a) \cap \mathsf{N}(y)$-block; hence B_j meets D_j directly, E_W lies in Δ_j, and a path in Δ_j from C_j to E_W, read backwards, lies in W and gives $C_j \subseteq W$ again. Consequently every hub G has $E_G \neq \emptyset$, and π is a permutation of the hubs.

Step 5: types, index pairs and contacts. Put $\rho = \varepsilon_a \vee \varepsilon_d$. By Step 2, $\rho \leq \Pi_1$, every class of ρ contained in a class Γ_G contains some A_i, $i \in G$, and a class of ρ not contained in any Γ_G is a plain class, which contains no T-block of a (Step 2) or of d (Step 4). By Lemma 1.1(a) and (c) (the blocks of $\theta_a(d)$ are $\mathsf{N}(a)$ and the sets U_K, K a class of ρ meeting $\operatorname{supp} a$, transversal iff K meets $\operatorname{supp} d$; and dually for $\theta_d(a)$), $a \bowtie d$ is therefore equivalent to: *for every group G, every class of ρ contained in Γ_G contains exactly one $\mathsf{T}(a)$-block and exactly one $\mathsf{T}(d)$-block.* This is proved for non-hub groups in Step 6 and for hubs in Step 7. Inside Γ_G the blocks of a are A_i ($i \in G$) and $\mathsf{N}(a)$-blocks, and the blocks of d are the *common* blocks (the $\mathsf{N}(c)$-blocks which are $\mathsf{N}(d)$-blocks; in a non-hub class these are all the N-blocks of c and of d in Γ_G, and $C_G = D_G$ is the union of the T-blocks of both), and, in a hub class, the hub C_G and the $\mathsf{T}(d)$-blocks D_m, $m \in \pi(G)$, whose union is the $\mathsf{N}(c)$-block E_G.

Types of blocks. By Step 3 every block Z of a either lies in $\operatorname{supp} y$, or lies in an $\mathsf{N}(y)$-block Q, which is then an $\mathsf{N}(b)$-block or a union of $\mathsf{T}(b)$-blocks. Call Z of N-*type* if it lies in an $\mathsf{N}(y)$-block which is an $\mathsf{N}(b)$-block (if Z is an $\mathsf{N}(a)$-block this means Z itself is an $\mathsf{N}(b)$- and $\mathsf{N}(y)$-block; if $Z = A_i$ it means that A_i lies in a bridge Q, an $\mathsf{N}(b)$- and $\mathsf{N}(y)$-block which is a union of $\mathsf{T}(a)$-blocks), and of b-*type* otherwise; a b-type block lies in $\operatorname{supp} y$ or in an $\mathsf{N}(y)$-block contained in $\operatorname{supp} b$, so by Step 3(b), (b′) it meets each C_m only inside $B_m \cap C_m$ and each D_m only inside $B_m \cap D_m$. For an $\mathsf{N}(b)$-block Q write $\delta(Q) = m$ if $Q \subseteq \Delta_m$ and $\lambda(Q) = k$ if $Q \subseteq \Lambda_k$ (undefined if the class of Q in $\varepsilon_c \vee \varepsilon_b$, resp. $\varepsilon_b \vee \varepsilon_d$, contains no T-block); an N-type block $Z \subseteq Q$ meets C_m only if $\delta(Q) = m$ and D_k only if $\lambda(Q) = k$. A block of a meeting $\operatorname{supp} c \cup \operatorname{supp} d$ at a point outside $\operatorname{supp} b$ is of N-type (Step 3(e), (d)).

In a non-hub class every point of $C_G = D_G$ lies in a unique $\mathsf{T}(c)$-block C_m and a unique $\mathsf{T}(d)$-block D_k (its *cell* (m, k)), and a point of C_G lying in $\operatorname{supp} b$ lies in $B_m \cap C_m \cap D_m$ for some m: its cell is *diagonal*. Hence a b-type block of a meets C_G only in diagonal cells, and since it meets at most one $\mathsf{T}(c)$-block (it lies in one class Γ_m), only in the cell (m, m); while an N-type block $Z \subseteq Q$ meets C_G only in points outside $\operatorname{supp} b$, in the cell $(\delta(Q), \lambda(Q))$.

In a hub class, with $C_G \subseteq \Lambda_j$ ($j \in G$) and $E_G \subseteq \Delta_{k'}$ ($k' \in \pi(G)$) as in Step 2: a b-type block meets C_G only in the sets $B_i \cap C_i$, $i \in G$, and E_G only in the sets $B_m \cap D_m$, $m \in \pi(G)$; an N-type block $Z \subseteq Q$ meets C_G only if $\lambda(Q) = j$, and then inside $C_{\delta(Q)}$, and meets E_G only if $\delta(Q) = k'$, and then inside $D_{\lambda(Q)}$.

Index pairs. For a common block E let $\lambda(E)$ be the index k with $E \subseteq \Lambda_k$ (undefined if the class of E in $\varepsilon_b \vee \varepsilon_d$ contains no $\mathsf{T}(d)$-block) and $\delta(E)$ the index m with $E \subseteq \Delta_m$ (undefined likewise). (i) *If E meets a b-type block of a at a point q, then either $q \in B_m$ for some m and $(\lambda(E), \delta(E)) = (m, m)$, or $q \notin \operatorname{supp} b$ and both indices are undefined.* Indeed, if $q \in B_m$ then E meets B_m. Otherwise q lies in $\operatorname{supp} y$ (a b-type block not contained in $\operatorname{supp} y$ lies in $\operatorname{supp} b$), in a $\mathsf{T}(y)$-block Y, which lies in an $\mathsf{N}(b)$-block $F \subseteq \operatorname{supp} y$ (Step 3), and the class K of $\varepsilon_y \vee \varepsilon_c$ with trace F is of the second kind and contains E (a block of c meeting Y). If $\delta(E) = m$ were defined, a path in Δ_m from E to C_m would contradict Step 3($*$). If $\lambda(E) = k$ were defined, a path in Λ_k from E to D_k shows by ($*$) that D_k lies in an $\mathsf{N}(c)$-block $E' \in K$, which lies in a class $\Delta_{k''}$ (Step 2, as in Step 3(a′)), and a path in $\Delta_{k''}$ from E' to $C_{k''}$ contradicts ($*$). (ii) If E meets an N-type block $Z \subseteq Q$ then $(\lambda(E), \delta(E)) = (\lambda(Q), \delta(Q))$ (as indices or as "undefined"), as E meets the b-block Q. (iii) Two common blocks meeting the same N-type $\mathsf{N}(a)$-block have the same pair. Consequently, along a path $E_0, F_1, E_1, \dots, F_r, E_r$ alternating between common blocks and $\mathsf{N}(a)$-blocks (consecutive blocks meeting; no hub involved), the pair is preserved across N-type $\mathsf{N}(a)$-blocks and, at a b-type $\mathsf{N}(a)$-block, the pairs of the two adjacent common blocks are each either diagonal ((m, m)) or undefined in both coordinates. In particular (*rigidity*): *if some common block of such a path, or some N-type $\mathsf{N}(a)$-block of it, has a defined non-diagonal pair (k, m), $k \neq m$, then every $\mathsf{N}(a)$-block of the path is of N-type and every block of the path has the pair (k, m); and a $\mathsf{T}(a)$-block met by a common block of the path is then of N-type with the pair (k, m) as well* (by (i) and (ii), which apply to $\mathsf{T}(a)$-blocks too).

Step 6: non-hub classes.

Proposition 5.27. *In the situation of Steps* 1–5*, let G be a non-hub group. Then every class of ρ contained in Γ_G contains exactly one $\mathsf{T}(a)$-block and exactly one $\mathsf{T}(d)$-block.*

Proof. Let $\mathcal{K}$ be the set of components of the graph on the $\mathsf{N}(a)$-blocks and the common blocks contained in Γ_G (blocks joined when they meet); each $\kappa \in \mathcal{K}$ is contained in a single class Γ_m, so it touches at most one $\mathsf{T}(a)$-block, A_m, and its $\mathsf{N}(a)$-blocks meet C_G only in C_m. Say that D_k is *attached* to A_i if A_i meets D_k, or if some κ touching A_i (a common block of κ meets A_i) contains an $\mathsf{N}(a)$-block meeting D_k. The classes of ρ inside Γ_G are generated by these adjacencies, so it suffices to prove:

(a) every κ contains $\mathsf{N}(a)$-blocks meeting at most one $\mathsf{T}(d)$-block;
(b) at most one $\mathsf{T}(d)$-block is attached to each A_i;
(c) at least one $\mathsf{T}(d)$-block is attached to each A_i.

Given (a)–(c): by (b), (c) the attachment defines a map $\tau\colon G \to G$; every D_k lies in a class of ρ containing some A_i (Step 5), and a ρ-path from D_k to A_i starts with D_k, a block of a meeting it, and, if that block is an $\mathsf{N}(a)$-block, continues inside its κ until the first T-block met, which by (a) is a $\mathsf{T}(a)$-block; so every D_k is attached to some A_i, τ is surjective, hence bijective; and then the ρ-class of A_i contains the T-blocks A_i and $D_{\tau(i)}$ only (a further $\mathsf{T}(a)$-block A_j in it would be reached through some D_k, which would be attached to both, and a further $\mathsf{T}(d)$-block through some κ or A, contradicting (a) or (b)), and every class of ρ in Γ_G is of this form.

(c): Take a path in Γ_i from A_i to C_i; its blocks of c are common blocks and its blocks of a are $\mathsf{N}(a)$-blocks, so they lie in one κ touching A_i, and the last block of a on it (or A_i itself) meets $C_i \subseteq C_G = D_G$, hence some D_k.

(b): Let D_k be attached to A_i. *Case A_i of b-type.* A direct contact $A_i \cap D_k \neq \emptyset$ is in the cell (i, i), so $k = i$. Otherwise κ touches A_i at a common block E whose pair is diagonal or undefined

(i), and contains an $\mathsf{N}(a)$-block F meeting D_k at a point of C_i. If F is of b-type, the contact is in the cell (i,i) and $k=i$. If F is of N-type, F is an $\mathsf{N}(b)$-block in $\Lambda_k \cap \Delta_i$, so the common block of κ adjacent to F on a path from E has the pair (k,i); but along the path from E all pairs are diagonal or undefined, so $k=i$. *Case $A_i \subseteq Q$ of N-type.* A direct contact lies in the cell $(i,\lambda(Q))$, so $k=\lambda(Q)$ and $\delta(Q)=i$. Otherwise the common block E of κ meeting A_i has the pair $(\lambda(Q),\delta(Q))$, and along the path from E to the contact block F the pair is preserved up to the first b-type $\mathsf{N}(a)$-block, if any. If there is none: F is of N-type with $(\lambda(F),\delta(F))=(k,i)$ equal to $(\lambda(Q),\delta(Q))$, or of b-type with $k=i$ and the pair $(\lambda(Q),\delta(Q))$ diagonal or undefined; if there is one, then $(\lambda(Q),\delta(Q))$ is diagonal or undefined and the contact gives $k=i$ (for an N-type F the adjacent pair (k,i) is diagonal). So every attached D_k is $D_{\lambda(Q)}$, and then $\delta(Q)=i$, or D_i, and then $\lambda(Q)=\delta(Q)$ or both are undefined; if both kinds occur then $\lambda(Q)=\delta(Q)=i$. Hence at most one $\mathsf{T}(d)$-block is attached.

(a): Let κ contain F meeting D_k and F' meeting D_l, in cells (γ,k) and (γ,l) where $\kappa \subseteq \Gamma_\gamma$. If F is of b-type then $k=\gamma$, and likewise for F'. If F' is of N-type, it is an $\mathsf{N}(b)$-block in $\Lambda_l \cap \Delta_\gamma$, and the common block next to F' on a path from F' to F has the pair (l,γ); following the path, if it contains no b-type block before F then either F is of N-type and $(k,\gamma)=(l,\gamma)$, or F is of b-type and the pair (l,γ) is diagonal, $l=\gamma=k$; if it contains a b-type block, the pair (l,γ) is diagonal, $l=\gamma$, and by symmetry (the case F of N-type gives $k=\gamma$ in the same way) $k=\gamma$. In all cases $k=l$. □

Step 7: hub classes.

Proposition 5.28. *In the situation of Steps 1–5, let G be a hub. Then $|\pi(G)|=|G|$, and every class of ρ contained in Γ_G contains exactly one $\mathsf{T}(a)$-block and exactly one $\mathsf{T}(d)$-block.*

Proof. Write $G'=\pi(G)$; by Steps 2 and 4, $E_G=D_{G'}$ is a nonempty $\mathsf{N}(c)$-block contained in Γ_G, hence in a single class Γ_e, $e\in G$, and in a single class $\Delta_{k'}$, $k'\in G'$; and $C_G \subseteq \Lambda_j$, $j\in G$. If $|G|=2$ we write $G=\{k,l\}$ with $l=j$, so that (cross wiring, Step 2) $C_k \subseteq \Delta_k \cap \Lambda_l$, $C_l \subseteq \Delta_l \cap \Lambda_l$ and $C_k \cap \operatorname{supp} b = \emptyset$. If $|G'|=2$ we write $G'=\{k',l'\}$, so that (cross wiring for G') $D_{k'} \subseteq \Delta_{k'} \cap \Lambda_{k'}$, $D_{l'} \subseteq \Delta_{k'} \cap \Lambda_{l'}$ and $D_{l'} \cap \operatorname{supp} b = \emptyset$. Let $\mathcal{N}$ be the set of components ν of the graph on the N-blocks of a and d inside Γ_G (the $\mathsf{N}(a)$-blocks, the common blocks and C_G; joined when they meet), and say that D_m ($m\in G'$) is *attached* to A_i ($i\in G$) if there is a path in the graph $\Gamma(a,d)$ from A_i to D_m all of whose interior blocks are N-blocks, i.e. A_i meets D_m, or some ν touching A_i contains an $\mathsf{N}(a)$-block meeting D_m. All paths considered below are shortest paths between their endpoints in the graph in question, hence simple, and an attachment path from A_i to D_m is always taken shortest, so that its interior is a shortest path in a component ν. A component ν not containing C_G lies in a single class Γ_i; a shortest path in a component ν between two blocks lying in the same class Γ_i (or a shortest attachment path with both ends in Γ_i) passes through C_G at most once, and if it does, both neighbours of C_G on it are $\mathsf{N}(a)$-blocks in Γ_i meeting C_G inside C_i, and the two parts of the path avoid C_G (each part lies in a single class of $\varepsilon_a \vee \varepsilon_c$, being a chain of blocks of a and of c); a path in ν between blocks in different classes Γ_k, Γ_l passes through C_G. We write $G=\{k,l\}$, $G'=\{k',l'\}$ as above; when $G'=G$ we have $k'=k$ and $l'=l$, because $E_G=D_G$ lies in $\Delta_{k'}$ and in Δ_k. By Step 5, inside Γ_G:

(i) a block of a meeting C_G in C_i is of N-type, contained in an $\mathsf{N}(b)$-block Q with $(\lambda(Q),\delta(Q))=(j,i)$, or it is of b-type, $i=j$ and the contact lies in $B_j \cap C_j$;
(ii) a block of a meeting $D_{k'}$ is of N-type with pair (k',k'), or of b-type with contact in $B_{k'} \cap D_{k'}$;
(iii) if $|G'|=2$, a block of a meeting $D_{l'}$ is of N-type with pair (l',k').

Claim 1: the $\mathsf{N}(a)$-blocks of a single component ν meet at most one $\mathsf{T}(d)$-block. Suppose $Z,Z'\in\nu$ meet D_m, $D_{m'}$ with $m\neq m'$, so $|G'|=2$, say $m'=l'$, $m=k'$. Both lie in Γ_e;

take a shortest path in ν from Z' to Z. If it avoids C_G, then by (iii) and rigidity all its blocks have the pair (l', k'), contradicting (ii) for Z. If it passes through C_G, the part from Z' to the neighbour Z_1 of C_G avoids C_G, so Z_1 has the pair (l', k') by rigidity, and Z_1 meets C_G inside C_e, so $(\lambda, \delta) = (j, e)$ by (i); thus $k' = e \in G \cap G'$, so $G' = G$, $|G| = 2$, $k' = k$, $e = k$, $l' = l = j$. Then the other neighbour Z_2 of C_G meets C_G inside C_k, so it is of N-type with pair (l, k) by (i) (as $C_k \cap \operatorname{supp} b = \emptyset$), and rigidity along the part from Z_2 to Z gives Z the pair (l, k), contradicting (ii).

Claim 2: at most one $\mathsf{T}(d)$-block is attached to each A_i. Suppose $D_{k'}$ and $D_{l'}$ are attached to A_i, so $|G'| = 2$. By (iii) the contact block of $D_{l'}$ (which is A_i itself or an $\mathsf{N}(a)$-block Z') has the pair (l', k'). *Case $G' \neq G$.* An attachment path from A_i to $D_{l'}$ cannot pass through C_G: the neighbour of C_G on the side of $D_{l'}$ would have the pair (l', k') by rigidity and $\delta = i'' \in G$ by (i), but $k' \notin G$. So by rigidity A_i is of N-type, $A_i \subseteq Q$ with $(\lambda(Q), \delta(Q)) = (l', k')$. Now consider an attachment path from A_i to $D_{k'}$. It does not start with a direct contact ($\lambda(Q) = l' \neq k'$), nor with C_G ($\lambda(Q) = l' \neq j$, as $l' \notin G$), so it starts with a common block meeting A_i, of pair (l', k'), and by rigidity all its blocks have this pair until it reaches C_G or the contact block of $D_{k'}$; the former is excluded as before ($k' \notin G$), the latter by (ii). *Case $G' = G$*, so $|G| = 2$, $k' = k$, $l' = l = j$. Suppose first $i = k$. If A_k meets D_l, then A_k is of N-type with pair (l, k) by (iii) and $e = k$. Otherwise take a shortest attachment path from A_k to D_l; its contact block has the pair (l, k). If the path passes through C_G, the neighbour of C_G on the side of the contact block has the pair (l, k) by rigidity, so by (i) it meets C_G inside C_k and lies in Γ_k, and so does the contact block, i.e. $e = k$; and the neighbour on the side of A_k then also lies in Γ_k, meets C_k and has the pair (l, k) by (i) ($C_k \cap \operatorname{supp} b = \emptyset$). If the path avoids C_G, all its blocks lie in Γ_k, so again $e = k$. In both cases rigidity along the part of the path next to A_k (or (i), if that part is empty and A_k meets C_G) shows that A_k is of N-type with pair (l, k). The attachment path from A_k to D_k then starts with the pair (l, k) (a direct contact is excluded by $\lambda = l \neq k$; a common block meeting A_k has the pair (l, k); and C_G, if it comes first or later, has both neighbours in Γ_k, of pair (l, k), as $e = k$), so by rigidity it never reaches a block of pair (k, k) or of b-type, contradicting (ii). Suppose now $i = l$. If the attachment path from A_l to D_l avoids C_G, then $e = l$, A_l is of N-type with pair (l, k) and, as C_G's neighbours in Γ_l have $\delta = l$ by (i), no path from A_l with pair (l, k) reaches C_G; so the attachment path from A_l to D_k keeps the pair (l, k) by rigidity and cannot reach a contact block as in (ii); a direct contact is excluded by $\lambda(Q) = l \neq k$. If it passes through C_G, then $e = k$ (a shortest path between blocks of Γ_l through C_G would have both neighbours of C_G in Γ_l, of b-type or with $\delta = l$ by (i), whereas rigidity from the contact block gives the neighbour on its side the pair (l, k)); a direct contact $A_l \cap D_k$ is then impossible ($A_l \subseteq \Gamma_l$, $D_k \subseteq \Gamma_k$), and an attachment path from $A_l \subseteq \Gamma_l$ to $D_k \subseteq \Gamma_k$ passes through C_G with its neighbour on the side of D_k in Γ_k, of pair (l, k) by (i), which by rigidity cannot reach a contact block as in (ii).

Consequences. Every D_m, $m \in G'$, is attached to some A_i: its class of ρ contains some A_i (Step 5), and on a shortest ρ-path from D_m to A_i the first T-block after D_m is reached through N-blocks of one component ν, which contains an $\mathsf{N}(a)$-block meeting D_m (unless the block following D_m is already a $\mathsf{T}(a)$-block), so by Claim 1 it is a $\mathsf{T}(a)$-block. Hence, by Claim 2, attachment is a partial map $\tau\colon G \rightharpoonup G'$ which is surjective, and $|\pi(G)| \leq |G|$ for every hub G. Since π is a permutation of the hubs (Step 4), $\sum_G |\pi(G)| = \sum_G |G|$, so $|\pi(G)| = |G|$ for every hub, and τ is a bijection $G \to G'$. In particular $|G'| = 2$ when $|G| = 2$.

Claim 3: every component ν touches at most one $\mathsf{T}(a)$-block. Suppose ν touches A_k and A_l, so $|G| = 2 = |G'|$. Then ν contains C_G, and a shortest path in ν from a block meeting A_k to a block meeting A_l passes through C_G; its neighbour of C_G on the side of A_k lies in Γ_k and meets C_k, so it is of N-type with pair (l, k) by (i), and by rigidity A_k is of N-type with pair (l, k) (this includes the case that A_k meets C_G directly, by (i)). *Case $e = k$.* The block $D_{k'}$ is attached to

some A_i, and every attachment path to $D_{k'}$ ends at a contact block as in (ii), of pair (k', k') or of b-type, lying in Γ_k. An attachment path from A_k: a direct contact would need $\lambda = l = k'$, impossible ($k' = k$ if $G' = G$, $k' \notin G$ otherwise); otherwise it starts with the pair (l, k), and if it passes through C_G both neighbours lie in Γ_k and have the pair (l, k); so by rigidity it cannot reach the contact block. A direct contact $A_l \cap D_{k'}$ is impossible ($A_l \subseteq \Gamma_l$), and an attachment path from $A_l \subseteq \Gamma_l$ to $D_{k'} \subseteq \Gamma_k$ passes through C_G with its neighbour on the side of $D_{k'}$ in Γ_k, of pair (l, k), and rigidity gives the same contradiction. So $D_{k'}$ is attached to no A_i, contradicting the consequences of Claims 1 and 2. *Case $e = l$.* The block $D_{l'}$ is attached to some A_i, through a contact block of pair (l', k') in Γ_l (iii). A direct contact $A_k \cap D_{l'}$ is impossible ($A_k \subseteq \Gamma_k$), and an attachment path from $A_k \subseteq \Gamma_k$ passes through C_G, whose neighbour on the side of $D_{l'}$ lies in Γ_l and meets C_l, so by (i) it is of b-type or has the pair (l, l), whereas rigidity from the contact block gives it the pair (l', k'), $l' \neq k'$: impossible. So $D_{l'}$ is attached to A_l, and by rigidity (the neighbours of C_G in Γ_l have $\delta = l$ by (i), and $k' = l$ is impossible: it would give $l \in G'$, $G' = G$, $k' = k$) the attachment path avoids C_G and A_l is of N-type with pair (l', k'); a direct contact $A_l \cap D_{l'} \neq \emptyset$ gives the same. But then A_l is not touched by ν: a block of ν meeting A_l is C_G (excluded, as $\lambda = l'$ and $\lambda = j = l$ would give $l \in G'$, $G' = G$, and then $\delta = k' = k$, so the contact would lie in $C_k \subseteq \Gamma_k$) or a common block of pair (l', k'), from which, by rigidity, no path in ν reaches a neighbour of C_G.

Conclusion. Let R be a class of ρ contained in Γ_G; it contains some A_i (Step 5). We claim that its T-blocks are exactly A_i and $D_{\tau(i)}$. Otherwise take a shortest path in $\Gamma(a, d)$ from $\{A_i, D_{\tau(i)}\}$ to another T-block; its interior consists of N-blocks. It cannot lead from A_i to a $\mathsf{T}(d)$-block (this would be attached to A_i, hence equal to $D_{\tau(i)}$), nor from A_i to a $\mathsf{T}(a)$-block $A_{i'}$ (Claim 3), nor from $D_{\tau(i)}$ to a $\mathsf{T}(a)$-block $A_{i'}$ ($D_{\tau(i)}$ would be attached to $A_{i'}$, so $i' = i$ as τ is injective), nor from $D_{\tau(i)}$ to a $\mathsf{T}(d)$-block (Claim 1). So every class of ρ in Γ_G contains exactly one $\mathsf{T}(a)$-block and one $\mathsf{T}(d)$-block. □

Theorem 5.29. *The condition* (Q) *holds in every $\mathcal{P}_X$: if $a \bowtie c \bowtie b \bowtie d$, $xa = ax$, $\theta_c(x) = \theta_d(x)$, $ya = ay$, $yb = by$ and $\theta_y(c) = \theta_y(b)$, then $a \bowtie d$.*

Proof. By Step 1 we may assume $x = u \leq a$. By Step 5 it suffices that every class of $\rho = \varepsilon_a \vee \varepsilon_d$ contained in a class Γ_G contains exactly one $\mathsf{T}(a)$-block and one $\mathsf{T}(d)$-block, which is Proposition 5.27 for the non-hub groups and Proposition 5.28 for the hubs. □

Corollary 5.30. (P_2) *and* (S) *hold in every $\mathcal{P}_X$, the latter with a computer-free proof of the hub case of Theorem 5.6; and $\mathfrak{P}_8^4$ and $\mathfrak{P}_8^5$ are minimal non-embeddable projection algebras.*

Proof. (Q)⇒(P_2)⇒(S) in every projection algebra (Proposition 5.32(a)); the hypotheses of (S) are those of (Q) with $x = y = \theta_a(cd)$. The two algebras violate (P_2), resp. (Q) (Section 5.8), and all their proper subalgebras embed. □

Remark 5.31. (1) Of the consequences of the hypotheses in projection algebras derived in Step 1, the proof uses only the normalisation $x = u \leq a$: Step 2 rests on the transport lemma and Lemma 1.1(a) applied to $\theta_c(u) = \theta_d(u)$ and $\theta_c(w) = \theta_d(w)$, and Step 3 on Lemma 1.1(a),(b) applied to $ya = ay$, $yb = by$ and $\theta_y(c) = \theta_y(b)$ (the identities $yc = ybc$, $ycb = yb$ are the semigroup form of Step 3(c), and $cd = dc$, $ud = ucbd$ are not needed). The identities $u \leq \theta_a(d)$ ($u * (d * a) = u$), $z \leq \theta_d(a)$ ($(u * c) * (a * d) = u * c$) and $\theta_{ya}(d) = ya$ ($d * (a * y) = a * y$), which also hold in every projection algebra satisfying the hypotheses (Prover9, `pa8/q2/p9/`), were used in earlier versions of the argument but are not needed: the fact that every class of ρ inside Γ_G contains a $\mathsf{T}(a)$-block is immediate from $\Pi_1 = \varepsilon_u \vee \varepsilon_d$ (Step 2). Whether $ya \leq \theta_a(d)$ follows as well is open (no countermodel of order ≤ 14). The identity $\theta_y(d) = \theta_y(b)$ holds in every instance in $\mathcal{P}_X$ but is not a consequence of the hypotheses in projection algebras, not even together with (S) (Mace4 finds 6-element countermodels); it is not used. (2) The hub-free case

($\operatorname{supp} c = \operatorname{supp} d$, Proposition 5.27 alone) contains the case (F1) of the proof of Theorem 5.6 (with $x = y = \theta_a(cd)$), for which Step 5 of that proof is the special case of (a), (b) in which all blocks of a are of b-type; the hub case of Theorem 5.6 is the case of Proposition 5.28 in which π is the identity and A_G is a bridge or $A_G = B_G$. There the bound $|G| \le 2$ was proved as the bound $p \le 2$ by a different argument and the rest was a finite computer check (`hubtypes.py`); the cross wiring of Step 2 and the rigidity of Step 5 replace it. (3) The hypothesis on y enters only through the classification of the blocks of a and b into $\mathsf{N}(y)$-blocks and unions of $\mathsf{T}(y)$-blocks and through the description of the classes of $\varepsilon_y \vee \varepsilon_c$ meeting $\operatorname{supp} y$ (Step 3); the hypothesis on x only through Step 2. Several further statements were verified on the instances in $\mathcal{P}(\mathcal{P}_m)$, $m \le 6$, before the proof was found and are now consequences of it or of its steps: every block of a meets at most one $\mathsf{T}(d)$-block, every block of d (and every hub) meets at most one $\mathsf{T}(a)$-block; replacing d by the projection d_1 with N-blocks $\mathsf{N}(c)$ and T-blocks D_k (k in non-hub groups) and C_i (i in hubs) gives instances (a, c, b, d_1) and (a, d_1, b, d) of the hypotheses (the second requires $\theta_y(d_1) = \theta_y(b)$, which we have not proved by hand); the hubs of size 2 have $\pi(G) = G$, exactly one of A_k, A_l meets C_G, namely the one of N-type with pair (l, k) found in Claim 3, and in $\mathcal{P}(\mathcal{P}_m)$, $m \le 6$, the other one meets its own D. The example of Step 2 shows that π can exchange singleton hubs.

5.12. **Independence of the conditions.** We now have the following list of conditions, all of which hold in every $\mathcal{P}_X$ or are conjectured to: (Q1), the triangle condition (Tri), (S), (R), and the seven of Proposition 5.19, (R′), (V), (P_1), (P_2), (Q), and the two of (e), which we call (E_6) (the one for $\mathfrak{P}_8^6$) and (E_1) (the one for $\mathfrak{P}_8^1$). Some of them are consequences of others.

Proposition 5.32. (a) *In every projection algebra, (Q) implies (P_2), and (P_2) implies (S); hence (Q) implies (S). In every projection algebra, (R′) implies (R) (Proposition 5.19(a)).*

(b) *Each of the eight conditions (Q1), Tri, (R′), (V), (P_1), (Q), (E_6), (E_1) fails in a finite projection algebra satisfying the other ten conditions of the list: $Q(K_2, \mathrm{const})$ for (Q1) (order 3), $\mathcal{P}(Q_0)$ for Tri (order 11), $\mathfrak{P}_8^3$ for (R′), $\mathfrak{P}_8^7$ for (V), $\mathfrak{P}_8^2$ for (P_1), $\mathfrak{P}_8^5$ for (Q), $\mathfrak{P}_8^6$ for (E_6), $\mathfrak{P}_8^1$ for (E_1). Consequently these eight conditions are independent: none of them follows from the conjunction of the other seven together with (S), (R), (P_2); in particular, for any two of them, neither implies the other even in the presence of (Q1), (S) and (R). The conditions (S), (R), (P_2) are redundant in the list, by (a).*

(c) *The witnesses of the redundant conditions: $\mathfrak{P}$ violates (S), (P_2), (Q) and (E_6) and satisfies the rest; $\mathfrak{P}_9$ violates (S), (P_2), (Q); $\mathfrak{P}_7$ violates (R) and (R′); $\mathfrak{P}_8^4$ violates (P_2) and (Q). Mace4 finds a projection algebra of order 9 (and none smaller) satisfying (Q1) and (E_6) but violating (S), (P_2) and (Q); so (E_6) does not imply (S) either. Altogether, for every ordered pair $X \ne Y$ of conditions in the list, X implies Y (in the presence of (Q1)) exactly in the cases of (a), (Q)⇒(P_2), (Q)⇒(S), (P_2)⇒(S), (R′)⇒(R); every other pair is separated by one of the algebras named in (b) and (c) or by this Mace4 model. The seven candidates of order 9 (Remark 5.22(3)) satisfy all eleven conditions.*

Proof. (a) Putting $y = x$ in (Q) gives exactly (P_2). For (P_2)⇒(S), let $a \bowtie c$, $b \bowtie c$, $b \bowtie d$, $cd = dc =: u$ and $\theta_a(u) = \theta_b(u)$ be the hypotheses of (S), and apply (P_2) to the path $p_0 = d$, $p_1 = b$, $p_2 = c$, $p_3 = a$ and $x = u$: $u \le d$ and $u \le c$, so u commutes with p_0 and p_2; $\theta_u(b) = ubu = u\,cbc\,u = ucu = u = \theta_u(c)$ since $u = uc = cu$ and $cbc = c$; and $\theta_{p_1}(u) = \theta_b(u) = \theta_a(u) = \theta_{p_3}(u)$. The conclusion $p_0 \bowtie p_3$ is $a \bowtie d$. (b) The violated sets were computed with `pa8/indep.py` (which also contains the Mace4 inputs used in (c)); the last assertion is immediate: a condition of the eight that followed from the others would hold in the witness listed for it. (c) By computation. □

Thus, as far as is currently known, the obstructions to embeddability form the list (Q1), Tri, (R′), (V), (P_1), (Q), (E_6), (E_1), pairwise and jointly independent, with (S)⊂(P_2)⊂(Q) (that is, (Q) is the strongest of the three) and (R)⊂(R′). Note that (V) neither implies nor is implied by any of the others, despite its simple form; and that among the eight only (Q1), Tri, (R′), (V), (P_1) and (Q) have been proved to hold in every $\mathcal{P}_X$ (Proposition 2.1, Theorems 4.2, 5.23, 5.20, 5.25, 5.29), together with the redundant (S), (P_2) and (R) (Theorems 5.6, 5.16; (R) is now also a corollary of (R′), and (S), (P_2) of (Q)).

5.13. **The order-9 candidates: (R″) and the family of 2-path conditions.** This subsection records what was learnt (October 2026) from the seven candidate algebras of order 9 (Remark 5.22(3); `pa8/cand9.json`), none of which violates a condition of the list of Section 5.12. The evidence level of everything stated here is "verified in $\mathcal{P}(\mathcal{P}_m)$ for $m \le 6$ by SAT, variables not assumed distinct, and satisfied by all 92 647 embeddable census algebras of order ≤ 9"; nothing in this subsection is proved for all $\mathcal{P}_X$. Code: `pa8/n4/`.

Length-4 path closing: negative. The natural guess that the path-closing family continues with paths of length 4 was tested first. On 16 000 sampled paths $p_0 \bowtie \ldots \bowtie p_4$ in $\mathcal{P}(\mathcal{P}_5)$ (10^8 negative instances, two auxiliaries, the 78-atom language of `pathclose3.py` without the $z = \theta_p(q)$ atoms, coincidences of variables allowed; hitting sets by a C program, `hit4.c`), the 84 minimal hypothesis sets of size ≤ 4 that force $p_0 \bowtie p_4$ on the sample all contain a chord $p_i \bowtie p_j$, except five, which fail in $\mathcal{P}(\mathcal{P}_4)$ as soon as variables may coincide. Of the seven candidates only one (candidate 1) has a chordless unclosed path of length 4, and no two-auxiliary condition valid in $\mathcal{P}(\mathcal{P}_5)$ explains it. So the obstructions do not grow with the path length, at least up to order 9 and two auxiliaries.

Unclosed 2-paths. Six of the seven candidates have an unclosed path of length 2, $a \bowtie c \bowtie b$ with $a \not\bowtie b$ (candidate 2 has none; its extracted identities have an order conclusion $a \le b$, as (V)). Candidate 6 is the graph semigroup $a - d - b$ over the 2×2 rectangular band $\{c, e, f, g\}$ (a over f, b over e, d over g) with two further elements h, i: $hd = dh$, $ib = bi$, $\theta_h(a) = \theta_h(d) = g$, $\theta_h(b) = c = \theta_i(a) = \theta_i(d)$, $\theta_i(b) = e$. Its greedily minimised identity (eleven atoms, nine variables) is this configuration written out through restrictions, and it reduces to a condition of exactly the shape of (R′):

$$(6)\qquad (\mathrm{R}'')\qquad a \bowtie c,\ b \bowtie c,\ xc = cx,\ \theta_x(a) = \theta_x(c),\ yb = by,\ \theta_x(b) = \theta_y(a) \implies a \bowtie b.$$

Compare (R′): $xc = cx$, $\theta_x(a) = \theta_x(c)$, $ya = ay$, $\theta_x(b) = \theta_y(b)$. (R″) is valid in $\mathcal{P}(\mathcal{P}_m)$ for $m \le 6$, holds in every embeddable census algebra of order ≤ 9, is violated by candidate 6 (with $x = h$, $y = i$) and by no other candidate of order 8 or 9; and it is independent of (R′): $\mathfrak{P}_8^3$ violates (R′) but not (R″), candidate 6 the other way round. The two remaining combinations of the same shape, $ya = ay$ with $\theta_x(b) = \theta_y(a)$ (equivalently $\theta_x(b) \le a$) and $yb = by$ with $\theta_x(b) = \theta_y(b)$ (equivalently $\theta_x(b) \le b$), are valid for $m \le 5$ but have no witness of order ≤ 9; the combinations with y commuting with c fail in $\mathcal{P}(\mathcal{P}_4)$. Why none of the earlier searches found (R″): the atom language of `pathclose3.py` (Section 5.7, Remark 5.22(5)) contains $\theta_x(p_i) = \theta_y(p_i)$ but not the cross atoms $\theta_x(p_i) = \theta_y(p_j)$, $i \ne j$. With these added (`pathclose4b.py`), a sampled search in $\mathcal{P}(\mathcal{P}_5)$ at length 2 finds (R″) and its mirror image as the only hypothesis sets of size ≤ 4 explaining an order-9 algebra, among about forty cross-atom sets valid for $m \le 5$ (whose status at $m = 6$ is being checked, `verify6_2.py`).

The other candidates. For candidates $0, 1, 3, 4, 5$ every assignment of two auxiliaries in the extended language is realised by a negative instance in $\mathcal{P}(\mathcal{P}_5)$ (`explain2.py`), so they need three auxiliaries or a richer language. The right language turned out to be the equalities among the elements and all terms $\theta_p(q)$ over the path a, c, b and three auxiliaries x, y, z: a hypothesis is the partition of this set of 36 symbols induced by an assignment in the candidate, weakened greedily

by splitting symbols off under the SAT test at $m = 4$ (`explain3b.py`). The terms that matter are $u = \theta_a(b) = aba \le a$ and $v = \theta_b(a) = bab \le b$, the "would-be" matching restrictions along the unclosed path; the conclusion $a \bowtie b$ is $u = a$ and $v = b$. The smallest explanation found for each unclosed path (`catalog.log`; all valid for $m \le 6$, none violated by an embeddable census algebra; "$aya = a$" is the half-matching $\theta_a(y) = a$):

	hypotheses besides $a \bowtie c \bowtie b$	conclusion	violated by
K0	$yc = cy$, $\theta_y(b) = v$, $bzb = b$, $\theta_z(a) = \theta_z(b)$, $\theta_y(z) = u$	$a \bowtie b$	cand. 0, $\mathfrak{P}_8^1$
K1a	$aya = a$, $\theta_y(b) = \theta_z(x)$, $\theta_x(b) = u = v$, $\theta_c(z) = \theta_x(z)$	$a \bowtie b$	cand. 1, $\mathfrak{P}_8^2$
K1b	$aya = a$, $\theta_z(c) = b$, $\theta_y(a) = \theta_y(x)$, $\theta_x(c) = \theta_z(x) = u$, $\theta_y(z) = v$	$a \bowtie b$	cand. 1
K1c	$bxb = b$, $\theta_y(a) = \theta_z(y)$, $\theta_c(z) = \theta_x(a) = \theta_y(x)$, $\theta_z(x) = v$	$a \bowtie b$	cand. 1, $\mathfrak{P}_8^2$
K3a	$\theta_z(c) = a$, $b \le y$, $\theta_x(c) = \theta_x(y)$, $\theta_y(b) = \theta_y(x)$, $\theta_x(b) = u = v$, $yz = zy$	$a \bowtie b$	cand. 3
K3b	$\theta_x(c) = a$, $b \le y$, $b \le z$, $\theta_y(c) = \theta_y(b)$, $\theta_z(c) = \theta_z(b)$, $\theta_y(z) = \theta_z(y) = u$, $\theta_x(b) = v$	$a \bowtie b$	cand. 3
K3c	$bxb = b$, $\theta_x(a) = \theta_x(b)$, $\theta_x(y) = \theta_y(z)$, $\theta_z(y) = u = v$, $zc = cz$	$a \bowtie b$	cand. 3, $\mathfrak{P}_8^1$
K4a	$aya = a$, $b \le z$, $yc = cy$, $\theta_z(y) = v$, $\theta_z(a) = u$	$a \bowtie b$	cand. 4, $\mathfrak{P}_8^6$
K4b	$aza = a$, $b \le y$, $zc = cz$, $\theta_z(y) = v$, $\theta_y(a) = u$	$a \bowtie b$	cand. 4
K5	$\theta_x(c) = a$, $bxb = b$, $zc = cz$, $\theta_z(x) = u$, $\theta_z(b) = v$	$a \bowtie b$	cand. 5
K6	$\theta_y(z) = \theta_z(a)$, $\theta_c(z) = \theta_y(a)$, $\theta_z(b) = v$	$a \bowtie b$	cand. 6
(R″)	as in (6)	$a \bowtie b$	cand. 6

Thus $\mathfrak{P}_8^1$ (the witness of (E_1)), $\mathfrak{P}_8^6$ (of (E_6)) and $\mathfrak{P}_8^2$ (of (P_1)) also violate 2-path conditions: the conditions (E_1), (E_6), whose shape looked unrelated to the path-closing ones, are path-closing conditions written in a language that cannot name u and v. Whether $\mathfrak{P}_8^4$, $\mathfrak{P}_8^5$ and the order-9 witness of (R′) violate 2-path conditions as well has not been checked. The members of the table are different conditions (each explains a different witness), and no master condition in a, b, c, u, v alone exists: the conjunction of all equalities among these five elements and their θ-terms common to the six instances does not force $a \bowtie b$ even in $\mathcal{P}(\mathcal{P}_4)$.

The friendly relation. The relation $\bowtie$ ("friendly" in East–Azeef Muhammed) is reflexive and symmetric but not transitive, and every condition of the list is a statement of local transitivity: $a \bowtie c \bowtie b$ closes under side hypotheses. Some facts about it, the first two verified in $\mathcal{P}(\mathcal{P}_m)$ for $m \le 5$ and the others valid in every regular $*$-semigroup:

(a) $\bowtie$ is not compatible with the operation θ in either argument (so it is not a tolerance *of the algebra* in the sense of Chajda and Zelinka): in $\mathcal{P}(\mathcal{P}_4)$, $p \bowtie p'$ fails to imply $\theta_q(p) \bowtie \theta_q(p')$ in 24 816 of 142 128 instances and $\theta_p(q) \bowtie \theta_{p'}(q)$ in 23 304.

(b) Its transitive closure is "same rank", i.e. $\mathcal{D}$ restricted to projections ($\mathcal{P}_3$–$\mathcal{P}_5$), which is not a congruence of $\mathcal{P}(\mathcal{P}_X)$ either.

(c) For $p \bowtie p'$ the maps $\theta_{p'} \colon [0,p] \to [0,p']$ and $\theta_p \colon [0,p'] \to [0,p]$ are inverse isomorphisms (for $x \le p$, $p(p'xp')p = (pp'p)x(pp'p) = pxp = x$, and symmetrically), and they preserve $\bowtie$: if $x \le p$ then $x \bowtie p'xp'$. Indeed $x = px = xp$ gives $xp'x = x(pp'p)x = xpx = x$, hence $x(p'xp')x = (xp'x)p'x = x$ and $(p'xp')x(p'xp') = p'(xp'x)p'xp' = p'xp'$.

(d) The down-sets form a presheaf on $(\mathcal{P}(S), \le)$: for $w \le x \le p$ and $y \le p$, $w(xyx)w = wyw$, so $y \mapsto xyx$ is a restriction map $[0,p] \to [0,x]$ compatible with composition; and the transports of (c) commute with these restrictions (for $x, y \le p$ and $p \bowtie q$: $q(xyx)q = (qxq)(qyq)(qxq)$, using $xqy = xpqpy = xy$). In $\mathcal{P}(\mathcal{P}_X)$ the values of this presheaf are concrete: $[0,p] \cong \mathcal{P}(\mathcal{P}_{T(p)})$, a restriction of p being a labelled partition of the set $T(p)$ of transversal blocks (Lemma 1.1(a)), and the transport along $p \bowtie q$ is induced by the bijection $T(p) \to T(q)$ of Lemma 1.1(c).

(e) Pairwise matched projections of $\mathcal{P}_X$ generate a subsemigroup all of whose elements have the same rank: a product along a chain of matched projections keeps the upper blocks

of its first factor and the lower blocks of its last; hence $tt^* = e$ for the first factor e, the subsemigroup is simple, and being finite it is completely simple. So the $\bowtie$-cliques play the role of the rectangular $\mathcal{D}$-classes of a band; but they overlap, and the obstructions are about how overlapping cliques may be glued.

In this language the embedding problem is a descent problem: an abstract projection algebra comes with the presheaf of (d) and the transports of (c); the assignment $Y \mapsto \mathcal{P}(\mathcal{P}_Y)$ is a functor of finite sets (bijections act, and for arbitrary p, q the map $\theta_q\colon [0,p] \to [0,q]$ is induced by the correspondence $T(p) \to$ (classes of $\varepsilon_p \vee \varepsilon_q$) $\leftarrow T(q)$), and representability asks that the abstract data be induced through this functor from a system $p \mapsto T(p)$ of finite sets. The hypotheses of all our conditions are statements that certain transports of certain restrictions agree (the index pairs of the proof of Theorem 5.29 are these transports), and the conclusions are that an edge of $\bowtie$ exists; holonomy around closed paths of matchings is non-trivial in $\mathcal{P}_X$ itself (a permutation of $T(a)$; e.g. $ad \neq acbd$ in Remark 5.22(5)), so the conditions are not flatness but agreement between the holonomy and the non-matched θ's. We know no literature in which projection algebras are set up this way.

6. Computations

All regular $*$-semigroups of order $n \leq 8$ were enumerated up to $*$-isomorphism by a backtracking search on multiplication tables with propagation of associativity, $(xy)^* = y^*x^*$ and $xx^*x = x$; for $n = 7, 8$ only the semigroups whose involution is not the identity were generated, since $x^* = x$ for all x forces commutativity and hence an inverse (Clifford) semigroup, which embeds by Wagner–Preston. For each non-inverse semigroup satisfying (Q1), a $*$-embedding into $\mathcal{P}_n$ was searched for, $n \leq 5$, by assigning images to projections first and forcing the images of all products.

order	classes	inverse	non-inverse, (Q1)	non-inverse, not (Q1)	of these, in $\mathcal{P}_5$
4	17	16	1	0	1
5	55	52	2	1	2
6	222	208	9	5	9
7	259†	196	35	28	35
8	1310†	970	178	162	178

† nontrivial involution only. The last column counts the non-inverse semigroups satisfying (Q1) that embed in $\mathcal{P}_5$.

Thus, up to order 8, a regular $*$-semigroup embeds in a partition monoid if and only if it satisfies (Q1) (the ones failing (Q1) do not embed, the others embed in $\mathcal{P}_5$, inverse ones by Wagner–Preston). The smallest known counterexample to sufficiency of (Q1) is the 61-element Q_0 of Theorem 4.19. The smallest known counterexample to sufficiency of (Q1) together with the triangle condition is the 19-element $\mathfrak{T}_{19}$ of Theorem 5.16 (before it, the 20-element $\mathfrak{Q}$ of Definition 5.2); so the smallest counterexample has order between 9 and 19 (the order-9 enumeration, some 10^7 tables, was not completed).

The ten $*$-congruences of $\mathcal{P}_3$ have quotients of orders 203, 179, 103, 79, 67, 43, 7, 3, 2, 1. The quotients of orders $179, 79, 43$ have a zero and are not inverse, so they fail (Q1); those of orders $7, 3, 2, 1$ are inverse; the two remaining proper quotients, of orders 103 and 67, satisfy (Q1) and are the semigroups of Theorem 4.19. For $\mathcal{P}_2$ all proper quotients satisfying (Q1) embed. For $\mathcal{P}_4$ the classification of East, Mitchell, Ruškuc and Torpey (Adv. Math. 2018) leaves only two proper $*$-congruences whose quotients have no zero (a quotient with a zero satisfying (Q1) is inverse, as every idempotent lies above the zero); they are μ, which identifies each partition of rank ≤ 1 with the rank-0 partition obtained by cutting its transversal block, and $\mu \vee \lambda_{S_2}$, which

moreover identifies the two elements of each rank-2 $\mathcal{H}$-class. We verified directly that these are $*$-congruences of $\mathcal{P}_4$ (with 2771 and 1810 classes) and that they are the only ones contained in the second. Both quotients satisfy (Q1) and fail the triangle condition, as they must: they contain the corresponding quotients of $\mathcal{P}_3$ (add the singletons $\{4\}, \{4'\}$). So the $*$-congruences of $\mathcal{P}_4$ give nothing new.

Projection algebras. Mace4 with isofilter enumerated all projection algebras (axioms (P1)′–(P5)′) satisfying (Q1) and the triangle clause, up to isomorphism: there are $2, 5, 16, 62$ of orders $2, 3, 4, 5$. Each was tested for embeddability into $\mathcal{P}(\mathcal{P}_m)$, $m \leq 5$, by a backtracking search which forces the image of every product of already assigned elements; all embed, in $\mathcal{P}(\mathcal{P}_2)$, $\mathcal{P}(\mathcal{P}_3)$ or $\mathcal{P}(\mathcal{P}_4)$ (3, 39, 20 of the 62 of order 5). Of order 6 there are exactly 296 (from 79332 Mace4 models before isofilter). All but one embed in $\mathcal{P}(\mathcal{P}_m)$ with $m \leq 5$: one in $\mathcal{P}(\mathcal{P}_2)$, 90 in $\mathcal{P}(\mathcal{P}_3)$, 173 in $\mathcal{P}(\mathcal{P}_4)$ and 31 in $\mathcal{P}(\mathcal{P}_5)$; the exception is $\mathfrak{P}$, which is also the only one of the 296 violating the wiring condition (S). So up to order 6 the projection algebras that embed in some $\mathcal{P}(\mathcal{P}_X)$ are exactly those satisfying (Q1), the triangle condition and (S). Of order 9, $\mathfrak{P}_9$ (Example 5.7) satisfies (Q1), the triangle condition and the quasi-identity of Theorem 5.3 but not (S), and does not embed in $\mathcal{P}(\mathcal{P}_m)$ for $m \leq 8$ (SAT). ($\mathfrak{P}$ does not embed in $\mathcal{P}(\mathcal{P}_m)$ for $m \leq 6$ by exhaustive search, nor for $m \leq 8$ by a SAT encoding of the join and product structure solved by CaDiCaL, $m = 8$ taking about 25 minutes, in agreement with Theorem 5.3.) Of order 7 there are 1 663 projection algebras satisfying (Q1) and (S) (Section 5.6; 295 of order 6); all satisfy the triangle condition, and all but $\mathfrak{P}_7$ embed in $\mathcal{P}(\mathcal{P}_m)$ with $m \leq 6$ (161 in $\mathcal{P}(\mathcal{P}_3)$, 924 in $\mathcal{P}(\mathcal{P}_4)$, 531 in $\mathcal{P}(\mathcal{P}_5)$, 46 in $\mathcal{P}(\mathcal{P}_6)$), while $\mathfrak{P}_7$ does not embed at all (Theorem 5.16; the SAT search confirms this for $m \leq 8$). Of order 8 there are 10 828 projection algebras satisfying (Q1), (S) and (R) (Lemma 5.18); all satisfy the triangle condition, 10 821 embed in $\mathcal{P}(\mathcal{P}_m)$ with $m \leq 7$, and the seven of Section 5.8 do not embed for $m \leq 8$ (whether they embed for larger m is open). Of order 9 there are 80 039, of which seven embed in no $\mathcal{P}(\mathcal{P}_m)$ with $m \leq 8$ and contain none of the seven of order 8 (Remark 5.22(3)). The graph semigroups $Q(G, \perp)$ with $|V| \leq 3$ and $\perp$ a proper colouring all embed in $\mathcal{P}_4$ (and Theorems 5.8, 5.12 settle all graph and layered graph semigroups).

Further checks: several completely simple semigroups with non-flat sandwich matrices (an $\mathcal{M}[\mathbb{Z}_2; 3, 3; P]$ embeds in $\mathcal{P}_4$; an $\mathcal{M}[\mathbb{Z}_3; 3, 3; P]$ and the 48-element $\mathcal{M}[\mathbb{Z}_3; 4, 4; P]$ of the previous paper do not embed in $\mathcal{P}_5$ but embed in $\mathcal{P}_{27}$ and $\mathcal{P}_{48}$ by Theorem 3.1); strong semilattices of Rees matrix semigroups with injective-on-projections structure maps (all embedded); $H \cup (A \times A)$ with various actions (all embedded).

7. General remarks and questions

Proposition 7.1. *A finite regular $*$-semigroup embeds in some partition monoid if and only if it satisfies every quasi-identity (in the language $\{\cdot, ^*\}$) that holds in all partition monoids.*

Proof. Necessity is clear. Conversely, a finite S satisfying all quasi-identities of the class $\{\mathcal{P}_X\}$ lies in the quasivariety it generates, so S embeds in a product $\prod_j A_j$ of ultraproducts A_j of partition monoids $\mathcal{P}_{X_{ij}}$, $i \in I_j$, over ultrafilters $\mathcal{U}_j$. As S is finite, finitely many factors suffice, and by Łoś's theorem each $*$-homomorphism $S \to A_j$ is, for $\mathcal{U}_j$-almost all i, induced by a $*$-homomorphism $S \to \mathcal{P}_{X_{ij}}$ with the same kernel. Taking one such i for each of the finitely many j and using $\mathcal{P}_X \times \mathcal{P}_Y \leq \mathcal{P}_{X \sqcup Y}$ gives an embedding of S into a partition monoid. □

So for finite semigroups the programme of finding quasi-identities is complete in principle; whether finitely many suffice is unclear. We have three pairwise independent ones: (Q1), the triangle condition and the wiring condition (S) of Theorem 5.6 (independence: Section 5), and a fourth one, the condition (R) of Theorem 5.16, independent of the other three; by Theorem 4.14 all the other matching-pattern conditions of Section 4 follow from the triangle condition, so they

add nothing; up to order 6 the first three describe the projection algebras embeddable in some $\mathcal{P}(\mathcal{P}_X)$ exactly, and up to order 7 the four do (Section 6). Whether the class of all (possibly infinite) embeddable regular $*$-semigroups is closed under ultraproducts, i.e. is a universal class, I do not know; the natural map from an ultraproduct of partition monoids to a partition monoid is not a homomorphism, because connectivity through the middle layer is not a first-order property.

Proposition 7.2. *For every set X, the map $\alpha \mapsto R_\alpha$, where*

$$R_\alpha = \{(A, B) \in 2^X \times 2^X : A \sqcup B' \text{ is a union of blocks of } \alpha\},$$

is an injective homomorphism of involuted semigroups from $\mathcal{P}_X$ into the semigroup of binary relations on 2^X with converse as involution.

Proof. $(A, C) \in R_\alpha \circ R_\beta$ iff there is B which is a union of lower blocks of α and of upper blocks of β, i.e. a union of classes of the middle join, with $A \sqcup B'$ and $B \sqcup C'$ unions of blocks; this is exactly the condition for $A \sqcup C'$ to be a union of blocks of $\alpha\beta$. Injectivity: the minimal nonempty A with $(A, \emptyset) \in R_\alpha$ are the upper nontransversal blocks, and the pairs (A, B) with A, B minimal and nonempty are the transversal blocks. □

Consequently every $*$-subsemigroup of a partition monoid is representable by binary relations with converse, so Schein's quasi-identities for representable involuted semigroups are necessary conditions. They seem, however, to carry no information here: every regular $*$-semigroup of order ≤ 5 is representable by relations on at most 5 points (checked by computer), including the 5-element band with zero which fails (Q1) (represent the 2×2 band by $A \times A$, $B \times B$ with $A \cap B \neq \emptyset$ and the zero by the empty relation). The point of $\mathcal{P}_X$ is that its relations R_α are *Boolean subalgebras* of $2^X \times 2^X$ containing (X, X); conversely every such subalgebra comes from a partition. Thus S embeds in $\mathcal{P}_X$ iff S has a faithful representation by relations on the Boolean algebra 2^X each of which is a subalgebra of $2^X \times 2^X$.

Question 7.3.

(1) Is embeddability of a finite regular $*$-semigroup determined by its projection algebra? All known obstructions, (Q1), the triangle condition, (S), (R) and those of Proposition 5.19, are projection-algebra quasi-identities, and Theorem 3.1 shows that groups and sandwich matrices impose nothing; but a regular $*$-semigroup is far from determined by its projection algebra.

(2) (Q1) and the triangle condition do *not* suffice, even together with all matching-pattern conditions (Corollary 5.4). What is the smallest counterexample (order 9 to 19)? Is there a common generalisation of (R′) and (V), or of (R′) and the remaining path-closing conditions (P_1), (Q), with a uniform proof? (The hub case of Theorem 5.6 now has a computer-free proof, Corollary 5.30.) Are there further wiring-type obstructions, e.g. with longer paths of matchings or with two commuting pairs? For graph semigroups and for layered graph semigroups the answer is complete (Theorems 5.8 and 5.12): (Q1) and (S) suffice, and the triangle condition is automatic; so new obstructions have to be sought in families whose projections have non-chain down-sets or large T-blocks (Remark 5.13(2)). More generally, is the quasivariety generated by the projection algebras $\mathcal{P}(\mathcal{P}_X)$ finitely based? Its members of order ≤ 6 are exactly the algebras satisfying (Q1), the triangle condition and (S), and (S) is needed only once, for $\mathfrak{P}$; of order 7 one further condition, (R), is needed, once ($\mathfrak{P}_7$); of order 8 seven further candidates appear (Section 5.8), and the quasi-identities they violate, valid in $\mathcal{P}(\mathcal{P}_m)$ for $m \leq 7$, are of at least three different shapes, so the list of obstructions is not closing up quickly. (V), (R′), (P_1) and (Q) are proved (Theorems 5.20, 5.23, 5.25, 5.29); are (E_6), (E_1) of Proposition 5.19 valid in every $\mathcal{P}_X$? (They are independent of each other and of the rest, Proposition 5.32.)

(3) Is there a conceptual description of the quasi-identities of Section 4, e.g. in terms of "two commuting families with a common meet-preserving matching must have the same meet"? Theorem 4.11 says that for finite X the pattern of matchings is irrelevant as long as it is not a star. The proof of Theorem 4.2 suggests a general principle: in $\mathcal{P}_X$, a projection is determined by its behaviour outside its support together with its T-blocks, and matched pairs cannot share a set which is transversal for one and nontransversal for the other.

(4) Are (R″) and the conditions K0–K6 of Section 5.13 valid in every $\mathcal{P}_X$? Their witnesses would be the first minimal non-embeddable projection algebras of order 9 beyond the (R′)-witness. More importantly: is there a combinatorial criterion deciding, for a configuration of auxiliaries attached to an unclosed path $a \bowtie c \bowtie b$ (a "wiring diagram": commutations, half-matchings, equalities of transports of restrictions), whether it forces $a \bowtie b$, with one uniform proof? The known proofs ((R), (R′), (P_1), (Q)) all run by transport along the matchings and propagation of an invariant along a join, which suggests that the criterion exists; the growth of the list (one auxiliary at order 7, two at order 8, two or three plus the terms u, v at order 9) suggests that the family is infinite, so that no finite basis exists and a parametrised one is the best possible. Related: is embeddability of a finite projection algebra in some $\mathcal{P}(\mathcal{P}_X)$ decidable? (For finite algebras a recursive basis of quasi-identities is equivalent to decidability of embeddability.)

(5) The infinite case. By Theorem 4.14 the triangle condition implies every non-star condition in every projection algebra, and by Theorem 4.17 the hexagon condition does not imply the triangle condition; so for infinite regular $*$-semigroups the triangle condition is the matching-pattern condition to use; but by Section 5 the wiring conditions are needed as well, and their infinite-case behaviour is unexplored. Does the hexagon condition imply the conditions on $k \geq 4$ indices in general (Remark 4.18(1))? Which pattern conditions are equivalent to the triangle condition in the variety of projection algebras (the path and the 2-cycle are; the hexagon is not)?

Files

The directory `embed/` contains the code: `pmon.py` (partition monoids), `enum_rss.py` (enumeration of regular $*$-semigroups), `embed.py` (embedding search), `analyse.py`, `congr.py` (congruences and quotients), `qsearch.py`, `hexsearch*.py`, `hexmin*.py`, `hexqi.py`, `qi.py` (searches for the hexagon configuration and quasi-identity tests), `hexagon.in`, `triangle.in`, `octagon.in` (Prover9/Mace4 inputs; the projection algebra is written with the binary operation $X * Y = \theta_Y(X)$, a device for Prover9 and Mace4, rather than with the unary operations θ_Y of East and Azeef Muhammed, arXiv:2301.04845; see Section 2), `digraph.py` (clauses and $\mathcal{P}_n$-checks for the matching-pattern conditions), `oct14.py`, `p4mu.py`, `octmin.py`, `octsub6.py` (the octagon experiments of Remark 4.3), `invariant.in`, `stab.in`, `period.in`, `uniform.in`, `derived.in` (Prover9/Mace4 inputs behind Section 4.1), `mace4parse.py`, `walk_hex.py`, `uniform_hex.py` (analysis of Mace4 models), `path.in`, `path2.in`, `square.in`, `square2.in`, `disjoint.in`, `disjoint2.in`, `disjoint3.in`, `sqfilter2.py`, `sqstar.py`, `digraph3.py`, `digraph4.py`, `nofull.py`, `nofull3.py`, `p4mu_disjoint.py` (Theorem 4.11 and Remark 4.15); `pa_enum.py`, `pa_embed2.py`, `pa_sat.py` (enumeration of projection algebras with (Q1) and the triangle clause, embedding into $\mathcal{P}(\mathcal{P}_m)$ by backtracking and by SAT), `graphsg.py`, `graphsg_survey.py`, `graphsg_thm.py` (the semigroups $Q(G, \perp)$, the survey of Remark 5.9(2) and the verification of the construction of Theorem 5.8), `layered.py`, `layered_search.py`, `layered_search2.py`, `layered_thm.py` (layered graph semigroups, Section 5.5), `pa7_q1S.in`, `pa7_driver.sh`, `pa7_post.py`, `fourth.py`, `fourth_pre.py` (the order-7 census, $\mathfrak{P}_7$, $\mathfrak{T}_{19}$, $\mathfrak{T}_{26}$ and the tests of (R), Section 5.6), `pathclose.py`, `pathclose_np.py`,

`pathclose4.py`, `pathclose5.py`, `pathclose_check.py`, `pc_prover.py`, `pc_mace_in.py`, `pc_mace.sh`, `pathclose2.py`, `pathclose2r.py`, `pc2_mace.py` (the search for path-closing conditions, Section 5.7), `pa8/extend.py`, `pa8/post8.py`, `pa8/cand8.py`, `pa8/cand8b.py`, `pa8/qi_extract.py`, `pa8/qi_all.py`, `pathclose3.py`, `pa8/pc3_hit.py`, `pa8/pc3_verify.py`, `pa8/pc1_classify.py`, `pa8/verify6.py`, `pa8/verify7.py`, `pa8/sat8.py`, `pa8/newconds.py`, `pa8/pass2_9.py`, `pa8/indep.py`, `pa8/run_indep.py`, `pa8/vexp.py` (the experiments behind the proof of Theorem 5.20), `pa8/p9/` (Prover9 inputs; `pa8/p9/conditions.in` collects the projection algebra axioms and all conditions of Section 5.12 with labels), `pa8/p1/` (instances of (P_1) in $\mathcal{P}(\mathcal{P}_4)$ and $\mathcal{P}(\mathcal{P}_5)$ and the block-level tests behind Theorem 5.25), `pa8/q/`, `pa8/gen/` (the reconnaissance of (Q) and the search for a common generalisation, Remark 5.22(5)), `pa8/rp/` (the experiments behind the proof of Theorem 5.23: `rp_inst.py` enumerates the instances of the hypotheses of (R') in $\mathcal{P}(\mathcal{P}_n)$, `rp_words.py`, `rp_min.py`, `rp_rel.py` the identities and relations they satisfy, `harness*.py` the block-level statements, `pa_*.in`, `g_*.in` the Prover9 inputs for Step 1), and the census files `pa8/ord`n`.json`, `pa8/post8.log` (the census by extension, the seven algebras of order 8 and the quasi-identities of Section 5.8), `m172*.py`, `qi2.py`, `wiringcheck.py` (tests of the wiring conditions in $\mathcal{P}_n$ and in given projection algebras), `hubtypes.py`, `hubcheck_rand.py` (the finite verification in the proof of Theorem 5.6), `wiring_*.in` (Prover9/Mace4 inputs relating (S), Theorem 5.3, (Q1) and the triangle clause; `mace_WnotS9.out` is $\mathfrak{P}_9$), `p4cong.py`, `p4cong2.py` (the $*$-congruences of $\mathcal{P}_4$) (Section 5; the Mace4 models are the files `mace_*.out`), `hinf2.py`, `hinf2_run.py`, `hinf2_pat.py`, `chain.py` (the unrolled hexagon of Theorem 4.17 and the chain models), `periodic.py`, `periodic3.py` (periodic projections of $\mathcal{P}_{\mathbb{Z}\times[m]}$ and the pattern searches of Remark 4.18(3)), `cs_embed.py` (the construction of Theorem 3.1), `normalband*.py`, `strongsl.py`, `examples*.py` (further test families), `relrep.py` (representations by binary relations), and the pickled enumerations `rss_*.pkl`. The directory `pa8/n4/` contains the code of Section 5.13 (`README.md` describes it: `pathclose4b.py`, `hit4.c`, `verify_hits.py`, `qi.py`, `explain2.py`, `explain3b.py`, `catalog.py`, `checkK.py`, `min9.py`, `mace_gen.py`; logs `catalog.log`, `explain3b_all.log`). The directory `pa8/q2/` contains the instance generators, the harness `hq.py`, the claim files `t*.py` (earlier reconnaissance) and `h1.py`–`h4.py` (the statements of Steps 2–7 of Section 5.11: cross wiring, the claims of Proposition 5.28, $\Pi_1 = \varepsilon_u \vee \varepsilon_d$, and the contact rules), and `NOTES_Q.md`.